

L-convex polyominoes and a colored-partition comparison

A proof of the A126764 asymptotic, three correction terms,
an all-orders radial expansion, and inverse asymptotics

Part II: the all-orders coefficient theorem

Part III: a complete transfer proof

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Abstract

Let a_n count fixed L-convex polyominoes of area n , with $a_0 = 1$. Starting from the established area generating function, we prove the Guttmann–Kotěšovec asymptotic recorded as a conjecture in OEIS A126764:

$$a_n \sim \frac{13\sqrt{2}}{768 n^{3/2}} \exp\left(\pi\sqrt{\frac{13n}{6}}\right).$$

The main device is a positive comparison with a colored-partition Euler product $R(q)$, given explicitly below. If $p_n = [q^n]R(q)$, we obtain the finite, coefficientwise bound $0 \leq p_n - 4a_n \leq 3(p_n - 2p_{n-1} + p_{n-2})$ for every $n \geq 3$. A higher finite-cut identity then proves three explicit correction terms, with relative remainder $O(n^{-7/4})$. Independently, a finite recurrence computes the complete radial Poincaré expansion of $\mathcal{A}(e^{-t})$ to any prescribed order. We also derive an asymptotic inverse, including the necessary distinction between a real interpolation and an integer threshold. Exact integer checks through $n = 2000$ and symbolic coefficient checks accompany the manuscript. No numerical fit enters the proofs.

Scope and status. The proofs below are conventional mathematical proofs, not Lean certificates or a claim of independent peer review. The exact enumerative generating function is a cited input; its asymptotic analysis is supplied here. The all-orders *radial* expansion is proved. An all-orders *coefficient* expansion, and exponentially small transseries sectors for a_n , are not claimed.

[write] Added 2 October 2026, batch 77P5. The box describes Part I. Part II (Sections 13–23), a second manuscript, proves the all-orders *coefficient* expansion (Theorem 14.1). Exponentially small transseries sectors remain unclaimed in both parts.

[write] Added 5 October 2026, batch 101. Since batch 101 the report has a third part, Part III (Sections 24–34). It proves the all-orders coefficient expansion again (Theorem 25.1), by a third route whose estimates are all written out. Exponentially small transseries sectors remain unclaimed in all three parts.

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1 The problem and the main results

1.1 The sequence and the source boundary

An L-convex polyomino is a convex polyomino in which any two cells can be joined inside the polyomino by a path with at most one turn. We count fixed polyominoes: translations are identified, but rotations and reflections are not generally identified. This is the convention in OEIS A126764 [1]. The initial terms are

$$1, 1, 2, 6, 15, 35, 76, 156, 310, 590, 1098, 1984, 3515, 6094, \dots$$

The zero-area term is a generating-function convention.

Castiglione, Frosini, Munarini, Restivo, and Rinaldi established an area generating function [2]; we use its explicit recurrence form as reproduced in Guttmann and Kotěšovec [3], equation (1.1). The latter authors proposed the leading asymptotic from a numerical analysis. The OEIS entry inspected for this report still labels that formula a conjecture [1]. A targeted search did not locate an earlier proof of the specific area asymptotic. This search is not a comprehensive priority certification.

The ProveIt repository [7] was inspected as the proposed research context. A targeted repository search for A126764 returned no matching file. The present manuscript is a standalone contribution: it does not assume any unverified theorem from another repository report. Enumeration by perimeter, and the asymptotics of 201-avoiding ascent sequences discussed alongside L-convex polyominoes in [3], are separate questions and are not asserted to be solved here.

[write] Added 2 October 2026, batch 75. This report is a single manuscript, printed in full: manuscript 05 of batch 75 of ProveIt’s incoming-reports intake, delivered as the archive A126764_Lconvex_Research.zip in commit 4b874cea0 (main file lconvex_asymptotics.tex, now article.tex) and placed in commit 6ea60e367 in the research-report collection, among the OEIS-sequence asymptotics reports. The manuscript names no ProveIt commit: its repository input was an overview and a targeted search for A126764 on 1 October 2026, so it has no pin. That search is still accurate for the repository it saw; the only match now is this report. Its PDF metadata records it as “prepared with ChatGPT”. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. The writing step changed only the labels (every label now carries the prefix lcp:), set the bibliography ragged-right, and added these [write] notes and two references [8, 9]; no statement, proof or number of the manuscript was altered. The polyomino report nearest in subject, polyomino-growth-finite-prefix-corrections [9], bounds Klarner’s growth constant $\lambda \leq 4.498$ for *all* polyominoes (exponential growth, with a Lean development Combinatorics/Polyominoes/KlarnerConstant for the weaker bound 9047/2000); the L-convex class grows only like $e^{B\sqrt{n}}$, so the two constants are unrelated and no theorem is shared. The report skew-partition-continued-fractions touches parallelogram polyominoes (A006958) only.

[write] Added 2 October 2026, batch 77P1. The 201-avoiding ascent sequences that the preceding paragraph sets aside are now treated in the sibling report a202062-ascent-201-enumeration (manuscript 43 of batch 77): it proves the Guttmann–Kotěšovec algebraic (cubic) generating function for A202062 and its all-orders coefficient asymptotics. The two problems share only the paper [3]; no theorem, constant or method of this report is used there, and nothing here changes.

[write] Added 2 October 2026, batch 77P5. The note above describes Part I, Sections 1–12 and Appendices A and B. Since batch 77 the report has a second part, Part II (Sections 13–23), printed in full from manuscript 31 of batch 77 (archive lconvex-asymptotics-reproducibility.zip, arrival commit 096ee7b87, placement commit 4f11bc9c0; no pin). It proves the all-orders coefficient expansion that Part I leaves open, by an exact decomposition $\mathcal{A} = RD^2 + (\text{an exponentially smaller remainder})$ in Part I’s letters. Its letters clash with Part I’s; Table 1 resolves them.

[write] **Added 5 October 2026, batch 101.** Part III (Sections 24–34) prints bundle Report 95 of batch 101 in full (archive `L_Convex_Polyomino_Area_Asymptotics_Source.zip`, arrival commit `60f54ea06`, placement commit `f7c612c72`; no pin). It proves the decomposition $\mathcal{A} = 1 + R^1 D^2 + (\text{a positive remainder})$ in Part I's letters by a duality argument, and the all-orders coefficient expansion by positivity, the common partition factor $(q; q)_N$ and Chebyshev interpolation. It knows Part I's manuscript but not Part II. Its letters clash with both parts; Table 4 resolves them.

1.2 Constants and coefficient asymptotics

Throughout, put

$$K = \frac{13\pi^2}{24}, \quad B = 2\sqrt{K} = \pi\sqrt{\frac{13}{6}}, \quad c = \frac{13\sqrt{2}}{768}. \quad (1.1)$$

The letter K denotes a constant, not a generating function.

Theorem 1.1 (Coefficient expansion). *As $n \rightarrow \infty$ through the positive integers,*

$$a_n = c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + \frac{e}{n} + \frac{f}{n^{3/2}} + O(n^{-7/4}) \right), \quad (1.2)$$

where

$$d = \frac{\sqrt{K}}{6} + \frac{3}{2\sqrt{K}} = \frac{B}{12} + \frac{3}{B}, \quad (1.3)$$

$$e = \frac{1}{2} + \frac{3}{4K} - \frac{35K}{72}, \quad (1.4)$$

$$f = \frac{755K^{3/2}}{1296} + \frac{175\sqrt{K}}{72} - \frac{5}{8\sqrt{K}}. \quad (1.5)$$

Numerically,

$$d = 1.0341052176262747\dots, \quad e = -1.9584764928967362\dots, \quad f = 12.550445779146218\dots$$

In particular, the leading asymptotic stated in the abstract holds.

There are two complementary proof routes. The first proves the leading formula and its $n^{-1/2}$ correction by an explicit positive squeeze. The second uses a finite rational approximation and a contour estimate; it proves all the terms in Theorem 1.1.

Theorem 1.2 (A finite coefficient squeeze). *Define*

$$R(q) = \frac{(-q; q^2)_\infty}{(q; q)_\infty^3} = \frac{(q^2; q^2)_\infty^2}{(q; q)_\infty^4 (q^4; q^4)_\infty} = \sum_{n \geq 0} p_n q^n, \quad J(q) = (1 - q)^2 R(q) = \sum_{n \geq 0} j_n q^n. \quad (1.6)$$

Then, for every $n \geq 3$,

$$0 \leq p_n - 4a_n \leq 3j_n = 3(p_n - 2p_{n-1} + p_{n-2}). \quad (1.7)$$

Both R and J have nonnegative integer coefficients. Moreover,

$$n \left(1 - \frac{4a_n}{p_n} \right) \longrightarrow \frac{K}{2} = \frac{13\pi^2}{48}. \quad (1.8)$$

The product R counts partitions with four colors available for an odd part, two for a part congruent to 2 modulo 4, and three for a multiple of 4. Thus the constant 13 comes from the average number $13/4$ of colors, not from a fit to a decimal growth constant. The factor $1/4$ in the comparison is forced by two boundary conditions of the recurrence. We do not assert a four-to-one combinatorial map; finding one is a natural further problem.

1.3 What the all-orders statement means

Let $t > 0$ and $q = e^{-t}$. We will prove

$$\mathcal{A}(e^{-t}) \sim \frac{t^{3/2}}{4(2\pi)^{3/2}} \exp\left(\frac{K}{t} - \frac{t}{6}\right) \mathcal{F}(t), \quad (1.9)$$

where $\mathcal{F}$ is a formally specified series beginning

$$\begin{aligned} \mathcal{F}(t) = & 1 - \frac{1}{2}t^2 + \frac{1}{2}t^3 - \frac{41}{48}t^4 + \frac{7}{4}t^5 - \frac{6317}{1440}t^6 \\ & + \frac{1529}{120}t^7 - \frac{1702513}{40320}t^8 + \dots \end{aligned} \quad (1.10)$$

Here $\sim$ is meant to all fixed algebraic orders along the positive real axis. It is not a claim that this series converges, nor that a real-axis expansion alone implies coefficient asymptotics. The complex estimate needed for the three coefficient corrections is proved separately.

2 An exact recurrence with a positive normalization

Write $(a; q)_m = \prod_{j=0}^{m-1} (1 - aq^j)$, and set $(q; q)_0 = 1$. The enumerative input is

$$\mathcal{A}(q) = 1 + \sum_{n \geq 1} \frac{q^n h_n(q)}{(q; q)_{n-1}^2 (1 - q^n)}, \quad (2.1)$$

where

$$h_0 = 1, \quad h_1 = 1, \quad h_{n+1} = 2h_n - (1 - q^n)^2 h_{n-1} \quad (n \geq 1). \quad (2.2)$$

Indeed, $h_n = f_{n-1}$ for $n \geq 1$ in the notation of [3], and $h_2 = 1 + 2q - q^2$. Introducing h_0 makes the recurrence valid already at $n = 1$.

The auxiliary solution

$$Q_0 = 1, \quad Q_1 = 2, \quad Q_{n+1} = 2Q_n - (1 - q^n)^2 Q_{n-1} \quad (2.3)$$

will produce the comparison product. For a general solution y_n of the same recurrence, define

$$b_n = \frac{y_n}{(q; q)_n^2}, \quad e_n = b_n - b_{n-1}, \quad T_n(q) = \frac{1}{(1 - q^n)^2} - 1.$$

We reserve d, e, f without subscripts for the constants in Theorem 1.1.

Lemma 2.1 (Positive transfer). *For $n \geq 2$,*

$$\begin{aligned} b_n &= \frac{b_{n-1} + e_{n-1}}{(1 - q^n)^2}, \\ e_n &= e_{n-1} + T_n(q)(b_{n-1} + e_{n-1}). \end{aligned} \tag{2.4}$$

Every entry of this linear transfer rule is a power series with nonnegative coefficients. Consequently, coefficientwise inequalities between two pairs (b_m, e_m) propagate to all later pairs and to their limiting slopes, whenever the slopes exist.

Proof. Dividing the recurrence by $(q; q)_{n-1}^2$ gives $(1 - q^n)^2 b_n = 2b_{n-1} - b_{n-2} = b_{n-1} + e_{n-1}$. Subtract b_{n-1} . Finally, $T_n = \sum_{r \geq 1} (r+1)q^{nr}$ has nonnegative coefficients. $\square$

For h , the initial normalized pair is

$$b_1 = (1 - q)^{-2}, \quad e_1 = (1 - q)^{-2} - 1,$$

and both entries are nonnegative. The same holds for Q , with $b_1 = 2(1 - q)^{-2}$ and $e_1 = 2(1 - q)^{-2} - 1$. Throughout, $F \preccurlyeq G$ means that every coefficient of $G - F$ is nonnegative. This is a stronger relation than $F(q) \leq G(q)$ for $0 < q < 1$. We will be careful not to multiply a coefficientwise inequality by a polynomial with negative coefficients.

3 Limiting slopes and the exact Euler product

3.1 Existence and analyticity of the slopes

Lemma 3.1. *For a solution with polynomial initial values and every $0 < r < 1$, there are analytic functions $s(q), b(q)$ on $|q| < 1$ such that, uniformly on $|q| \leq r$,*

$$y_n(q) = s(q)n + b(q) + O_r(n^2 r^n).$$

In particular, the normalized limiting slope

$$\mathcal{L}_y(q) = \lim_{n \rightarrow \infty} \left(\frac{y_n}{(q; q)_n^2} - \frac{y_{n-1}}{(q; q)_{n-1}^2} \right) = \frac{s(q)}{(q; q)_\infty^2} \tag{3.1}$$

exists locally uniformly. The same limit exists coefficientwise as a formal power series.

Proof. Put $v_j = 2q^j - q^{2j}$. Twice summing $y_{j+1} - 2y_j + y_{j-1} = v_j y_{j-1}$ gives

$$y_n = y_0 + n(y_1 - y_0) + \sum_{j=1}^{n-1} (n-j)v_j y_{j-1}.$$

On $|q| \leq r$, $|v_j| \leq 3r^j$. Divide by $n+1$ and apply the elementary finite-product version of discrete Gronwall's inequality. The resulting bound is a constant times $\prod_{j \geq 1} (1 + 3jr^j) < \infty$. Thus $y_n = O_r(n+1)$. The increments $y_{n+1} - y_n$ have a uniformly convergent sum of successive differences, with exponentially small tails. One more summation yields the displayed affine expansion. Since $(q; q)_n / (q; q)_\infty = 1 + O_r(r^n)$, normalization gives (3.1). Finally, in (2.4) the change in e_n has valuation at least n . Each fixed coefficient therefore stabilizes. $\square$

Write $L = \mathcal{L}_h$. If

$$C(z; q) = \sum_{n \geq 0} \frac{h_n(q)}{(q; q)_n^2} z^n, \quad C_j(q) = C(q^j; q), \quad (3.2)$$

then C has radius of convergence at least 1 in z and a possible double pole at $z = 1$ with leading coefficient L . Equation (2.1) becomes

$$\mathcal{A} = 1 + C_1 - C_2. \quad (3.3)$$

Multiplying the normalized recurrence by z^n and summing, including the initial condition $h_1 = 1$, yields

$$(1 - z)^2 C(z; q) - 2C(qz; q) + C(q^2 z; q) = -z. \quad (3.4)$$

Taking $z \rightarrow 1$ is now legitimate and gives

$$L = 2C_1 - C_2 - 1, \quad \mathcal{A} = L - C_1 + 2. \quad (3.5)$$

The pole term must not be dropped in this limit.

3.2 Solving the auxiliary generating function

Define $G(z) = \sum_{n \geq 0} Q_n z^n / (q; q)_n$. Directly from (2.3),

$$(1 - z)^2 G(z) = (1 + qz^2) G(qz), \quad G(0) = 1.$$

Iteration gives the exact identity

$$G(z) = \frac{(-qz^2; q^2)_\infty}{(z; q)_\infty^2}. \quad (3.6)$$

At $z = 1$, the coefficient of $(1 - z)^{-2}$ is $(-q; q^2)_\infty / (q; q)_\infty^2$. It follows, either by subtracting the principal part or by Lemma 3.1, that

$$\mathcal{L}_Q = \frac{(-q; q^2)_\infty}{(q; q)_\infty^3} = R(q). \quad (3.7)$$

The second expression for R in (1.6) follows from $(-q; q^2)_\infty = (q^2; q^2)_\infty^2 / ((q; q)_\infty (q^4; q^4)_\infty)$. In particular,

$$R(q) = \prod_{j \geq 1} (1 - q^j)^{-b_j}, \quad b_j = 4 - 2\mathbf{1}_{2|j} + \mathbf{1}_{4|j}. \quad (3.8)$$

This proves the colored-partition interpretation stated earlier.

4 A coefficientwise proof of the leading conjecture

4.1 An exact positive defect

Let $v_n = Q_n - 2h_n$. Then $v_0 = -1$, $v_1 = 0$, $v_2 = (1 - q)^2$. Its normalized pair at index 2 is

$$b_2 = e_2 = (1 - q^2)^{-2}.$$

The positive transfer lemma shows that its slope

$$E = R - 2L \quad (4.1)$$

has nonnegative coefficients. Combining (3.3) and (3.5) gives the exact decomposition

$$R - 4\mathcal{A} + 6 = E + 2C_2. \quad (4.2)$$

Both terms on the right are nonnegative power series. The constant 6 only corrects the chosen zero-area terms.

4.2 A positive majorant for the defect

We prove the bounds

$$0 \preccurlyeq E \preccurlyeq J, \quad 0 \preccurlyeq C_2 \preccurlyeq J + q + 2q^2. \quad (4.3)$$

These statements require a little care: a real-axis bound by $(1 - q)^2 R$ would not by itself imply a bound on coefficients.

First multiply the normalized Q solution by $(1 - q)^2$, and denote the resulting pair by $(\tilde{b}_n, \tilde{e}_n)$. At $n = 2$,

$$\begin{aligned} \tilde{b}_2 &= \frac{3 + 2q - q^2}{(1 - q^2)^2}, \\ \tilde{e}_2 &= \frac{1 + 2q + 3q^2 - 2q^4}{(1 - q^2)^2}. \end{aligned}$$

Subtracting the normalized pair for v gives

$$\frac{2 + 2q - q^2}{(1 - q^2)^2}, \quad \frac{2q + 3q^2 - 2q^4}{(1 - q^2)^2}.$$

Both are nonnegative power series: use $2 + 2q - q^2 = (1 + 2q) + (1 - q^2)$ and $3q^2 - 2q^4 = q^2 + 2q^2(1 - q^2)$. The transfer lemma and passage to slopes prove $E \preccurlyeq J$. The displayed pair for $\tilde{b}_2, \tilde{e}_2$ itself is nonnegative by the same decompositions.

Next, every coefficient of $(1 - q^n)h_n/(q; q)_n^2$ is nonnegative. For $n = 1$ this is immediate; for $n \geq 2$ it follows by removing only one of the two factors $(1 - q^n)$ in the denominator in (2.4). Hence $C_2 - C_3 \geq 0$. Setting $z = q$ in (3.4) gives

$$2C_2 - C_3 = (1 - q)^2 C_1 + q, \quad C_2 \preccurlyeq (1 - q)^2 C_1 + q.$$

It remains to compare the last expression with J .

Let $C_j^Q = \sum_{n \geq 0} Q_n q^{jn}/(q; q)_n^2$. For the Q solution the initial-condition term in (3.4) is zero, so $R = 2C_1^Q - C_2^Q$. The solution $u_n = Q_n - h_n$ has $u_0 = 0, u_1 = 1$; after normalization and multiplication by $(1 - q)^2$ its initial pair at 1 is $(1, 1)$. Consequently,

$$(1 - q)^2(C_1^Q - C_1) \geq 0.$$

Also

$$(1 - q)^2(C_1^Q - C_2^Q) = \sum_{n \geq 1} q^n(1 - q^n)\tilde{b}_n \geq -2q^2.$$

For completeness, the $n = 1$ summand is $2q - 2q^2$. For $n = 2$ the remaining factor is $(3 + 2q - q^2)/(1 - q^2) = 1 + (2 + 2q)/(1 - q^2)$, which is nonnegative. For $n \geq 3$, positivity follows from the nonnegative pair at index 2 and (2.4). These two inequalities imply $(1 - q)^2 C_1 \preccurlyeq J + 2q^2$, establishing (4.3).

Taking coefficients of (4.2) now proves (1.7). In particular, the finite bound is independent of any asymptotic approximation.

4.3 Why the squeeze is asymptotically sharp enough

The analytic estimates in Section 6 imply

$$p_n = 4c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + O(n^{-1}) \right), \quad (4.4)$$

$$j_n \sim \frac{K}{n} p_n. \quad (4.5)$$

Therefore the squeeze alone gives

$$a_n = c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + O(n^{-1}) \right).$$

This already proves the full proposed leading constant and the first correction. The further work below is needed for e and f , not for the leading conjecture.

5 Finite cuts: an exact rational approximation

5.1 General positive remainder series

The discrete Wronskian of h and Q is

$$h_m Q_{m+1} - h_{m+1} Q_m = (q; q)_m^2. \quad (5.1)$$

At $m = 0$ both sides are 1; the recurrence multiplies each succeeding Wronskian by $(1 - q^m)^2$, proving the identity by induction. Define

$$E_m = h_m R - Q_m L \quad (m \geq 1). \quad (5.2)$$

Thus $E_1 = E$.

Lemma 5.1 (A positive tail with a small radial bound). *The series E_m has nonnegative coefficients. For real $0 < q < 1$,*

$$0 \leq E_m(q) \leq (q; q)_m^2 R(q). \quad (5.3)$$

Furthermore, for each fixed m ,

$$C_{m+1}(e^{-t}) = O_m(t^{2m} R(e^{-t}) + 1) \quad (t \downarrow 0). \quad (5.4)$$

Proof. For $n \geq m$, the solution $w_n = h_m Q_n - Q_m h_n$ has $w_m = 0$, $w_{m+1} = (q; q)_m^2$. Its normalized pair at $m+1$ is $(1 - q^{m+1})^{-2}$ in both coordinates. Hence it is positive and its limiting slope, E_m , is positive.

For the real bound, let $z_m = 0, z_{m+1} = 1$ satisfy the original recurrence for later indices. Then $w_n = (q; q)_m^2 z_n$. For real q , the increments of Q_n are positive and nondecreasing:

$$Q_{n+1} - Q_n = (Q_n - Q_{n-1}) + (2q^n - q^{2n}) Q_{n-1}.$$

Thus $Q_n \geq n+1$ and $\Delta Q_m = Q_m - Q_{m-1} \geq 1$. A second Wronskian calculation yields

$$\frac{s_z}{s_Q} = Q_m \sum_{k=m}^{\infty} \frac{\prod_{j=m+1}^k (1 - q^j)^2}{Q_k Q_{k+1}} \leq \frac{1}{\Delta Q_m} \leq 1.$$

To obtain the middle inequality, use $Q_k \geq Q_m + (k - m)\Delta Q_m$ and telescope the reciprocals of the resulting arithmetic progression. Since normalization divides both slopes by $(q; q)_\infty^2$, (5.3) follows.

Finally, positivity gives $C_1 \leq C_1^Q \leq R$ on the real axis. Indeed, $Q - h$ is positive after normalization, and $R - C_1^Q = C_1^Q - C_2^Q \geq 0$. From (3.4) and $C_{j+1} \geq C_{j+2}$ on the real axis,

$$C_{j+1} \leq (1 - q^j)^2 C_j + q^j.$$

Iterate this inequality a fixed number m of times. The leading coefficient is $(q; q)_m^2 = O_m(t^{2m})$, and the finitely many remaining terms are bounded as $t \downarrow 0$. $\square$

5.2 Eliminating the auxiliary evaluations

Put $U_m = Q_m - h_m$, and let polynomials P_m be specified by

$$P_0 = P_1 = 0, \quad P_{j+1} = 2P_j - (1 - q^j)^2 P_{j-1} + q^j.$$

Iteration of (3.4) gives

$$C_{m+1} = Q_m C_1 - U_m(L + 1) - P_m.$$

Solving this identity together with (5.2) and $\mathcal{A} = L - C_1 + 2$ proves the following exact formula.

Proposition 5.2 (Finite-cut identity). *Where $Q_m(q) \neq 0$,*

$$\mathcal{A}(q) = \left(\frac{h_m}{Q_m} \right)^2 R - \frac{h_m E_m}{Q_m^2} - \frac{C_{m+1}}{Q_m} + 2 - \frac{U_m + P_m}{Q_m}. \quad (5.5)$$

After multiplication by Q_m^2 , this is an identity of analytic functions on the entire open unit disk and of formal power series.

The square in the first term is the precise source of the factor $1/4$: for every fixed $m \geq 1$, $h_m(1) = 2^{m-1}$ and $Q_m(1) = 2^m$. The identity also provides an algebraic expansion algorithm and a route to rigorous coefficient bounds.

For the coefficient theorem we use only $m = 3$:

$$\begin{aligned} h_3 &= 1 + 4q - q^4, & Q_3 &= 4 + 4q + 2q^2 - 2q^4 = -2(q + 1)(q^3 - q^2 - 2), \\ U_3 + P_3 &= 3 + 2q + 3q^2 - q^4, & F_3(q) &= \left(\frac{1 + 4q - q^4}{4 + 4q + 2q^2 - 2q^4} \right)^2. \end{aligned} \quad (5.6)$$

The cubic $q^3 - q^2 - 2$ has no zero in $|q| \leq 1$. A zero there would satisfy $|q|^2|q - 1| = 2$; equality in the bound $|q|^2|q - 1| \leq 2$ forces $q = -1$, which does not solve the equation. Thus $F_3 R$ is analytic in $|q| < 1$, with only a possible additional boundary pole at $q = -1$.

6 Saddle estimates for the comparison product

6.1 The local modular expansion

The classical eta transformation, in its Euler-product form, is

$$(e^{-z}; e^{-z})_\infty = \sqrt{\frac{2\pi}{z}} \exp\left(-\frac{\pi^2}{6z} + \frac{z}{24}\right) (e^{-4\pi^2/z}; e^{-4\pi^2/z})_\infty, \quad \Re z > 0. \quad (6.1)$$

The square root is the branch positive for positive real z . This follows from the usual modular transformation of Dedekind's eta function; see [5], equations 27.14.12–27.14.14. Applying it at $z, 2z, 4z$ gives, uniformly in any fixed closed sector $|\arg z| < \pi/2$,

$$R(e^{-z}) = \left(\frac{z}{2\pi}\right)^{3/2} \exp\left(\frac{K}{z} - \frac{z}{6}\right) \left(1 + O(e^{-\kappa/|z|})\right) \quad (6.2)$$

for an appropriate sector-dependent $\kappa > 0$. In particular, there is no omitted algebraic power-series correction in (6.2) beyond $\exp(-z/6)$.

6.2 A minor-arc bound rather than a Tauberian shortcut

Let $r = e^{-t}$. The colored-product form (3.8) implies

$$\frac{|R(re^{i\theta})|}{R(r)} \leq \exp\left[-\frac{c_0}{t} \min\left\{1, \frac{\theta^2}{t^2}\right\}\right] \quad (-\pi \leq \theta \leq \pi) \quad (6.3)$$

for all sufficiently small t and some $c_0 > 0$. Here and below a constant such as c_0 is unrelated to c in (1.1).

To verify (6.3), restrict the product to integers $N \leq j \leq 2N$, with N comparable to $1/t$. Then r^j stays in a compact subinterval of $(0, 1)$, and

$$\log \frac{|1 - r^j e^{ij\theta}|}{1 - r^j} \geq c_1(1 - \cos(j\theta)).$$

In the product for $|R(re^{i\theta})|/R(r)$, every omitted factor is at most 1. The elementary trigonometric-sum estimate

$$\sum_{j=N}^{2N} (1 - \cos(j\theta)) \geq c_2 N \min\{1, N^2 \theta^2\}$$

finishes the proof. For $|\theta| \leq 1/N$ it follows from the quadratic bound for $1 - \cos x$; for an intermediate compact range of $N\theta$ it follows by comparison with its strictly positive integral; for larger $N\theta$ use the geometric-sum bound for $\sum e^{ij\theta}$. This argument also covers the neighborhood of $\theta = \pi$.

Consequently a rational factor with at most fixed-order poles on the unit circle does not create a competing dominant saddle. The product's exponential suppression outweighs such polynomial losses.

6.3 The transfer calculation

For a local expansion

$$H(e^{-z}) \sim \lambda z^{3/2} e^{K/z} \sum_{j \geq 0} s_j z^j,$$

with the preceding minor-arc control, Cauchy's integral is localized at $t = \sqrt{K/n}$ and angular width $t^{3/2}$. Termwise saddle integration gives

$$[q^n]H(q) \sim \frac{\lambda K}{2\sqrt{\pi}} n^{-3/2} e^{2\sqrt{Kn}} \sum_{\ell \geq 0} n^{-\ell/2} \sum_{j+k=\ell} s_j K^{j/2} \frac{\beta_k(j)}{(2\sqrt{K})^k}, \quad (6.4)$$

where

$$\beta_k(j) = \frac{(-1)^k}{k!8^k} \prod_{r=1}^k ((5+2j)^2 - (2r-1)^2), \quad \beta_0(j) = 1. \quad (6.5)$$

For clarity, (6.4) asserts a finite-order expansion whenever a corresponding finite-order local expansion is available.

Here is a direct way to check the constants. The model integral for $z^{3/2+j}e^{K/z}$ is the Bessel-type contour expression

$$\left(\frac{K}{n}\right)^{(5/2+j)/2} L_{-5/2-j}(2\sqrt{Kn}).$$

Its large positive-argument saddle expansion is $e^x/\sqrt{2\pi x}$ times the polynomial coefficients (6.5) in powers of x^{-1} . Alternatively those same coefficients follow by substituting a power series into the Bessel differential equation. The coefficient $\lambda K/(2\sqrt{\pi})$ in (6.4) follows immediately. One need not identify the original generating function with a Bessel function: only this local contour model is used.

For a fully controlled remainder, take a central arc $|\theta| \leq t^{3/2-\delta}$ with $0 < \delta < 1/6$. On the complement, (6.3) makes the integral smaller than any algebraic multiple of its leading saddle value. On the central arc, put $\theta = t^{3/2}u$; the real part of the quadratic term is a negative constant times u^2 , and Taylor remainders are bounded by a polynomial in u times that Gaussian. Truncation and integration are therefore valid to every fixed order. This supplies the error justification behind (6.4).

For $H = R$, use $\lambda = (2\pi)^{-3/2}$ and $s_j = (-1/6)^j/j!$. This proves (4.4). Multiplication by $(1 - e^{-z})^2 = z^2 + O(z^3)$ gives (4.5). It also gives

$$p_n = 4c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + \frac{\frac{1}{2} + \frac{3}{4K} + \frac{K}{72}}{n} + O(n^{-3/2}) \right). \quad (6.6)$$

The comparison product itself has an all-orders coefficient expansion; this fact is not being inferred for $\mathcal{A}$ without additional estimates.

7 Proof of the three correction terms

7.1 Bounding the finite-cut remainder on the whole circle

Proposition 7.1. *With F_3 from (5.6),*

$$a_n = [q^n](F_3 R) + O(p_n n^{-7/4}). \quad (7.1)$$

Proof. Take $q = re^{i\theta}$ with $r = e^{-t}$ and $t = \sqrt{K/n}$. By positivity and Lemma 5.1,

$$|E_3(q)| \leq E_3(r) = O(t^6 R(r)), \quad |C_4(q)| \leq C_4(r) = O(t^6 R(r) + 1).$$

Since the cubic factor of Q_3 is bounded away from zero on the closed unit disk, there are uniform estimates

$$\int_{-\pi}^{\pi} \frac{d\theta}{|Q_3(re^{i\theta})|^2} = O(t^{-1}),$$

$$\int_{-\pi}^{\pi} \frac{d\theta}{|Q_3(re^{i\theta})|} = O(\log(1/t)).$$

The first follows from $\int_{-\pi}^{\pi} |1 + re^{i\theta}|^{-2} d\theta = 2\pi/(1 - r^2)$; the second follows by integrating $(t^2 + (\theta - \pi)^2)^{-1/2}$ near the single boundary zero. The polynomials h_3, U_3, P_3 are uniformly bounded.

Apply Cauchy's coefficient formula to the three remainder terms in (5.5) with $m = 3$. Their total is

$$O(r^{-n}\{t^5 R(r) + t^6 R(r) \log(1/t) + \log(1/t) + 1\}).$$

The terms without $R(r)$ are exponentially smaller than the main coefficient scale. Since the saddle estimate gives $p_n \asymp r^{-n} R(r) t^{3/2}$, the dominant error is $O(p_n t^{7/2}) = O(p_n n^{-7/4})$. $\square$

7.2 Extracting the coefficients

A finite rational expansion at $q = 1$ gives

$$4F_3(e^{-z}) = 1 - \frac{1}{2}z^2 + \frac{1}{2}z^3 + O(z^4).$$

Together with (6.2), this yields

$$(F_3 R)(e^{-z}) = \frac{z^{3/2} e^{K/z}}{4(2\pi)^{3/2}} \left(1 - \frac{z}{6} - \frac{35z^2}{72} + \frac{755z^3}{1296} + O(z^4) \right), \quad (7.2)$$

up to a locally exponentially small modular remainder. The rational factor has only fixed-order boundary poles, so the minor-arc argument applies. Using (6.4) through $\ell = 3$ gives (1.3)–(1.5); its own truncation error is $O(n^{-2})$ relative to the leading term. The larger remainder in Proposition 7.1 is $O(n^{-7/4})$, proving Theorem 1.1.

For an explicit audit of the algebra, the relative n^{-1} coefficient is

$$\underbrace{\frac{3}{4K}}_{j=0, k=2} + \underbrace{\frac{1}{2}}_{j=1, k=1} - \underbrace{\frac{35K}{72}}_{j=2, k=0}.$$

The relative $n^{-3/2}$ coefficient is

$$\underbrace{-\frac{5}{8\sqrt{K}}}_{j=1, k=2} + \underbrace{\frac{175\sqrt{K}}{72}}_{j=2, k=1} + \underbrace{\frac{755K^{3/2}}{1296}}_{j=3, k=0}.$$

There is no $j = 0, k = 3$ contribution: the Bessel polynomial for that half-integer order terminates after $k = 2$.

Subtracting four times (1.2) from (6.6) proves (1.8). Thus the lost proportion in the colored-partition comparison is asymptotically $K/(2n)$, substantially smaller than the explicit upper bound $3K/n$ supplied by the elementary squeeze.

8 A proved all-orders radial expansion

8.1 A compatible family of rational approximants

Define $D(q) = L(q)/R(q)$ for real $0 < q < 1$. For every fixed $m \geq 1$, (5.2) and (5.3) imply

$$D(e^{-t}) = \frac{h_m(e^{-t})}{Q_m(e^{-t})} + O_m(t^{2m}). \quad (8.1)$$

The denominator tends to 2^m , so it is nonzero near the positive real limit. The Taylor expansions of these rational functions are therefore compatible: the coefficient of t^j is fixed once $2m > j$. Let $\mathcal{D}(t)$ be their unique formal limit.

Theorem 8.1 (Radial expansion to arbitrary fixed order). *The expansion (1.9) holds with*

$$\mathcal{F}(t) = 4\mathcal{D}(t)^2, \quad \mathcal{D}(t) = \lim_{m \rightarrow \infty}^{\text{formal}} \frac{h_m(e^{-t})}{Q_m(e^{-t})}. \quad (8.2)$$

For each fixed m ,

$$\frac{\mathcal{A}(e^{-t})}{R(e^{-t})} = \left(\frac{h_m(e^{-t})}{Q_m(e^{-t})} \right)^2 + O_m(t^{2m}). \quad (8.3)$$

The coefficients through t^8 are those in (1.10).

Proof. Divide the finite-cut identity (5.5) by R . For fixed m , the denominator $Q_m(e^{-t})$ stays bounded away from zero. The E_m/R and C_{m+1}/R terms are $O_m(t^{2m})$ by Lemma 5.1; the remaining rational expression divided by R is smaller than every power of t by (6.2). This proves (8.3). Compatibility and the product expansion give (8.2). Ordinary rational series arithmetic then gives the displayed coefficients. $\square$

For example, setting $w = 1 - q$ gives the useful intermediate expression

$$\mathcal{D} = \frac{1}{2} - \frac{1}{8}w^2 - \frac{5}{32}w^4 + \frac{1}{8}w^5 - \frac{65}{128}w^6 + \frac{37}{32}w^7 - \frac{2053}{512}w^8 + O(w^9).$$

Replacing w by $1 - e^{-t}$ and squaring reproduces (1.10). The accompanying symbolic checker verifies these operations exactly over the rationals.

8.2 Why this is not yet an all-orders coefficient theorem

Increasing m makes the positive radial tails smaller, but the complex behavior of $1/Q_m$ must still be controlled. For general m , one cannot simply assume that all zeros of Q_m lie outside the unit disk. No general zero classification is assumed or established here. A meromorphic rational approximant can be useful on a contour, but poles inside that contour require attention if one interprets its integral as an ordinary Taylor coefficient.

The choice $m = 3$ deliberately avoids these issues: the factorization in (5.6) proves the required zero exclusion exactly. Consequently the first three correction terms are supported by a full complex argument. Coefficients at later orders obtained by purely formal transfer of (1.9) are candidates until an equally valid remainder argument is supplied. No exponentially small contribution from other roots of unity, nor a complete transseries with all such sectors, is claimed in this report.

[write] Added 2 October 2026, batch 77P5: the gap is closed in Part II. Part II proves the all-orders coefficient theorem (Theorem 14.1): for every fixed M , a_n has an expansion in powers of $(n - 1/6)^{-1/2}$ with remainder $O(n^{-M/2})$ and algorithmically determined coefficients. It does not control the zeros of Q_m . It avoids them: the series $D = L/R$ of this section is a partial theta function (Remark 16.2), and $\mathcal{A} = R D^2$ plus a remainder whose coefficients are exponentially smaller (Proposition 16.1, Corollary 17.2), with whole-circle bounds for both factors. The later coefficients called “candidates” above are thereby proved. The statement about exponentially small sectors and a complete transseries still stands for both parts.

[write] Added 5 October 2026, batch 101. Part III proves the same theorem again (Theorem 25.1), also without controlling the zeros of Q_m : it uses the positivity of the generating function off the real axis, a

damping factor from the finite product $(q; q)_N$, and an interpolation lemma that carries this section's radial expansion into the saddle window (Section 28). That transfer uses only a complete real-axis expansion, analyticity and the positive representation of Section 28, so this section's radial expansion, combined with it, already gives every order. The zeros of Q_m remain a separate question (Section 33, item 6).

9 Inverse asymptotics and integer thresholds

9.1 A real inverse requires an interpolation convention

The sequence itself is defined only on integers. To state a real inverse accurately, choose an interpolation that preserves the proved asymptotic accuracy. Put

$$M(x) = c x^{-3/2} e^{B\sqrt{x}} (1 - dx^{-1/2} + ex^{-1} + fx^{-3/2})$$

for sufficiently large real x , where the parenthesis is positive. For an integer k , set $r_k = \log a_k - \log M(k)$. Theorem 1.1 gives $r_k = O(k^{-7/4})$. For $x = k + \theta$, $0 \leq \theta \leq 1$, define

$$\log \tilde{a}(x) = \log M(x) + (1 - \theta)r_k + \theta r_{k+1}. \quad (9.1)$$

This continuous interpolation agrees with a_k at integers and obeys

$$\log \tilde{a}(x) = \log M(x) + O(x^{-7/4}).$$

It is eventually strictly increasing: the derivative of $\log M$ is $B/(2\sqrt{x}) + O(x^{-1})$, while the piecewise-linear residual has slope $O(x^{-7/4})$. Denote its eventual inverse by $v(y)$. Other interpolations with the same asymptotic error give the same expansion below to the stated accuracy.

This convention matters beyond low orders. Direct linear interpolation of $\log a_k$, without first subtracting $\log M$, introduces a curvature error of order $x^{-3/2}$, which can obscure the third logarithmic correction. Equation (9.1) avoids that ambiguity.

9.2 Lambert W and the correction terms

Define

$$\begin{aligned} D_0 &= Bd = \frac{B^2}{12} + 3, \\ \alpha_2 &= B^2(e - d^2/2), \quad \alpha_3 = B^3(f + de - d^3/3), \\ \ell &= \log \frac{y}{cB^3}, \quad s_0 = -3W_{-1}\left(-\frac{1}{3}e^{-\ell/3}\right). \end{aligned} \quad (9.2)$$

Here W_{-1} is the real lower branch, which tends to $-\infty$ as its argument tends to 0 from below. It is selected because the inverse index tends to $+\infty$.

Theorem 9.1 (Asymptotic inverse). *For the interpolation (9.1), as $y \rightarrow \infty$,*

$$v(y) = \frac{1}{B^2} \left[s_0^2 + 2D_0 + \frac{2(3D_0 - \alpha_2)}{s_0} + \frac{18D_0 - 6\alpha_2 - D_0^2 - 2\alpha_3}{s_0^2} \right] + O(s_0^{-5/2}). \quad (9.3)$$

In particular, a formula involving only logarithms is

$$v(y) = \frac{\ell^2 + 6\ell \log \ell + 9(\log \ell)^2 + 18 \log \ell + 2D_0}{B^2} + O\left(\frac{(\log \ell)^2}{\ell}\right). \quad (9.4)$$

Proof. Set $s = B\sqrt{v(y)}$. Taking logarithms in the coefficient expansion and using the interpolation error gives

$$\ell = s - 3 \log s - \frac{D_0}{s} + \frac{\alpha_2}{s^2} + \frac{\alpha_3}{s^3} + O(s^{-7/2}). \quad (9.5)$$

The definition of s_0 is equivalent to $s_0 - 3 \log s_0 = \ell$. Put $s = s_0 + \delta$ and compare successive inverse powers of s_0 . Since $1 - 3/s_0$ is bounded away from zero,

$$\delta = \frac{D_0}{s_0} + \frac{3D_0 - \alpha_2}{s_0^2} + \frac{9D_0 - 3\alpha_2 - D_0^2 - \alpha_3}{s_0^3} + O(s_0^{-7/2}).$$

Squaring and dividing by B^2 gives (9.3). For (9.4), direct substitution in (9.5) yields

$$s = \ell + 3 \log \ell + \frac{9 \log \ell + D_0}{\ell} + O((\log \ell)^2 / \ell^2),$$

and another squaring completes the calculation. □

If $N(y) = \min\{n \geq n_0 : a_n \geq y\}$ for a sufficiently large fixed n_0 , then exactly

$$N(y) = \lceil v(y) \rceil.$$

The error term belongs *inside* that ceiling. Uniformly discarding it can produce an incorrect integer when the approximation is very close to a threshold. A safe coarser statement is that the right-hand side of (9.4), without its remainder, differs from $N(y)$ by $O(1)$. Away from integers by more than a proved error bound, rounding would be unambiguous; this article does not supply a numerical universal constant for that error bound.

[write] Added 2 October 2026, batch 75: an instance of repository results. The inversion of this section is an instance of the Lambert-centred inversion apparatus of the transseries-and-inversion volume [8], and no novelty is claimed for the method. The core $s_0 - 3 \log s_0 = \ell$ is `p0:thm:lambert-core` with $a = 1$, $b = -3$; its branch rule selects W_{-1} , as above. Equation (9.5) is the core plus a power series in $1/s$, and the correction δ is the all-orders reversion around the core, `p0:thm:lambert-centered`, truncated at the order the forward remainder allows. The shape is that of the partition-number chapter of the same volume, `p3:sec:lambert` with `p3:thm:core-correction`, where the factor n^{-1} in $p(n)$ gives $b = -2$; here $n^{-3/2}$ gives $b = -3$. The rule $N(y) = \lceil v(y) \rceil$ and the warning that follows it are part (1) and part (2) (the separation condition) of `p0:thm:staircase`, specialized as in `p3:thm:staircase`. The generic arithmetic of those two parts is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; those statements concern an arbitrary monotone interpolation and give no formal status to Theorem 9.1. What is specific to this report is the forward expansion that feeds the inversion and the explicit interpolation (9.1).

10 Reproducible exact and numerical checks

10.1 What was computed

The attached `verify.py` uses only Python's standard library and unbounded integers. It computes $a_0, \dots, a_{2000}$ from the normalized recurrence, and independently computes p_n from the colored Euler product. Division by $(1 - q^m)^2$ is performed by two exact cumulative-sum passes. Thus the computation avoids floating-point cancellation in the original polynomial recurrence.

The checker confirms the initial 37 OEIS terms and four separately read b-file anchors, at $n = 50, 100, 200, 500$ [4]. It then verifies the slope identity, the positive defect decomposition, both majorants in (4.3), and the squeeze for every integer $3 \leq n \leq 2000$. The third finite-cut identity, after clearing its polynomial denominator, and positivity of its auxiliary tails are checked as well. The output CSV contains all computed coefficients, not just a selected plot. This is an exact finite test, not a formal proof of the infinite statements.

[write] Added 5 October 2026, batch 101. Since batch 101 all 2001 terms of the b-file have been compared: Part III's checker matches $a_0, \dots, a_{2000}$ with a snapshot that is byte-identical to the live b-file (checked at intake on 5 October 2026), and this checker's coefficients agree with them (Section 24.2). Run with `-nmax 8000` on a copy at intake, it also passed all its checks, the squeeze included, through $n = 8000$, and reproduced Part III's coefficients there.

The optional `verify_symbolic.py` uses SymPy for rational symbolic arithmetic and mpmath for high-precision numerical evaluations. It checks Wronskians, the factorization of Q_3 , the radial coefficients through t^8 , and the transfer coefficients d, e, f . All included assertions passed in the supplied run.

10.2 Accuracy of the asymptotic approximants

Let

$$A_0(n) = cn^{-3/2}e^{B\sqrt{n}}, \quad A_1 = A_0(1 - d/\sqrt{n}), \quad A_2 = A_1 + A_0e/n, \quad A_3 = A_2 + A_0f/n^{3/2}.$$

The table reports $a_n/A_j(n)$; a ratio approaching 1 is the desired behavior. Rounded display values are not used by the checker.

n	a_n/A_0	a_n/A_1	a_n/A_2	a_n/A_3
50	0.836729727	0.980057860	1.027184142	0.984290725
100	0.885790139	0.987955090	1.010017535	0.995767525
200	0.920485946	0.993103974	1.003708062	0.998875075
500	0.950776376	0.996878614	1.000989560	0.999807957
1000	0.965688969	0.998335826	1.000361244	0.999950135
2000	0.976025253	0.999128395	1.000130943	0.999987161

More terms need not improve a short truncation at a modest index: $n = 50$ illustrates this. At $n = 2000$, the four-term relative error is approximately 1.284×10^{-5} . The data also show

$$n(1 - 4a_n/p_n) = 1.739699, 1.940375, 2.110009, 2.286402, 2.386995, 2.463821$$

at the six listed indices, tending toward the proved limit $K/2 = 2.673017858628 \dots$

10.3 Testing the inverse at known sequence values

For $y = a_n$, the compatible inverse is exactly n . The leading Lambert approximation is s_0^2/B^2 ; the corrected one is the bracketed expression in (9.3).

n	Leading inverse minus n	Corrected inverse minus n
50	-0.598588435	-0.028604645
100	-0.560169154	-0.011441486
200	-0.530798365	-0.004479985
500	-0.502617905	-0.001257271
1000	-0.487447241	-0.000471103
2000	-0.476247422	-0.000174032

The nonvanishing limiting offset in the uncorrected Lambert approximation is explained by the constant term $2D_0/B^2$ in the inverse theorem. These values are numerical checks of already derived formulas.

11 Further research questions and concrete next steps

1. **An all-orders coefficient expansion.** Use the general finite-cut identity (5.5) to extend the $O(n^{-7/4})$ remainder to arbitrary powers. The outstanding issue is not generating formal coefficients; it is uniform contour control of Q_m^{-1} and treatment of possible interior poles of rational approximants. A theorem organizing admissible cuts would turn the radial algorithm into an all-orders coefficient algorithm.

[write] Added 2 October 2026, batch 77P5: answered. Part II proves the all-orders coefficient expansion (Theorem 14.1) by a different route, an exact product-square decomposition, without organizing finite cuts. A theorem on admissible cuts, or on the zeros of Q_m , remains a separate question.

2. **Explicit numerical error constants.** Make the product minor-arc constants and the cubic lower bound in the $m = 3$ proof effective. This would give certified numerical intervals for a_n and safe integer inversion, rather than asymptotic O statements with unspecified constants.
3. **A combinatorial explanation of the factor four.** Construct a direct relation between four copies of the L-convex class and the colored partitions counted by R . The proven inequality $p_n \geq 4a_n$ for $n \geq 1$ is suggestive, but an injection or decomposition is not supplied by the analytic proof. A description of the defect class could explain the limit $K/2$ combinatorially.
4. **The singularities of the boundary series.** Study $D(q) = L/R$ as an analytic object, not just a radial formal series. The recurrence, Wronskians, and positive tails give several exact representations. Determine its domain of continuation and the role of zeros of the finite denominators Q_m .

[write] Added 2 October 2026, batch 77P5: partly answered. $D = L/R$ equals the partial theta function $1 - \sum_{k \geq 0} q^{k(k+1)}(1 - q^{2k+1})/(1 + q^{2k+1})$ on the whole disk $|q| < 1$ (Remark 16.2, from Part II's Proposition 16.1). Its continuation and the zeros of Q_m remain open; Part II's open problem 5 continues the question.

5. **Beyond-all-orders terms.** Separate exponentially small contributions to $\mathcal{A}(e^{-t})$ from the modular product and from the recurrence boundary corrections. Determine which roots of unity contribute to a coefficient transseries and whether the radial series has a useful summability or resurgence structure. None of these properties follows just from the computed rational coefficients.

[write] Added 2 October 2026, batch 77P5. Part II separates an exact remainder whose coefficients are $O(n^{1/4}e^{\pi\sqrt{2n}})$, exponentially smaller than a_n (Corollary 17.2); the bound is deliberately coarse. The actual exponentially small contributions remain open (Part II, open problem 2).

6. **A reusable theorem for recurrence-defined area series.** Abstract the positive-transfer and finite-cut mechanism to other recurrences $y_{n+1} = a(q^n)y_n - b(q^n)y_{n-1}$ whose auxiliary generating functions are Euler products. State conditions under which a positive boundary defect is small enough for coefficient transfer. The present proof gives a worked example with all analytic and positivity obligations visible.
7. **Z-convex and higher-convexity area enumeration.** Investigate whether the numerically studied Z-convex area asymptotics [6] admit a related positive comparison. This is a proposed application, not an assertion that the L-convex recurrence or constant extends unchanged. Area and perimeter must remain distinct parameters.
- [write] Added 2 October 2026, batch 77P5. Part II's open problem 4 asks the analogous question for its product-square decomposition.
- [write] Added 5 October 2026, batch 101. The series data of [6] are flagged as wrong by its first author (the note in that entry and in Section 31), so any comparison would have to start from corrected Z-convex data.
8. **Formalization and OEIS integration.** Formalize the normalized transfer, coefficientwise order, Wronskian, and finite-cut identity first. These are algebraic and should be separated from the eta transformation and complex saddle analysis. An OEIS update should cite a reviewed version of the coefficient proof, rather than only a numerical output or the all-orders radial statement.

12 Conclusion and claim ledger

The principal result is a proof of a specific recorded conjecture, not a conjecture inferred from additional terms. The established generating function leads to a positive normalized recurrence; one auxiliary initial condition leads exactly to a colored-partition Euler product. Two boundary conditions produce the factor $1/4$. A positive coefficient squeeze proves the leading asymptotic and its first correction. A finite cut at index three, together with a full-circle error estimate, proves two further corrections and a relative $O(n^{-7/4})$ remainder.

The following boundaries summarize what can be reused without confusion.

Statement	Status in this manuscript
Area generating function and its enumerative interpretation	Cited input from prior work
Colored-partition product, positive squeeze, finite-cut identity	Proved here from the recurrence
Leading coefficient asymptotic and three correction terms	Proved with a complex remainder estimate
All-orders expansion for $\mathcal{A}(e^{-t})$, $t > 0$	Proved to every fixed radial order
Inverse expansion for a specified compatible interpolation	Proved; integer rounding treated separately
All-orders coefficient expansion and exponentially small sectors	Further research; not claimed
Finite checks through $n = 2000$ and symbolic arithmetic	Executed checks, not formal certification
Independent peer review or Lean verification	Not performed in this report

[write] Added 2 October 2026, batch 77P5. The ledger is Part I's. Its row "All-orders coefficient expansion and exponentially small sectors" is now half answered: Part II proves the all-orders coefficient expansion (Theorem 14.1); exponentially small sectors are still not claimed. The last two rows hold for Part II as well.

Part II. All-order asymptotics and inversion

Sections 1–12 and Appendices A and B form the original report (Part I) of 1 October 2026. Part I is unchanged except for dated [write] pointers to this addition (after the scope box, in Section 1.1, at the end of Section 8.2, in research questions 1, 4, 5 and 7 of Section 11, and after the claim ledger of Section 12), a line in the title, contents entries for the two parts, two macros in the preamble, one bibliography entry used only by Part II, and notes in two existing entries. No number of Part I moves.

13 Part II: source, scope and notation

This section was written when Part II was added. Sections 14–22 print the manuscript; Section 23 records the provenance of both parts and the choices the merge made.

13.1 Source and status

Sections 14–22 print, in full, manuscript 31 of batch 77 of ProveIt’s incoming-reports intake, titled “All order asymptotics and inversion for L convex polyominoes” and dated 1 October 2026. It arrived as the archive `lconvex-asymptotics-reproducibility.zip` in commit `096ee7b87` and was placed in commit `4f11bc9c0`. Its author line is empty and it names no tool; like every manuscript of the intake it is treated as AI-assisted research output. It names no ProveIt commit and does not mention the repository, so it has no pin, and it was not written against Part I.

It is not a version of Part I’s manuscript. Only 40 of its 380 distinct non-blank source lines occur in Part I’s source. Its proof architecture is different: an exact product-square decomposition and a Hankel–Bessel transfer, where Part I uses a positive coefficient squeeze and finite cuts. The proof note that its shipped audit reviewed is a different file from Part I’s manuscript. The package ships three internal reviews (`02-all-orders-mathematical-audit.md`, `02-all-orders-final-source-review.md`, `02-all-orders-computational-review.md`). They are the delivery’s own checks, not external peer review. Part II is unrefereed. Nothing in it is formalized in Lean or Rocq, and its place in the collection confers no formal status on it.

What was rechecked at intake and in this write. Independently of the delivery:

- $a_0, \dots, a_{120}$ from the original form (14.1) agree with the shipped coefficient file, and that file agrees entry by entry with the column a_n of Part I’s `data/coefficients.csv` for all $0 \leq n \leq 2000$;
- $c_0, \dots, c_4$ follow from (14.6) with $b_0, \dots, b_4$ read from (18.1), the h_r of Section 20 are $(2\sqrt{\kappa})^r c_r$, and the unshifted re-expansion gives exactly Part I’s d, e, f (Remark 14.2), all with exact SymPy arithmetic;
- $B = D$ and $A = PB^2 + R$, with R from the literal $E(q)$, at $q = 0.3, 0.6$ and $0.3 + 0.4i$, to 40 digits (discrepancies 10^{-38} to 10^{-41});
- the identity of Remark 16.2 between Part I’s L/R^1 and Part II’s B , at the same three points;
- the entries of Table 2, the leading constant, the monotonicity injection (by hand), and the log-concavity expansion numerically (Corollary 19.1);
- the shipped replay, core and optional parts (Section 21).

The whole-circle estimates (Lemmas 17.1 and 17.3) and the contour deformation in Section 19 were read but not re-derived in detail; they are written at the level of standard saddle-point estimates.

13.2 What Part II adds, and what stays in Part I

Part II proves, from the same cited area generating function:

- the exact decomposition $A = PBD + R$ and the identity $B = D$, a partial theta function, hence $A = PB^2 + R$ exactly (Propositions 15.2 and 16.1);
- whole-circle bounds and an exponentially smaller exact remainder (Lemma 17.1, Corollary 17.2);
- **the all-orders coefficient theorem** (Theorem 14.1) in the shifted index $N = n - 1/6$, with coefficients given by a finite algorithm. This closes the gap that Part I states in Section 8.2 and research question 1 of Section 11;
- eventual strict log-concavity with a two-term expansion of $\log(a_n^2/(a_{n-1}a_{n+1}))$ (Corollary 19.1);
- an injection proving that a_n is nondecreasing;
- a Lambert-W inverse in shifted coordinates, logarithmic inverse polynomials, and shrinking integer brackets for every order (Section 20).

Only Part I has the coefficientwise squeeze $0 \leq p_n - 4a_n \leq 3j_n$ for every $n \geq 3$ with its colored-partition reading, the deficit limit $K/2$, the finite-cut identity and its $m = 3$ route, the interpolation $\nu(y)$ with the exact rule $N(y) = \lceil \nu(y) \rceil$, and the tables of Section 10.

Several results are proved in both parts, by different arguments; each part keeps its own proof.

- The leading asymptotic and the corrections d, e, f of Theorem 1.1: Part I by squeeze and finite cut, with remainder $O(n^{-7/4})$; Part II as the case $M = 4$ of Theorem 14.1, with remainder $O(n^{-2})$ (Remark 14.2).
- The radial expansion (1.9) with the coefficients (1.10) of Theorem 8.1: it follows from (16.2), and the coefficients are four times those of $D(e^{-z})^2$ in (18.1) (Remark 16.2).
- The local eta expansion (6.2) is (18.3); the minor-arc bound (6.3) is Lemma 17.3, for the same product; the transfer (6.4)–(6.5) is (19.2)–(14.6) in the shifted index.
- The generating function (2.1) is (14.1), with Part I's h_n equal to the manuscript's f_{n-1} ; the initial terms are printed in both parts.

13.3 Notation across the two parts

Part II keeps the manuscript's letters. The shipped data files and reviews use them (for example the keys `A_equals_P_B_squared_plus_R` and `initial_R` of `02-all-orders-exact-identities.json`, and `kappa` in the forward and inverse series), and renaming would break that correspondence. Many of these letters mean something else in Part I. In Sections 14–22 a letter has its Part II meaning. Where a [write] note in Part II refers to a Part I object whose letter clashes, it adds the superscript I: $R^I, K^I, B^I, D^I, L^I, y^I$. No symbol of either manuscript was renamed and no normalization was changed. Table 1 lists every clash. The last column gives the tempting false reading.

Table 1: Letters of Part II, the same object in Part I, and the clash to avoid.

Part II	Meaning in Part II	Same object in Part I	Not to be confused with
κ	$13\pi^2/24$	$K, (1.1)$	Part I's κ in (6.2), an unspecified sector-dependent exponent
C	$13\sqrt{2}/768$	c	Part I's series $C(z; q)$, C_j , (3.2)
N	$n - 1/6$	—	Part I's threshold $N(y)$, and its block length $N \asymp 1/t$ in the minor-arc proof
$P(q)$	$(-q; q^2)_\infty / (q; q)_\infty^3$, (15.3), (18.2)	$R(q)$, (1.6)	Part I's polynomials P_m ; Part II's P_j
$R(q)$	exact remainder $1 + E(q)/(q; q)_\infty$, Proposition 15.2	none	Part I's $R(q)$, which is Part II's P : $A = PB^2 + R$ does <i>not</i> say $\mathcal{A} = R^1 D^2 + R^1$
p_n	$[q^n]P(q)B(q)^2$, Corollary 17.2; $p_n \sim a_n$	none	Part I's $p_n = [q^n]R^1(q) = [q^n]P(q)$, with $p_n \sim 4a_n$
$E(q)$	the double sum (15.6)	none	Part I's defect $E = R - 2L$, (4.1), and tails E_m , (5.2)
$B(q)$	$1 - \sum_l u_l$, the partial theta function (16.1)	$D(q) = L/R$ of Section 8.1, exactly (Remark 16.2)	Part I's constant $B = 2\sqrt{K} = \pi\sqrt{13/6}$
$D(q)$	the series of (15.3); $D = B$	$D(q) = L/R$: the same function, not only the same letter	Part II's bracket constants D_M
K	$C(2\sqrt{\kappa})^3$, (20.1)	cB^3	Part I's $K = 13\pi^2/24$, which is Part II's κ
L, w	$\log(X/K)$, $\log L$	$\ell = \log(y/(cB^3))$ of (9.2)	Part I's slope series $L = \mathcal{L}_h$, (3.1)
X	target value of the inverse	y of Section 9	—
y	$2\sqrt{\kappa(n - 1/6)} = B^1\sqrt{N}$	unshifted analogue $s = B\sqrt{v}$	Part I's y , the target value
y_0	Lambert root (20.4)	s_0 of (9.2), the same number when $X = y^1$	—
$n_*(X)$	$\min\{n : a_n \geq X\}$	$N(y)$ of Section 9	—
F_M, x_M	real model (20.7) and its large inverse	analogues of $M(x)$ and $v(y)$	Part I's rational $F_3(q)$, (5.6), and series $\mathcal{F}(t)$
M	truncation order	—	Part I's model function $M(x)$
h_n, H, G	$f_{n-1}/(q; q)_n$, its generating function, the product (15.1)	Part I's $h_n = f_{n-1}$ lacks the factor $1/(q; q)_n$; G is Part I's G , (3.6)	h_r of Section 20
h_r, ℓ_r	coefficients of (20.2) and of its logarithm, (20.3)	D_0, α_2, α_3 play this role for s	h_n above; Part I's ℓ
b_m	Taylor coefficients of $D(e^{-z})^2$, (14.5)	$\mathcal{F}(t) = 4 \sum b_m t^m$, (8.2)	Part I's $b_n = y_n/(q; q)_n^2$ and color multiplicities b_j , (3.8)
c_r	coefficients of (14.4)	unshifted: $1, -d, e, f$	Part I's c and c_0 ; the local constants $c, c_1, \dots, c_4, C_1$ of Part II's proofs
u_l, T_j	summands of B in (15.3); summands of D (proof of Lemma 18.1)	—	Part I's $u_n = Q_n - h_n$ and $T_n(q)$

Part II	Meaning in Part II	Same object in Part I	Not to be confused with
P_j	logarithmic inverse polynomials	—	$P(q)$; Part I's P_m
a, b, S, V	local to the proof of Proposition 16.1 ($a = i\sqrt{q}$)	—	$a_n; b_m$; Part I's s
z, t	auxiliary variable in Section 15; from Section 18 on $q = e^{-z}$, $t = \Re z$	same two uses in Part I	saddle radius $t = \sqrt{\kappa/N}$ here, $t = \sqrt{K/n}$ in Part I
λ, α	$\sqrt{\kappa N}; 3/2 + m$	—	Part I's amplitude λ in (6.4); α_2, α_3
d_r	local to the proof of Corollary 19.1	—	Part I's d

Two further letters are shared with the same meaning: f_n (the recurrence (14.2); Part I's h_{n+1}) and the branch W_{-1} .

13.4 The manuscript's abstract

Let a_n count fixed L -convex polyominoes of area n , with the empty object included. Starting from the known exact area generating function, we prove the asymptotic equivalent conjectured by Guttmann and Kotěšovec and recorded in OEIS A126764. An exact decomposition into an eta quotient times the square of a partial theta function, plus an exponentially smaller coefficient remainder, supplies every fixed algebraic order. The correction coefficients are given by a finite algorithm. We also give a branch-explicit Lambert W inverse, logarithmic inverse polynomials, and shrinking intervals for the integer threshold inverse before rounding. The proof includes whole-circle estimates; a positive-real singular expansion alone is not used as a coefficient-transfer argument. Exact coefficient checks and a portable replay accompany the analysis.

14 The sequence and the main theorem

An L -convex polyomino is a convex polyomino in which any two cells can be joined internally by a path with at most one right-angle turn. We count fixed polyominoes, identifying translations but keeping rotations and reflections distinct. Its area generating function is

$$A(q) = \sum_{n \geq 0} a_n q^n = 1 + \sum_{n \geq 1} \frac{q^n f_{n-1}(q)}{(q; q)_{n-1}^2 (1 - q^n)}, \quad (14.1)$$

where

$$f_{-1} = f_0 = 1, \quad f_n = 2f_{n-1} - (1 - q^n)^2 f_{n-2} \quad (n \geq 1). \quad (14.2)$$

Here $(a; q)_j = \prod_{h=0}^{j-1} (1 - aq^h)$. Formula (14.1) is the Castiglione–Frosini–Munarini–Restivo–Rinaldi area formula [2], in the normalization reproduced in [3, Eq. (1.1)]. The first terms are

$$1, 1, 2, 6, 15, 35, 76, 156, 310, 590, 1098, 1984, 3515, 6094, \dots$$

Set

$$\kappa = \frac{13\pi^2}{24}, \quad C = \frac{13\sqrt{2}}{768}, \quad N = n - \frac{1}{6}. \quad (14.3)$$

Theorem 14.1. For every fixed integer $M \geq 1$,

$$a_n = CN^{-3/2}e^{2\sqrt{\kappa}N} \left(\sum_{r=0}^{M-1} c_r N^{-r/2} + O(N^{-M/2}) \right). \quad (14.4)$$

The coefficients are algorithmically determined as follows. Define the formal asymptotic coefficients b_m by

$$\left(\sum_{j \geq 0} \frac{(-1)^j e^{-zj(j-1)/2} (e^{-z}; e^{-z})_j}{(-e^{-z}; e^{-2z})_{j+1}} \right)^2 \sim \sum_{m \geq 0} b_m z^m. \quad (14.5)$$

To determine $b_0, \dots, b_{M-1}$, it suffices to retain $0 \leq j < M$ on the left and take ordinary Taylor coefficients. Then

$$c_r = \sum_{\substack{m+k=r \\ 0 \leq k \leq m+2}} 4b_m \kappa^{m/2} \frac{(-1)^k (m+2+k)!}{(m+2-k)! k! (4\sqrt{\kappa})^k}. \quad (14.6)$$

In particular,

$$\begin{aligned} c_0 &= 1, & c_1 &= -\frac{3}{2\sqrt{\kappa}}, & c_2 &= \frac{3}{4\kappa} - \frac{\kappa}{2}, \\ c_3 &= \frac{\sqrt{\kappa}(\kappa+5)}{2}, & c_4 &= -\frac{45}{8} - \frac{15\kappa}{4} - \frac{41\kappa^2}{48}. \end{aligned} \quad (14.7)$$

The $M = 1$ case proves the OEIS conjecture

$$a_n \sim \frac{13\sqrt{2}}{768} n^{-3/2} \exp\left(\pi\sqrt{\frac{13n}{6}}\right). \quad (14.8)$$

For readers preferring the unshifted variable, the first correction is

$$a_n = Cn^{-3/2}e^{2\sqrt{\kappa}n} \left[1 - \left(\frac{\sqrt{\kappa}}{6} + \frac{3}{2\sqrt{\kappa}} \right) n^{-1/2} + O(n^{-1}) \right]. \quad (14.9)$$

The shift $N = n - 1/6$ keeps the higher formulas shorter.

Remark 14.2 (write, 2 October 2026: consistency with Part I). Take $M = 4$ in Theorem 14.1 and re-expand in the unshifted index. With $\kappa = K^1$ this gives

$$a_n = Cn^{-3/2}e^{2\sqrt{\kappa}n} \left(1 - \frac{d}{\sqrt{n}} + \frac{e}{n} + \frac{f}{n^{3/2}} + O(n^{-2}) \right),$$

with exactly the d, e, f of (1.3)–(1.5): the three symbolic differences vanish (exact SymPy arithmetic, at intake and again in this write). So Theorem 14.1 re-proves Theorem 1.1 with the sharper relative remainder $O(n^{-2})$, by a route independent of Part I's squeeze and finite cut, and continues it to every order. Equation (14.9) is Part I's first correction $-d$, and (14.8) is the leading asymptotic proved in both parts. Part I keeps its own statement and proof.

Literature status and prefactor. The inspected OEIS entry [1] explicitly labels (14.8) conjectural. The published 2023 paper [3] describes its coefficient analysis as conjectural, and the 2024 paper [6, p. 19] repeats that status. The introduction of [3] and its quoted numerical value $0.023938510821419\dots$ agree with C above, while its final display in Section 2 prints the reciprocal. Our derivation fixes the normalization independently. Targeted searches through 1 October 2026 did not locate an earlier proof or an all-orders expansion; this is a scoped search result, not an exhaustive claim of priority.

[write] Added 2 October 2026, batch 77P5. The statement that the final display of Section 2 of [3] prints the reciprocal of C was *not* checked at intake. The shipped mathematical audit (item 13) says the same, but it is the delivery's own review. Part I cites [3] for the conjecture and its equation (1.1) only. As in Part I, the search statement is scoped, and no priority is claimed. Part I's own proof of the leading term and three corrections (Theorem 1.1) predates this manuscript in the repository, though the two were written independently.

[write] Added 5 October 2026, batch 101. Now checked: at intake on 5 October 2026 the final display of Section 2 of [3] (arXiv v3) was found to have the amplitude $1/C$, while the decimal constant quoted there is C , as Part II says. See the note in Section 25.

15 An exact decomposition of the generating function

All q -series in this section converge for $|q| < 1$. Introduce

$$h_n = \frac{f_{n-1}}{(q; q)_n}, \quad H(z) = \sum_{n \geq 0} h_n z^n \quad (|z| < 1), \quad G(z) = \frac{(-qz^2; q^2)_\infty}{(z; q)_\infty^2}. \quad (15.1)$$

The letter z here is an auxiliary generating-function variable; later $q = e^{-z}$ will be explicitly stated.

Lemma 15.1. *The following identity holds for $|z| < 1$:*

$$H(z) = G(z) \left[1 - \sum_{j \geq 0} \frac{q^j z (q^{j+1} z; q)_\infty^2}{(-q^{2j+1} z^2; q^2)_\infty} \right]. \quad (15.2)$$

Proof. The recurrence gives

$$(1 - z)^2 H(z) = (1 + qz^2) H(qz) - z,$$

whereas G satisfies the corresponding homogeneous equation. Division by G and iteration toward $q^j z \rightarrow 0$ give (15.2). To justify convergence directly, put $\epsilon_k = 2q^k - q^{2k}$. First differences in (14.2) yield

$$f_n = 1 + \sum_{k=1}^n (n - k + 1) \epsilon_k f_{k-2}.$$

Divide by $n + 2$ and apply the discrete Gronwall inequality; the relevant product converges because $\sum k |\epsilon_k| < \infty$. Thus $f_n = O_q(n)$, uniformly on compact q -subdisks. Since $(q; q)_n$ has a nonzero limit, H is analytic for $|z| < 1$. All subsequent rearrangements are absolutely convergent, initially on compact subdisks. $\square$

Define

$$\begin{aligned} P(q) &= \frac{(-q; q^2)_\infty}{(q; q)_\infty^3}, \\ u_l(q) &= \frac{q^l(q^{l+1}; q)_\infty^2}{(-q^{2l+1}; q^2)_\infty}, \\ D(q) &= \sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2} (q; q)_j}{(-q; q^2)_{j+1}}. \end{aligned} \quad B(q) = 1 - \sum_{l \geq 0} u_l(q), \quad (15.3)$$

Formula (15.2) gives

$$H(q^{j+1}) = G(q^{j+1}) \left(B + \sum_{l=0}^j u_l \right). \quad (15.4)$$

Proposition 15.2. *There is an exact identity*

$$A(q) = P(q)B(q)D(q) + R(q), \quad (15.5)$$

where $R(q) = 1 + (q; q)_\infty^{-1} E(q)$ and

$$E(q) = \sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2}}{(q; q)_j} \sum_{l=0}^j \frac{q^l (q^{l+1}; q)_{j-l}^2}{(-q^{2l+1}; q^2)_{j-l+1}}. \quad (15.6)$$

Proof. Euler's q -binomial theorem gives

$$\frac{1}{(q; q)_{n-1}} = \frac{(q^n; q)_\infty}{(q; q)_\infty} = \frac{1}{(q; q)_\infty} \sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2 + nj}}{(q; q)_j}.$$

Substitute this in $A - 1 = \sum_{n \geq 1} q^n h_n / (q; q)_{n-1}$ and interchange sums. The constant piece vanishes because

$$\sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2}}{(q; q)_j} = (1; q)_\infty = 0.$$

For the homogeneous part of (15.4), use

$$G(q^{j+1}) = \frac{(-q; q^2)_\infty}{(q; q)_\infty^2} \frac{(q; q)_j^2}{(-q; q^2)_{j+1}}.$$

The remaining product simplifies, for $l \leq j$, to

$$G(q^{j+1}) u_l = \frac{q^l (q^{l+1}; q)_{j-l}^2}{(-q^{2l+1}; q^2)_{j-l+1}},$$

which proves the proposition. $\square$

16 The square identity and a partial theta representation

Proposition 16.1. *One has*

$$B(q) = D(q) = 1 - \sum_{k \geq 0} q^{k(k+1)} \frac{1 - q^{2k+1}}{1 + q^{2k+1}}. \quad (16.1)$$

Consequently

$$A(q) = P(q)B(q)^2 + R(q). \quad (16.2)$$

Proof. Set $a = i\sqrt{q}$, $b = -i\sqrt{q}$, so $a + b = 0$, $ab = q$. Both choices of square root give the same symmetric expressions. We use the convention

$${}_2\phi_1(a, b; c; q, z) = \sum_{n \geq 0} \frac{(a, b; q)_n}{(c, q; q)_n} z^n.$$

For the transformation argument assume $q \neq 0$; the identities at $q = 0$ follow by continuity. Heine's first transformation [10, 17.6.6] applied to ${}_2\phi_1(a, b; q; q, q)$ gives

$$S := 1 - B = \sum_{n \geq 0} \frac{b^n}{1 - aq^n} = \sum_{n, k \geq 0} a^k b^n q^{nk}. \quad (16.3)$$

Indeed, the defining sum of the u_l is

$$\frac{(q; q)_\infty^2}{(a, b; q)_\infty} {}_2\phi_1(a, b; q; q, q).$$

After Heine's transformation, the series ratio is $(a; q)_n / (aq; q)_n = (1 - a) / (1 - aq^n)$; the product prefactor cancels with $1 - a$. Formula (16.3) is symmetric in a, b .

Let $V = (1 + q)D$ and let V_q denote the same summation with an extra factor q^n in term n . Fine's second transformation [10, 17.6.10], in the form

$$(1 - z) {}_2\phi_1(q, \alpha q; \beta q; q, z) = \sum_{n \geq 0} \frac{(\beta/\alpha; q)_n (-\alpha z)^n q^{n(n+1)/2}}{(\beta q, zq; q)_n},$$

with $\alpha = a/q^2$, $\beta = a/q$, $z = b$, yields

$$\frac{V - aV_q}{1 - a} = (1 - b) {}_2\phi_1(q, a/q; a; q, b) = (1 - b)[1 + (b - 1)S]. \quad (16.4)$$

For the second equality, split off $n = 0$ and use

$$\frac{(a/q; q)_n}{(a; q)_n} = \frac{1 - a/q}{1 - aq^{n-1}}, \quad ab = q.$$

Swap a, b in (16.4). Multiplying the original and swapped equations by $1 - a$ and $1 - b$ and adding gives $2V = 2(1 + q)(1 - S)$. Hence $D = B$.

Finally split the double sum (16.3) into its diagonal and two off-diagonal triangles. On a diagonal difference d , $a^d + b^d$ vanishes for odd d , while it is $2(-1)^m q^m$ for $d = 2m$. Therefore

$$S = \sum_{k \geq 0} q^{k(k+1)} \left(1 + 2 \sum_{m \geq 1} (-1)^m q^{m(2k+1)} \right),$$

and summing the geometric progression gives (16.1). $\square$

Remark 16.2 (write, 2 October 2026: Part I's boundary series is the partial theta function). For $|q| < 1$, Part I's series $D^I(q) = L^I(q)/R^I(q)$ of Section 8.1, with L^I the normalized limiting slope (3.1) of the solution $h_n^I = f_{n-1}$, equals $B(q) = D(q)$. Indeed, in (15.2) the bracket is analytic for $|z| < |q|^{-1/2}$ (the zeros of $(-q^{2j+1}z^2; q^2)_\infty$ have $|z| \geq |q|^{-1/2}$, and the factor q^j makes the sum converge) and equals $B(q)$ at $z = 1$, while G has in $|z| < |q|^{-1}$ only the double pole at $z = 1$, with $(1-z)^2 G(z) \rightarrow (-q; q^2)_\infty / (q; q)_\infty^2$. Hence

$$\frac{f_{n-1}}{(q; q)_n} = [z^n]H(z) = \frac{(-q; q^2)_\infty B(q)}{(q; q)_\infty^2} n + O_q(1),$$

so the slope of Lemma 3.1 is $s(q) = (-q; q^2)_\infty B(q) / (q; q)_\infty$, and $L^I = s / (q; q)_\infty^2 = R^I B$. A 40-digit evaluation at $q = 0.3, 0.6$ and $0.3 + 0.4i$ agrees to 10^{-37} . Consequences: Part I's formal limit $\mathcal{D}(t)$ in (8.2) is the Poincaré expansion of $B(e^{-t})$ given by Lemma 18.1, and the coefficients (1.10) are four times those of $D(e^{-z})^2$ in (18.1); Part I's rational approximants h_m/Q_m in (8.1) approximate this partial theta function; and Theorem 8.1 also follows from (16.2), since $P = R^I$ and R is exponentially smaller on the radius by Lemma 17.1. This answers part of Part I's research question 4 (Section 11): D^I has the closed form (16.1) on the whole unit disk. Its continuation beyond $|q| < 1$ and the zeros of the finite denominators Q_m are not addressed.

17 Global estimates and the exponentially smaller part

These bounds hold on the entire coefficient-extraction circle, not only on the positive real radius. Let $r = e^{-t}$, $t > 0$, and $|q| = r$.

Lemma 17.1. *As $t \downarrow 0$, uniformly on $|q| = r$,*

$$\begin{aligned} |B(q)| &= O\left(t^{-1} \log \frac{1}{t}\right), \\ |R(q)| &= O\left(t^{-1/2} e^{\pi^2/(2t)}\right). \end{aligned} \tag{17.1}$$

Proof. From (16.1),

$$|B(q)| \leq 1 + \sum_{k \geq 0} r^{k(k+1)} \frac{1 + r^{2k+1}}{1 - r^{2k+1}}.$$

Split at $k \asymp t^{-1/2}$ and $k \asymp t^{-1}$. The small- k part is bounded by a harmonic sum times t^{-1} , and the Gaussian tails give the stated estimate.

In (15.6), bound the finite numerator products by $(-r; r)_\infty^2$, the finite denominator products from below by $(r; r^2)_\infty$, and $\sum_{l=0}^j r^l$ by $(1-r)^{-1}$. Euler's identity gives

$$\sum_{j \geq 0} \frac{r^{j(j-1)/2}}{(r; r)_j} = (-1; r)_\infty = 2(-r; r)_\infty.$$

Thus

$$|R(q)| \leq 1 + \frac{2(-r; r)_\infty^3}{(1-r)(r; r)_\infty(r; r^2)_\infty}. \tag{17.2}$$

The standard eta-product estimates

$$(r; r)_\infty \sim \sqrt{\frac{2\pi}{t}} e^{-\pi^2/(6t)},$$

$$(-r; r)_\infty \sim 2^{-1/2} e^{\pi^2/(12t)}, \quad (r; r^2)_\infty \sim \sqrt{2} e^{-\pi^2/(12t)}$$

prove the second bound. Crucially, $\pi^2/2 < \kappa = 13\pi^2/24$. $\square$

Corollary 17.2. *Let $p_n = [q^n]P(q)B(q)^2$. Then*

$$a_n - p_n = O\left(n^{1/4} e^{\pi\sqrt{2n}}\right). \quad (17.3)$$

After Theorem 14.1, this also gives

$$\frac{a_n - p_n}{a_n} = O\left(n^{7/4} e^{-\pi(\sqrt{13/6} - \sqrt{2})\sqrt{n}}\right). \quad (17.4)$$

Proof. Cauchy's estimate and Lemma 17.1 bound the coefficient of R by $O(t^{-1/2} \exp(nt + \pi^2/(2t)))$. Choose its own optimizing radius $t = \pi/\sqrt{2n}$. This is independent of the principal saddle used below. $\square$

This is an exponentially smaller remainder for an *exact principal generating function*. It is not a claim that the algebraic series in Theorem 14.1 converges, nor a complete transseries resolving all smaller exponential scales.

Lemma 17.3. *There are constants $c, t_0 > 0$ such that for $0 < t < t_0$ and $-\pi \leq \theta \leq \pi$,*

$$\frac{|P(e^{-t+i\theta})|}{P(e^{-t})} \leq \exp\left[-c \min\left(\frac{\theta^2}{t^3}, \frac{1}{t}\right)\right]. \quad (17.5)$$

Proof. Each numerator factor satisfies $|1 + q^{2j+1}| \leq 1 + r^{2j+1}$. For denominator factors on the consecutive block $t^{-1} \leq j \leq 2t^{-1}$,

$$|1 - r^j e^{ij\theta}|^2 = (1 - r^j)^2 + 2r^j(1 - \cos j\theta).$$

Since r^j stays in a compact subinterval of $(0, 1)$, taking logarithms gives an upper bound $-c_1 \sum_j (1 - \cos j\theta)$ for the product ratio. For $|\theta| \leq c_2 t$, the quadratic cosine bound gives $c_3 \theta^2/t^3$. For $|\theta|/t$ in a fixed compact interval away from zero, rescaled Riemann sums converge to the strictly positive integral $\int_1^2 (1 - \cos(ux)) dx$. For $|\theta| \geq C_1 t$, choose C_1 large and use $|\sum_j e^{ij\theta}| \leq 1/|\sin(\theta/2)|$ to obtain a lower bound c_4/t . These ranges cover the circle. $\square$

[write] Added 2 October 2026, batch 77P5. The proofs of Lemmas 17.1 and 17.3 are sketched at the level of standard estimates (the split of the harmonic sum, and the Riemann-sum compactness step); items 6–8 of the shipped mathematical audit spell them out. Since $P = R^1$, Lemma 17.3 is Part I's bound (6.3) for the same product, proved by the same argument; both texts are kept.

[write] Added 5 October 2026, batch 101. Part III proves Theorem 14.1 again (Theorem 25.1) without either lemma. Off the real axis it uses only the positivity of the coefficients of A , a low-index part that is exponentially smaller, and a damping bound for the finite product $(q; q)_N$ whose three ranges are written out ((28.4), (28.6), (28.7)); in the saddle window it uses an interpolation lemma instead of local expansions of B and P (Section 24.3).

18 Uniform local expansions

From now on put $q = e^{-z}$, with $t = \Re z > 0$ and $|\Im z| \leq t/8$.

Lemma 18.1. *For every fixed M , $D(e^{-z})$ has a uniform Poincaré expansion through degree $M - 1$, obtained by Taylor expanding only its summands $0 \leq j < M$. Its remaining tail is $O(z^M)$.*

Proof. Write T_j for the j th summand of D . Then

$$\frac{T_{j+1}}{T_j} = -q^j \frac{1 - q^{j+1}}{1 + q^{2j+3}}.$$

Assume $0 < t \leq 0.1$ and put $x = (j + 1)t$. If $x \leq 1$, then $|(2j + 3)\Im z| \leq (2 + t)/8 < \pi/2$, so the denominator has real part at least one. Also $|1 - e^{-(j+1)z}| \leq (j + 1)|z|$. Therefore the ratio has modulus at most

$$e^{0.1} \frac{\sqrt{65}}{8} x e^{-x} \leq \frac{e^{0.1} \sqrt{65}}{8e} < \frac{2}{3}.$$

If $x \geq 1$, the bound is

$$e^t \frac{e^{-x}(1 + e^{-x})}{1 - e^{-2x}} = \frac{e^t}{e^x - 1} \leq \frac{e^{0.1}}{e - 1} < \frac{2}{3}.$$

Since $T_M = O(z^M)$ for fixed M , geometric summation proves the uniform tail estimate. $\square$

The first coefficients are

$$\begin{aligned} D(e^{-z}) &= \frac{1}{2} - \frac{z^2}{8} + \frac{z^3}{8} - \frac{11z^4}{48} + \frac{15z^5}{32} - \frac{6737z^6}{5760} + \frac{6451z^7}{1920} - \frac{888521z^8}{80640} + O(z^9), \\ D(e^{-z})^2 &= \frac{1}{4} - \frac{z^2}{8} + \frac{z^3}{8} - \frac{41z^4}{192} + \frac{7z^5}{16} - \frac{6317z^6}{5760} + \frac{1529z^7}{480} - \frac{1702513z^8}{161280} + O(z^9). \end{aligned} \quad (18.1)$$

These are exact rational coefficients, derived from finite Taylor arithmetic.

The eta quotient identity

$$P(q) = \frac{(q^2; q^2)_\infty^2}{(q; q)_\infty^4 (q^4; q^4)_\infty} \quad (18.2)$$

and eta inversion [10, 23.18.5] imply uniformly in the same wedge

$$P(e^{-z}) = \left(\frac{z}{2\pi}\right)^{3/2} \exp\left(\frac{\kappa}{z} - \frac{z}{6}\right) \left(1 + O(e^{-c/t})\right), \quad (18.3)$$

with the principal power of z . For example, the standard product formula is

$$(e^{-z}; e^{-z})_\infty = \sqrt{\frac{2\pi}{z}} \exp\left(-\frac{\pi^2}{6z} + \frac{z}{24}\right) (1 + O(e^{-c/t})).$$

Substitution at $z, 2z, 4z$ proves both the constant and the shift in (18.3).

19 Coefficient extraction to every fixed order

Proof of Theorem 14.1. Use Cauchy's coefficient integral on $q = e^{-t+i\theta}$ with $t = \sqrt{\kappa/N}$ and separate the exact remainder R . Lemma 17.1 makes its coefficient exponentially smaller than the result. Let

$$\theta_0 = t^{3/2} \log(1/t).$$

On $|\theta| > \theta_0$, Lemma 17.3, together with the polynomial bound for B , makes PB^2 smaller than every algebraic relative order. On the remaining arc set $z = t - i\theta$. Lemma 18.1 and (18.3) give, for any fixed M ,

$$P(e^{-z})B(e^{-z})^2e^{nz} = (2\pi)^{-3/2}e^{\kappa/z+Nz} \left(\sum_{m=0}^{M-1} b_m z^{3/2+m} + O(|z|^{3/2+M}) \right), \quad (19.1)$$

up to an exponentially small term. The real part of the phase has Gaussian decrease $-c\theta^2/t^3$; hence the integrated remainder is a relative $O(t^M) = O(N^{-M/2})$.

For explicit coefficients, consider a monomial z^α in (19.1), with $\alpha = 3/2 + m$. Deform the truncated vertical saddle segment to the corresponding arc of $|z| = t$, adding radial connectors. Put $\lambda = \sqrt{\kappa N}$. On the circle the phase has real part $2\lambda \cos \phi$. On either connector, writing $|z|/t = s \in [1, \sec \phi]$, it is $\lambda \cos \phi(s + s^{-1})$. At the endpoint angle both are at most $2\lambda - c \log^2(1/t)$. Extending the circle to a Hankel contour cut along the negative real axis therefore changes the integral only by a superalgebraically small relative amount. The negative rays $z = -s$, $s \geq t$, have phase $-\kappa/s - Ns \leq -2\lambda$.

The resulting Hankel integral is

$$\frac{1}{2\pi i} \int_{\mathcal{H}} z^\alpha e^{\kappa/z+Nz} dz = \left(\frac{\kappa}{N} \right)^{(\alpha+1)/2} I_{-\alpha-1}(2\sqrt{\kappa N}). \quad (19.2)$$

One can verify (19.2) by expanding $e^{\kappa/z}$ and applying the reciprocal-Gamma Hankel formula term by term; $|z| \geq t$ ensures convergence. For fixed half-integral order the growing-exponential expansion of I [10, 10.40.1] terminates. Its coefficient at order k is

$$\frac{(-1)^k (m+2+k)!}{(m+2-k)! k! (4\sqrt{\kappa N})^k}, \quad 0 \leq k \leq m+2.$$

Collecting total order $m+k=r$ gives (14.6). The leading constant is

$$\frac{(2\pi)^{-3/2} b_0 \kappa}{2\sqrt{\pi}} = \frac{13\sqrt{2}}{768}, \quad b_0 = \frac{1}{4},$$

as required. □

[write] Added 2 October 2026, batch 77P5. The transfer (19.2)–(14.6) is Part I's (6.4)–(6.5) written in $N = n - 1/6$: with $m = j$, $\beta_k(j)/(2\sqrt{\kappa})^k$ equals the coefficient displayed above, and the factor $\exp(-z/6)$ of (18.3), which Part I expands into its s_j , is absorbed by the shift. Part I applies the transfer to R^1 and to $F_3 R^1$; Part II applies it to PB^2 , whose complement R is exponentially smaller, and that is what yields every order.

Corollary 19.1. *The sequence is eventually strictly log-concave: $a_n^2 > a_{n-1}a_{n+1}$ for all sufficiently large n . More precisely,*

$$\log \frac{a_n^2}{a_{n-1}a_{n+1}} = \frac{\sqrt{\kappa}}{2N^{3/2}} - \frac{3}{2N^2} + O(N^{-5/2}). \quad (19.3)$$

Proof. Take $M = 5$ in Theorem 14.1 and expand the logarithm. For fixed real constants $d_1, \dots, d_4$ this gives

$$\log a_n = g(N) + O(N^{-5/2}), \quad g(u) = \log C + 2\sqrt{\kappa}u - \frac{3}{2} \log u + \sum_{r=1}^4 d_r u^{-r/2}.$$

Taylor expansion of this explicit smooth function gives

$$g(N+1) - 2g(N) + g(N-1) = -\frac{\sqrt{\kappa}}{2N^{3/2}} + \frac{3}{2N^2} + O(N^{-5/2}).$$

The second difference of the three remainder values is still $O(N^{-5/2})$ by the triangle inequality; no derivative bound on the asymptotic remainder is assumed. Negating the resulting second difference proves (19.3), whose leading term is positive. $\square$

[write] Added 2 October 2026, batch 77P5. From the exact coefficients through $n = 2000$: $a_n^2 > a_{n-1}a_{n+1}$ for $3 \leq n \leq 1999$, and it fails only at $n = 1, 2$; the difference between $\log(a_n^2/(a_{n-1}a_{n+1}))$ and the two displayed terms, times $N^{5/2}$, is 0.674 at $n = 500$ and 0.593 at $n = 1999$. These are numerical observations, not part of the proof; no effective threshold is claimed.

20 Continuous inverse models and integer thresholds

All inverse statements below concern the increasing large branch. Put

$$y = 2\sqrt{\kappa(n-1/6)}, \quad K = C(2\sqrt{\kappa})^3, \quad L = \log(X/K), \quad w = \log L. \quad (20.1)$$

The transformed expansion has the form

$$X \sim K \frac{e^y}{y^3} \left(1 + \sum_{r \geq 1} \frac{h_r}{y^r} \right), \quad h_r = (2\sqrt{\kappa})^r c_r. \quad (20.2)$$

Its first coefficients are

$$\begin{aligned} h_1 &= -3, & h_2 &= 3 - 2\kappa^2, \\ h_3 &= 4\kappa^3 + 20\kappa^2, & h_4 &= -\frac{41\kappa^4}{3} - 60\kappa^3 - 90\kappa^2. \end{aligned}$$

If $\log(1 + \sum h_r y^{-r}) \sim \sum \ell_r y^{-r}$, then

$$\begin{aligned} \ell_1 &= -3, & \ell_2 &= -\frac{3}{2} - 2\kappa^2, \\ \ell_3 &= 4\kappa^3 + 14\kappa^2, & \ell_4 &= \frac{9}{4} - 42\kappa^2 - 48\kappa^3 - \frac{47\kappa^4}{3}. \end{aligned} \quad (20.3)$$

20.1 A branch explicit Lambert inverse

The leading equation $X = Ke^{y_0}/y_0^3$ has the exact large solution

$$y_0 = -3W_{-1}\left(-\frac{1}{3}(K/X)^{1/3}\right). \quad (20.4)$$

The branch W_{-1} is essential; it gives $y_0 \rightarrow \infty$. Reverting (20.2) around y_0 gives

$$y = y_0 + \frac{3}{y_0} + \frac{21/2 + 2\kappa^2}{y_0^2} + \frac{45/2 - 8\kappa^2 - 4\kappa^3}{y_0^3} + O(y_0^{-4}), \quad n = \frac{1}{6} + \frac{y^2}{4\kappa}. \quad (20.5)$$

These are inverse asymptotic models, not an assertion that a discrete sequence has a canonical noninteger inverse.

20.2 Logarithmic inverse polynomials

Substitute

$$y = L + 3 \log L + \sum_{j \geq 1} P_j(\log L) L^{-j}$$

into $L = y - 3 \log y + \sum_{r \geq 1} \ell_r y^{-r}$. At each new order P_j enters linearly with coefficient one. This determines every polynomial recursively. In particular,

$$\begin{aligned} y = L + 3w + \frac{9w + 3}{L} + \frac{-27w^2/2 + 18w + 21/2 + 2\kappa^2}{L^2} \\ + \frac{27w^3 - 189w^2/2 - 9w + 45/2 - 12\kappa^2w - 8\kappa^2 - 4\kappa^3}{L^3} + O\left(\frac{w^4}{L^4}\right). \end{aligned} \quad (20.6)$$

After any fixed order J , the remainder in the continuous inverse model for y is

$$O\left(\frac{(\log L)^{J+1}}{L^{J+1}}\right).$$

Squaring and dividing by 4κ converts this to an index-scale error. The Lambert model and the logarithmic polynomials describe the same reversion in different coordinates.

20.3 Shrinking integer brackets before rounding

For $M \geq 1$, define the explicit real function

$$F_M(x) = C(x - 1/6)^{-3/2} e^{2\sqrt{\kappa(x-1/6)}} \sum_{r=0}^{M-1} c_r(x - 1/6)^{-r/2}. \quad (20.7)$$

It is positive and increasing for sufficiently large x , with

$$\frac{d}{dx} \log F_M(x) = \frac{\sqrt{\kappa}}{\sqrt{x}} + O(x^{-1}).$$

Let $x_M(X)$ be its unique large solution of $F_M(x) = X$, and let $n_*(X) = \min\{n : a_n \geq X\}$.

Corollary 20.1. *For each fixed $M \geq 2$, there are constants $D_M, X_M > 0$ such that, for $X \geq X_M$,*

$$\lceil x_M(X) - \delta_M(X) \rceil \leq n_*(X) \leq \lceil x_M(X) + \delta_M(X) \rceil, \quad \delta_M(X) = D_M x_M(X)^{-(M-1)/2}. \quad (20.8)$$

Thus the real bracket width tends to zero. If it does not cross an integer, the threshold is determined exactly by rounding its endpoints.

Proof. Theorem 14.1 gives $a_n/F_M(n) = 1 + O(n^{-M/2})$. The derivative above converts this relative error to an index displacement $O(n^{-(M-1)/2})$. Enlarge its implied constant to obtain strict comparisons on either side of $x_M \pm \delta_M$.

The sequence is nondecreasing: choose the topmost maximum-width row and adjoin one cell immediately to its left outside the bounding box. Row intervals of an L -convex polyomino are nested, since otherwise opposite extreme cells have no internal L path. Thus the chosen row spans all columns, and the new cell connects to every old cell through that row using at most one turn. The enlarged row is uniquely longest, so removing its new leftmost cell recovers the original. This is an injection. Eventual strict increase also follows from the two-term expansion, because the positive order- $n^{-1/2}$ term in a_{n+1}/a_n dominates its order- n^{-1} uncertainty. Monotonicity and the two comparisons prove (20.8). $\square$

The constants in (20.8) are asymptotic existence constants. This report does not supply numerically certified values or a finite starting index. A small numerical model error by itself is therefore not a certified rounding rule, especially at a target $X = a_n$.

[write] Added 2 October 2026, batch 77P5: an instance of repository results. As for Part I's inverse (the note in Section 9.2), this section is an instance of the transseries-and-inversion volume [8], and no novelty is claimed for the method. The core $y_0 - 3 \log y_0 = L$ is `p0:thm:lambert-core` with $a = 1, b = -3$ and the branch W_{-1} ; the corrections in (20.5) and the polynomials (20.6) are the reversion of `p0:thm:perturbed-inversion`, in the Lambert-centred form `p0:thm:lambert-centered`; Corollary 20.1 is part (2), the separation condition, of `p0:thm:staircase`, applied to the model F_M with the model error moved into δ_M .

Relation to Part I's Theorem 9.1. Since $K = c(B^1)^3$, Part II's L at $X = y^1$ is Part I's ℓ , and $y_0 = s_0$. Part II's y is $B^1 \sqrt{n-1/6}$, while Part I's s is $B^1 \sqrt{v}$; hence $s = y + (B^1)^2/(12y) + \dots$, and $y = y_0 + 3/y_0 + \dots$ gives $s = s_0 + D_0/s_0 + \dots$ with $D_0 = (B^1)^2/12 + 3$, Part I's first correction. The two inverses invert different real functions (Part II's model F_M , Part I's interpolation $\tilde{a}$), so they agree to the orders both control, not term by term beyond them. Part II's $n_*(X)$ is Part I's $N(y)$; with monotonicity proved for every n , Part II needs no starting index n_0 in the definition. The shipped data/`02-all-orders-inverse-series.json` also records a fourth Lambert-centred coefficient, $(47\kappa^4/3 + 36\kappa^3 + 45/4)/y_0^4$, and a fourth logarithmic polynomial. They come from formal substitution (residual zero through order 4) and are not printed in the manuscript.

[write] Added 5 October 2026, batch 101. Part III sharpens the monotonicity: the same injection, with the cell adjoined on the right, misses the vertical bar of area $n+1$, so $a_{n+1} \geq a_n + 1$ for every $n \geq 1$ (Section 29). It also proves that $n - n_0(a_n) \rightarrow 1/6 + 36/(13\pi^2) < 1/2$ for the leading Lambert root $n_0 = y_0^2/(4\kappa)$ (in this part's letters, y_0 at $X = a_n$), so rounding $y_0^2/(4\kappa)$ recovers n for all large n (Remark 29.1). Its brackets (29.5) are those of Corollary 20.1 with the order shifted by one.

21 Exact checks and reproducible calculations

The proofs above do not infer their constants from numerical fits. For a separate check, set $g_n = f_{n-1}/(q; q)_n^2$. Then

$$(1 - q^n)^2 g_n = 2g_{n-1} - g_{n-2}, \quad g_{-1} = g_0 = 1, \quad A(q) = 1 + \sum_{n \geq 1} q^n (1 - q^n) g_n. \quad (21.1)$$

Truncated power series division by $1 - q^n$ is a simple shifted addition; using it twice gives exact integer coefficients in $O(N^2)$ arithmetic operations up to area N . The replay computes through $N = 2000$ and matches the first 37 displayed OEIS values.

Table 2 gives $(F_M(n)/a_n) - 1$ for selected truncations. More terms improve these examples substantially, but a Poincaré expansion is a fixed-order statement as $n \rightarrow \infty$; indefinitely increasing M at a fixed small n need not improve accuracy. The package contains exact coefficient generation,

n	$M = 1$	$M = 3$	$M = 5$	$M = 9$
100	$8.8963 \cdot 10^{-2}$	$-9.3690 \cdot 10^{-3}$	$-1.7838 \cdot 10^{-3}$	$-5.2974 \cdot 10^{-4}$
200	$5.8496 \cdot 10^{-2}$	$-3.4962 \cdot 10^{-3}$	$-3.4199 \cdot 10^{-4}$	$-2.6476 \cdot 10^{-5}$
500	$3.4317 \cdot 10^{-2}$	$-9.3753 \cdot 10^{-4}$	$-3.7809 \cdot 10^{-5}$	$-4.9192 \cdot 10^{-7}$
1000	$2.3243 \cdot 10^{-2}$	$-3.4290 \cdot 10^{-4}$	$-7.0411 \cdot 10^{-6}$	$-2.3623 \cdot 10^{-8}$
2000	$1.5900 \cdot 10^{-2}$	$-1.2444 \cdot 10^{-4}$	$-1.2961 \cdot 10^{-6}$	$-1.1150 \cdot 10^{-9}$

Table 2: Signed relative error of the shifted forward model.

rational expansion arithmetic, inverse reversion checks, and independent evaluations of the exact decomposition at real and nonreal points. These checks test transcription and normalization; the complex bounds and coefficient-transfer proof supply the asymptotic theorem. A separate audit records the precise source version reviewed. Neither these computational checks nor that review constitute machine-checked formal verification or external peer review.

[write] Added 2 October 2026, batch 77P5: files and rerun. In the repository the replay is `code/02-all-orders-replay.py` (delivered as `code/replay.py`); its recorded outputs are the files `data/02-all-orders-*.json`, and the “separate audit” is the three `02-all-orders-*.md` reviews. The replay reads its reference fixture from `./data/coefficients-2000.txt` relative to itself, a file shipped as `data/02-all-orders-coefficients-2000.txt`, so it runs only in the delivered layout; the report’s README recreates that layout on a scratch copy. The fixture agrees entry by entry with the column a_n of Part I’s `data/coefficients.csv`, $0 \leq n \leq 2000$: the two parts computed the same 2001 integers by different recurrences. On 2 October 2026 that recipe passed here: the core replay in 6 s, and the replay with the optional SymPy and mpmath parts in 78 s. All five outputs agreed with the shipped records apart from line endings (Windows writes CRLF).

[write] Added 5 October 2026, batch 101. Since batch 101 the fixture is known to be the complete OEIS b-file: Part III’s snapshot of that b-file, byte-identical to the live file on 5 October 2026, equals the fixture plus a final newline (Section 24.2). Part III extends the exact coefficients to $n = 8000$.

22 What remains open

1. **Effective constants and thresholds.** Quantifying all remainder bounds would turn the shrinking asymptotic brackets into certified finite algorithms with explicit starting ranges.
2. **Smaller exponential scales.** The exact remainder bound (17.3) is deliberately coarse. Determining the actual dominant singular behavior of R , and then secondary contributions from PB^2 , would be needed for a complete exponentially improved expansion.
3. **Large correction order.** The growth of the b_m and c_r , optimal truncation, and possible resummation are separate problems. The fixed-order proof does not settle them.
4. **Further convexity classes.** The decisive step here is the exact product-square decomposition. Whether comparable transformations can be derived from area generating functions for other

constrained polyomino classes is a concrete direction; this result alone makes no assertion about their asymptotics.

5. **The partial theta factor.** The Lambert, partial theta, and rapidly convergent q -series forms of B may permit sharper arithmetic or analytic descriptions. No modularity claim for B is required here.

[write] **Added 2 October 2026, batch 77P5.** These questions overlap Part I's research questions of Section 11: item 1 continues Part I's question 2 (explicit error constants), item 2 its question 5 (beyond-all-orders terms), item 4 its question 7 (Z-convex and other classes), and item 5 its question 4, since B is Part I's D^I (Remark 16.2). Part I's question 1 is answered by Theorem 14.1.

23 Provenance of the two parts, and where the merge had to choose

This section was written when Part II was added.

[write] **Added 5 October 2026, batch 101.** Since batch 101 the report is built from three manuscripts. The third, Part III, has its own provenance section, Section 34.

Two manuscripts. The report is built from two manuscripts, each printed in full.

- **Part I** (Sections 1–12, Appendices A and B): manuscript 05 of batch 75, “L-convex polyominoes and a colored-partition comparison”, archive A126764_Lconvex_Research.zip, arrival commit 4b874cea0, placed in 6ea60e367 and written in 936896776. It contributed the squeeze, the finite-cut proof of three corrections, the radial theorem, the interpolated inverse and the exact and numerical checks of Section 10. It names no ProveIt commit (its repository input was an overview and a targeted search), so it has no pin.
- **Part II** (Sections 14–22): manuscript 31 of batch 77, “All order asymptotics and inversion for L convex polyominoes”, archive lconvex-asymptotics-reproducibility.zip, arrival commit 096ee7b87, placed in 4f11bc9c0. It contributed the exact product-square decomposition, the all-orders coefficient theorem, log-concavity, monotonicity by injection, and the shifted inverse with integer brackets. It names no ProveIt commit and does not mention the repository, so it has no pin either.

Where the merge had to choose.

- *Placement.* Part II is printed after Part I's closing Section 12 and before Part I's appendices, so no section, theorem or equation number of Part I moves. Part I's Appendix A remains the reproduction appendix of Part I; Part II's reproduction notes are in Section 21.
- *Letters.* Part II keeps the manuscript's letters, which the shipped data and reviews use; the clashes with Part I are resolved by Table 1 and by the superscript I in write notes, not by renaming (Section 13.3). The manuscript's macros `\e`, `\ii` and `\dd` were mapped to Part I's `\ee` and two new preamble macros; three unused macros of the manuscript were dropped.
- *Duplicated results.* Results proved in both manuscripts keep both proofs, with cross-references (Section 13); no proof was shortened.

- *Bibliography.* The manuscript's entries for Castiglione–Frosini–Munarini–Restivo–Rinaldi, Guttmann–Kotěšovec, Guttmann–Massazza and OEIS A126764 cite the same works as Part I's entries [2, 3, 6, 1] and were merged into them (the page reference for Guttmann–Massazza is noted in that entry). Its DLMF entry cites other sections and a stated version, so it is kept as a separate entry [10].
- *Additions.* Section 13, this section, the remarks headed “write” and the [write] notes in Part II are editorial; Remark 16.2 is a write-phase observation, with its proof and a numerical check. No statement, proof or number of either manuscript was changed; labels of Part II carry the prefix lcp:ao:.
- *Not shipped.* The delivered PDF of manuscript 31, its build script, its delivery README, its checksum manifest MANIFEST.json with the package verifier, the replay's own copy of its coefficient output (equal to the shipped fixture plus a final newline) and the record of its PDF build. The archive survives in the arrival commit; retrieve it with `git show 096ee7b87:docs/incoming/lconvex-asymptotics-reproducibility.zip`.

Part III. All-order area asymptotics: a complete transfer proof

Parts I and II (Sections 1–23 and Appendices A and B) are unchanged except for dated [write] notes of 5 October 2026 (after the scope box, in Section 1.1, at the end of Section 8.2, in Section 10.1, in research question 7 of Section 11, in Sections 14, 17, 20, 21 and 23), a line in the title, a contents entry for this part, and a note in the bibliography entries `castiglione` and `zconvex`. No number of Part I or Part II moves.

24 Part III: source, scope and notation

This section was written when Part III was added. Sections 25–32 print the manuscript; Section 33 collects its unproved claims and open directions, and Section 34 records the provenance of Part III and the choices the write made.

24.1 Source and status

Sections 25–32 print, in full, bundle Report 95 of ProveIt’s incoming-reports intake (batch 101), titled “All Order Area Asymptotics for L Convex Polyominoes” and dated 2 October 2026. It arrived as the archive `L_Convex_Polyomino_Area_Asymptotics_Source.zip` in commit `60f54ea06` and was placed in commit `f7c612c72`. Its author line, and the author field of its PDF, read “Research note prepared with OpenAI”; like every manuscript of the intake it is treated as AI-assisted research output. It names no ProveIt commit and does not mention the repository, so it has no pin. Its only input from this report’s history is Part I’s manuscript, which it cites as a separate unpublished manuscript of 1 October 2026 [11]. It does not know Part II, which was written into the report on 2 October 2026 (commits `4f11bc9c0` and `424734dc4`). Part III is unrefereed. Nothing in it is formalized in Lean or Rocq, and its place in the collection confers no formal status on it. The manuscript’s Sections 1–7 and its Appendix A are Sections 25–32 here; a section number written out in its text (“Sections 2–4” in Section 30) uses its own numbering, so that its Sections 2–4 are Sections 26–28.

The manuscript’s own limitations are printed with it and remain in force: it gives no complete transseries and identifies no exponentially small term; its remainder constants are not effective, so its rounding statements are eventual and the onset 556 of Section 30 is an observation, not a proved cutoff; thresholds for arbitrary values need the ceiling brackets of Section 29; the coefficients $a_{2001}, \dots, a_{8000}$ are generated, not compared with published data; its literature search was targeted, and W. R. G. James’s 2005 thesis was not checked in full; it does not claim the first proof of the leading equivalent; its Bessel expressions are asymptotic shorthands, not convergent sums; and it claims no journal publication.

24.2 What was rechecked at intake and in this write

Independently of the delivery, on 5 October 2026:

- the delivered suite was rerun on a copy in the delivered layout (Python 3.14.4, SymPy 1.14.0): the multiplier expansion $d_0, \dots, d_8$ and the agreement of the two formulas for h through order 8; the

Gaussian-saddle check through z^{-4} and the formal inverse through Y^{-4} ; the exact coefficients through $n = 8000$ with all 2001 b-file terms and the literal expansion through $n = 400$; the diagnostics through $n = 8000$; and the rounding scan. All passed, and every recorded output was reproduced apart from line endings and timing fields;

- the b-file snapshot that the suite compares with is byte-identical to the live OEIS b-file (downloaded at intake, SHA-256 d036795c...), and it equals Part II's fixture data/02-all-orders-coefficients-2000.txt plus a final newline;
- Part I's code/verify.py, a different implementation, was run on a copy with `-nmax 8000`: all its assertions, among them the squeeze $0 \leq p_n - 4a_n \leq 3j_n$ of Theorem 1.2, hold through $n = 8000$, and its a_n equal the shipped data/03-transfer-exact-exact-coefficients_8000.json for every $0 \leq n \leq 8000$;
- the identities of Section 26 were evaluated at $q = 0.3$ and 0.6 to 40 digits: h against the alternating series (26.15) and against Part II's B ; $A - 1 - Mh^2 - R$; the remainder R of (26.13) against Part II's $E(q)/(q; q)_\infty$; and the bound (26.14). The discrepancies are at most $4 \cdot 10^{-28}$;
- in this write: $c_0, \dots, c_4$ from (25.6) and (27.4), and their exact (SymPy) equality with Part I's $1, -d, e, f$ and with Part II's $c_0, \dots, c_4$ re-expanded at $N = n - 1/6$ (Remark 24.1); the Al-Salam-Chihara identification (Remark 26.3) against the basic hypergeometric definition; the contour identity behind Lemma 28.1, numerically; the value of η ; and $a_{n+1} - a_n \geq 1$ for $1 \leq n < 8000$ in the shipped data, with equality only at $n = 1$.

The transfer argument of Section 28 was checked line by line at intake (the positivity induction, the low-index bound, the damping product, the interpolation constants, the saddle window and the Bessel collection) and again in this write, which spells out the last step in Lemma 28.1. An independent check of the write, made afterwards, is recorded at the end of Section 34.

24.3 What Part III adds, what is proved twice, and three proofs of the main theorem

Part III proves, from the same cited area generating function:

- the exact decomposition $A = 1 + Mh^2 + R$ with a positive series for the remainder and two real-axis bounds (Proposition 26.1). The identity itself is Part II's (16.2): Part III's M is Part II's P , its h is Part II's $B = D$, and its R is Part II's $R - 1 = E/(q; q)_\infty$. **New** are the duality proof (26.11), the positive formula (26.13), the sign $R \geq 0$ on $(0, 1)$ and the bounds (26.14);
- the multiplier h by reduction of order, with the explicit tail bound (26.8) and a proof that $h > 0$, and by the alternating series of Proposition 26.2, which is Part II's series D in (15.3) because $(-q; q^2)_{j+1} = (1 + q)(-q^3; q^2)_j$. Its alternating error bound is new;
- the complete real-axis expansion (25.7): Part I's Theorem 8.1 again, by a third route;
- **a third proof of the all-orders coefficient theorem** (Theorem 25.1), by positivity, a common partition factor and Chebyshev interpolation (Section 28). Every estimate of this route is written out;
- **the strict increase** $a_{n+1} \geq a_n + 1$ for every $n \geq 1$ (Section 29); Part II proves that a_n is nondecreasing, and strictly increasing only eventually;
- **the limit** $n - n_0(a_n) \rightarrow \eta = 1/6 + 36/(13\pi^2)$ for the leading Lambert model, hence its eventual exact-range rounding rule (Remark 29.1), with a finite scan through $n = 8000$;

- the coefficients in the variable $z = \beta\sqrt{n}$, the logarithmic coefficients $\lambda_1, \dots, \lambda_4$, a power-log inverse in $Y = \log y$, and threshold brackets (the brackets are Part II's Corollary 20.1, with the order shifted by one);
- exact coefficients through $n = 8000$ and a comparison with all 2001 b-file terms.

Three proofs of the coefficient asymptotics. The leading equivalent and the corrections d, e, f of Theorem 1.1 are now proved three times, and the all-orders expansion twice. The routes differ at every step.

Table 3: The three proofs of the coefficient asymptotics.

Step	Part I	Part II	Part III
Orders proved	leading term and d, e, f ; relative remainder $O(n^{-7/4})$ (Theorem 1.1)	every order, in $N = n - 1/6$ (Theorem 14.1); $O(n^{-2})$ at $M = 4$	every order, in n (Theorem 25.1); $O(n^{-2})$ at $m = 4$
Decomposition	squeeze (1.7) and finite cut (5.5) at $m = 3$	$A = PB^2 + R$, by Heine and Fine (Proposition 16.1)	$A = 1 + Mh^2 + R$, by duality (Proposition 26.1)
Whole circle	positive tails E_3, C_4 and explicit integrals of $ Q_3 ^{-1}$, $ Q_3 ^{-2}$ (Proposition 7.1)	bounds for $ B $ and $ R $ (Lemma 17.1) and minor arcs for P (Lemma 17.3), both sketched	positivity of A , a low part L_N and a damping factor D_N from the common factor $(q; q)_N$ (28.6), (28.7), proved in full
Saddle window	eta expansion of R^I and Taylor expansion of the rational F_3	local expansions of D and P in a wedge (Lemma 18.1, (18.3))	Chebyshev interpolation of the real expansion (28.8)–(28.10): no complex estimate of h or M
Extraction	Bessel transfer (6.4)	Hankel–Bessel (19.2)	Bessel (28.11)–(28.13), completed by Lemma 28.1

Proved in two or three parts. Each part keeps its own proof.

- The leading equivalent: Part I (Theorems 1.1 and 1.2), Part II (Theorem 14.1, $M = 1$), Part III (Theorem 25.1, $m = 1$).
- The corrections d, e, f : the three parts, with the coefficients equal exactly (Remark 24.1).
- The all-orders coefficient expansion: Parts II and III. Their coefficients determine each other (Remark 24.1).
- The radial expansion: Part I's Theorem 8.1 (rational approximants h_m/Q_m), Part II (from (16.2) and Lemma 18.1; Remark 16.2) and Part III (from (26.12) and the tail bound (26.8)). The coefficients agree: $\sum_j d_j t^j = e^{-t/6} \mathcal{F}(t)$ with Part I's $\mathcal{F}$ of (1.10).
- The product generating function: Part I's (3.6), Part II's G in (15.1), Part III's (26.3). The discrete Wronskian: (5.1) and (26.6).
- Part I's $D^I = L/R^I$, Part II's $B = D$ and Part III's h are one function: Part II's Remark 16.2, and Part III's definition (26.7) with Proposition 26.2. Part III's Lambert series (32.1) is Part II's (16.3), by the same Heine transformation.

- The eta expansion: (6.2), (18.3), (27.2). The minor-arc damping: Part I's (6.3) and Part II's Lemma 17.3 for the infinite product, Part III's (28.6) for the finite product $(q; q)_N$, by the same three-range argument.
- Monotonicity by an injection: Part II adjoins a cell on the left and proves $a_{n+1} \geq a_n$; Part III adjoins it on the right and, noting that the vertical bar is not in the image, proves $a_{n+1} \geq a_n + 1$.
- The Lambert inverse: Part I's s_0 , Part II's y_0 and Part III's z_0 are the same number at the same target; all three are instances of `p0:thm:lambert-core` [8]. Part III's brackets (29.5) are Part II's (20.8).

Only Part I has the coefficientwise squeeze with its colored-partition reading, the deficit limit, the finite-cut identity, the interpolation $v(y)$ and the tables of Section 10. Only Part II has the partial theta form (16.1), the coefficient bound (17.3) for its remainder, and eventual log-concavity.

24.4 Notation across the three parts

Part III keeps the manuscript's letters, as Part II did: its shipped scripts and data use them (for example `b_in_K` and `lambda_in_K` in `data/03-transfer-symbolic_results.json`). In Sections 25–32 a letter has its Part III meaning. Where a [write] note in Part III refers to a clashing object of another part, it adds the superscript I or II. No symbol was renamed and no normalization was changed. Table 4 lists every clash; the last column gives the tempting false reading. The manuscript itself reuses several letters (rows marked “internal”).

Table 4: Letters of Part III, the same object in Parts I and II, and the clash to avoid.

Part III	Meaning in Part III	Same object in Parts I and II	Not to be confused with
$A(q)$ K	the area generating function $13\pi^2/24$	Part I's $\mathcal{A}(q)$; Part II's $A(q)$ Part I's K ; Part II's κ	Part III's constant $\mathcal{A}$ below Part II's $K = C(2\sqrt{\kappa})^3$, which is Part III's $\mathcal{A}$
C	$1/(4(2\pi)^{3/2})$, the real-axis amplitude (25.4)	written out in Part I's (1.9)	Part II's $C = 13\sqrt{2}/768$, which is Part III's δ ; Part I's series $C(z; q)$; the constants C_0, C_M of Sections 28–29
δ	$13\sqrt{2}/768$	Part I's c ; Part II's C	Part I's correction δ in the proof of Theorem 9.1; Part II's δ_M
β	$2\sqrt{K} = \pi\sqrt{13/6}$	Part I's constant B ; Part II's $2\sqrt{\kappa}$	Part II's function $B(q)$; Part III's constant B (Section 28)
$\mathcal{A}$ p_n, p_∞	$\delta\beta^3$ (Section 29) $(q; q)_n, (q; q)_\infty$	Part I's cB^3 ; Part II's K Part I's $(q; q)_n$	Part I's generating function $\mathcal{A}(q)$ Part I's $p_n = [q^n]R^1 (\sim 4a_n)$; Part II's $p_n = [q^n]PB^2 (\sim a_n)$
u_n Q_n	f_{n-1} , with $u_0 = 1$ the auxiliary solution (26.2)	Part I's h_n Part I's Q_n , the same	Part I's $u_n = Q_n - h_n$; Part II's u_l the inverse corrections Q_r of Section 29 (internal)
V_n	$2q^n - q^{2n}$	Part I's v_j in the proof of Lemma 3.1; Part II's ϵ_k	Part I's $v_n = Q_n - 2h_n$; Part III's v_n of Section 28

Part III	Meaning in Part III	Same object in Parts I and II	Not to be confused with
$P(z)$	$\sum Q_n z^n / p_n$, (26.3)	Part I's and Part II's $G(z)$	Part II's $P(q)$, which is Part III's M ; Part I's P_m ; Part III's $P_r(T)$ and its order P in Section 28 (internal)
$D(q)$	$(-q; q^2)_\infty / p_\infty$, the limit of the increments of Q_n , (26.5)	the slope $s(q)$ of Lemma 3.1 for $y = Q$	Part I's $D = L/R$ and Part II's $D = B$, which are Part III's h ; the constant D in $k = \lceil D \log(1/t) \rceil$ (internal); Part I's D_0 ; Part II's D_M
$h(q)$	the multiplier (26.7)	Part I's $D = L/R$; Part II's $B = D$	Part I's h_n ; Part II's h_n and h_r
$B_Q(z), A_Q$	$\sum Q_n z^n / p_n^2$; $B_Q(q) - B_Q(q^2) = Mh$	$B_Q(q^j)$ is Part I's C_j^Q	Part II's $B(q)$; Part III's constant B (internal)
$M(q)$	the eta quotient (27.1)	Part I's $R(q)$; Part II's $P(q)$	Part I's model $M(x)$; Part II's order M ; the bound M of the interpolation lemma and the orders M of Section 29 (internal)
$R(q)$	the remainder (26.13), ≥ 0 on $(0, 1)$	Part II's $R - 1 = E/(q; q)_\infty$	Part I's $R(q)$, which is Part III's M ; Part II's R itself; the integer R in (29.1) (internal)
w_r	$B_Q(q^{r+1})$, Section 26	—	$w_n = v_n - v_{n-1}$ of Section 28 (internal); Part II's $w = \log L$
d_j	coefficients of the real expansion (25.7): $\sum d_j t^j = 4e^{-t/6} h(e^{-t})^2$	$e^{-t/6} \mathcal{F}(t)$ of Part I; $4e^{-t/6} \sum b_m t^m$ with Part II's b_m	the variable $d = 1 - q$ of Section 27 (internal); Part I's constant d ; Part II's local d_r
c_r	unshifted coefficients (25.6)	Part I's $1, -d, e, f$ for $r \leq 3$	Part II's c_r , which are shifted (Remark 24.1); Part I's c
v_n, w_n	$u_n/(q; q)_n$ and $v_n - v_{n-1}$ (Section 28)	v_n is Part II's h_n	Part I's v_n ; Part III's w_r
a_j	$2r^j - r^{2j}$ in Section 28	V_j at $q = r$	the coefficients a_n (internal)
L_N, H_N, G_N, D_N	low and high parts of (28.2), the numerator of H_N , the damping factor	—	the truncation index L of Section 28 (internal); Part I's slope L ; Part II's L ; $G(z)$ of Parts I and II; $D(q)$
N	$\lfloor 1/t \rfloor$ in Section 28; the threshold $N(y)$ in Section 29	$N(y)$ is Part I's $N(y)$ and Part II's n_*	Part II's $N = n - 1/6$
E	$3 + \log 2$, (28.4)	—	Part I's defect E ; Part II's $E(q)$
B, D, L, P	constants of the interpolation step in Section 28	—	the functions $B_Q, D(q), L_N, P(z)$ (internal)
z	$\beta\sqrt{n}$ in Section 29; $q = e^{-z}$ in Section 28; a generating-function variable in Sections 26 and 32	Part II's $y = \beta\sqrt{n - 1/6}$ is the shifted analogue	the three uses (internal)
b_r	$c_r \beta^r$, (29.2)	—	$b_n(q)$ of Section 30 (internal; the source says so); Part II's b_m ; Part I's b_n, b_j
λ_r	coefficients of $\log(1 + \sum b_r u^r)$	the unshifted analogue of Part II's ℓ_r (not equal)	Part II's $\lambda = \sqrt{\kappa N}$; Part I's amplitude λ

Part III	Meaning in Part III	Same object in Parts I and II	Not to be confused with
y, Y, T	target value; $\log y$; $3 \log Y - \log \mathcal{A}$	Part I's y ; Part II's X ; $Y = L + \log \mathcal{A}$ with Part II's L	Part II's y and w ; Proposition 26.2; the Chebyshev T_k ; $y = \Im z$ in (28.9) (internal); Part I's $T_n(q)$
$P_r(T)$	power-log inverse polynomials (29.4)	—	Part II's $P_j(\log L)$, polynomials in another variable
z_0, n_0	Lambert root; $z_0^2/(4K)$	Part I's s_0 ; Part II's y_0	Part I's starting index n_0 in $N(y)$
F_M, r_M	logarithmic model and its root (Section 29)	r_M plays the role of Part II's x_M	Part II's F_M , a non-logarithmic shifted model; Part I's F_3 ; the radius r
$S(z), H(z)$	$\sum u_n z^n / p_n$ and S/P (Section 32)	S is Part II's $H(z)$; H is the bracket of (15.2)	Part II's $S = 1 - B$ and H
a, b	$\pm i\sqrt{q}$ (Section 32)	the same in Part II	$a_n; b_r$

24.5 Unshifted and shifted coefficients

Part III expands in powers of $n^{-1/2}$, Part II in powers of $N^{\text{II}} = n - 1/6$. The following remark converts between them.

Remark 24.1 (write, 5 October 2026: the conversion). Put $x = n^{-1/2}$, so that $N^{\text{II}} = n(1 - x^2/6)$, and let c_s^{II} be Part II's coefficients (14.7) (with $\kappa = K$ and $C^{\text{II}} = \delta$). Then the coefficients c_r of Theorem 25.1 are the Taylor coefficients of

$$\sum_{r \geq 0} c_r x^r = \sum_{s \geq 0} c_s^{\text{II}} x^s \left(1 - \frac{x^2}{6}\right)^{-(3+s)/2} \exp\left(\frac{2\sqrt{K}}{x} \left(\sqrt{1 - \frac{x^2}{6}} - 1\right)\right), \quad (24.1)$$

where the exponent $-\sqrt{K}x/6 - \sqrt{K}x^3/144 - \dots$ is a power series in x without constant term; c_r depends only on $c_0^{\text{II}}, \dots, c_r^{\text{II}}$. For example

$$c_1 = c_1^{\text{II}} - \frac{\sqrt{K}}{6}, \quad c_2 = c_2^{\text{II}} - \frac{\sqrt{K}}{6} c_1^{\text{II}} + \frac{K}{72} + \frac{1}{4}.$$

At the level of the real expansions the same shift reads $\sum_j d_j t^j = 4e^{-t/6} \sum_m b_m^{\text{II}} t^m$, with Part II's b_m of (14.5): Part II absorbs the factor $e^{-z/6}$ of the eta quotient into e^{nz} , since $e^{nz-z/6} = e^{N^{\text{II}}z}$.

Proof. Fix m . As $n \rightarrow \infty$,

$$(N^{\text{II}})^{-(3+s)/2} = n^{-3/2} x^s \left(1 - \frac{x^2}{6}\right)^{-(3+s)/2}, \quad e^{2\sqrt{K}N^{\text{II}}} = e^{2\sqrt{K}n} \exp(2\sqrt{K}(\sqrt{N^{\text{II}}} - \sqrt{n})),$$

with $2\sqrt{K}(\sqrt{N^{\text{II}}} - \sqrt{n})$ equal to the exponent in (24.1). Every factor is analytic at $x = 0$, and $O((N^{\text{II}})^{-m/2}) = O(x^m)$. Hence Theorem 14.1 with $M = m$ gives $a_n = \delta n^{-3/2} e^{2\sqrt{K}n} (\sum_{r < m} \tilde{c}_r x^r + O(x^m))$, with $\tilde{c}_r$ the Taylor coefficients of the right side of (24.1). Theorem 25.1 gives the same expansion with c_r . An asymptotic expansion in powers of x is unique, so $\tilde{c}_r = c_r$ for $r < m$, and m is arbitrary. The examples are the first two Taylor coefficients, using $(1 - x^2/6)^{-3/2} = 1 + x^2/4 + O(x^4)$. The last statement follows from $h = B^{\text{II}} = D^{\text{II}}$ (Proposition 26.2 and (15.3)) and (27.3). $\square$

The formula was also checked symbolically for $r \leq 4$ (exact SymPy arithmetic). The values are:

r	c_r (Part III, unshifted)	value	c_r^{II} (Part II, shifted)	value
0	1	1	1	1
1	$-\frac{\sqrt{K}}{6} - \frac{3}{2\sqrt{K}} = -d$	-1.03410521762627	$-\frac{3}{2\sqrt{K}}$	-0.64874695296076
2	$\frac{1}{2} + \frac{3}{4K} - \frac{35K}{72} = e$	-1.95847649289674	$\frac{3}{4K} - \frac{K}{2}$	-2.53272698896975
3	$\frac{755K^{3/2}}{1296} + \frac{175\sqrt{K}}{72} - \frac{5}{8\sqrt{K}} = f$	12.5504457791462	$\frac{\sqrt{K}(K+5)}{2}$	11.9607911105086
4	$\frac{5}{16K} - \frac{175}{32} - \frac{3775K}{864} - \frac{29375K^2}{31104}$	-55.7596598453122	$-\frac{45}{8} - \frac{15K}{4} - \frac{41K^2}{48}$	-50.0848008875789

The manuscript prints c_1 and c_2 (Theorem 25.1) and $b_r = c_r \beta^r$ for $r \leq 4$ (Section 29); the closed form of c_4 above is b_4/β^4 . The column c_1, c_2, c_3 is Part I's $-d, e, f$ of (1.3)–(1.5).

24.6 The manuscript's abstract

The area generating function for L-convex polyominoes admits an exact decomposition into an explicit eta quotient times the square of a regular multiplier, together with a positive remainder that is exponentially smaller on the positive real axis. The decomposition follows from a product generating function for an auxiliary continuant recurrence and a duality of its ordinary generating function. It proves a complete real-axis singular expansion. A positive representation with a common finite partition factor supplies uniform minor-arc damping; an elementary interpolation lemma transports the real expansion into the saddle window. Together these arguments prove the coefficient equivalent conjectured by Guttman and Kotěšovec and extend it to every algebraic order. We give effective formulas for the coefficients, logarithmic inversion with an integer-threshold qualification, and exact numerical checks against the published coefficient data.

25 The enumeration problem and the result

An L-convex polyomino is a polyomino in which any two cells can be joined inside the polyomino by a path with at most one right-angle bend. We use the fixed area-counting convention of OEIS A126764 and include the empty object. Write

$$A(q) = \sum_{n \geq 0} a_n q^n = 1 + q + 2q^2 + 6q^3 + 15q^4 + \cdots$$

Castiglione, Frosini, Munarini, Restivo and Rinaldi [2] proved the exact generating function

$$A(q) = 1 + \sum_{k \geq 0} \frac{q^{k+1} f_k(q)}{(q; q)_k^2 (1 - q^{k+1})}, \quad (25.1)$$

where

$$f_0 = 1, \quad f_1 = 1 + 2q - q^2, \quad f_k = 2f_{k-1} - (1 - q^k)^2 f_{k-2}. \quad (25.2)$$

Here $(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j)$ and $(a; q)_\infty$ denotes the corresponding convergent infinite product when $|q| < 1$. Guttman and Kotěšovec [3] numerically conjectured

$$a_n \sim \frac{13\sqrt{2}}{768} n^{-3/2} \exp\left(\pi \sqrt{\frac{13n}{6}}\right). \quad (25.3)$$

Their introduction and the OEIS entry give the constant in (25.3); the concluding display in the paper prints its reciprocal. The exact data and the derivation below select (25.3).

[write] Added 5 October 2026, batch 101. The statement about the reciprocal was checked at intake on 5 October 2026 against arXiv:2109.09928v3: the final display of Section 2 of [3] is $3 \cdot 2^8 e^{\pi \sqrt{13n/6}} / (13\sqrt{2} n^{3/2})$, whose amplitude $3 \cdot 2^8 / (13\sqrt{2})$ is $1/\delta$, while the decimal constant quoted in that section is $\delta = 0.0239385 \dots$; the paper's notation c versus $1/c$ around its equation (22) is inconsistent in the same way. The amplitude in its equation (5) and in the OEIS entry is δ . This also settles the unchecked statement of Part II (note in Section 14).

Set

$$K = \frac{13\pi^2}{24}, \quad C = \frac{1}{4(2\pi)^{3/2}}, \quad \delta = \frac{CK}{2\sqrt{\pi}} = \frac{13\sqrt{2}}{768}. \quad (25.4)$$

Theorem 25.1. *For every fixed positive integer m ,*

$$a_n = \delta n^{-3/2} e^{2\sqrt{Kn}} \left(\sum_{r=0}^{m-1} c_r n^{-r/2} + O_m(n^{-m/2}) \right), \quad c_0 = 1. \quad (25.5)$$

The constants c_r are given effectively by

$$c_r = \sum_{j=0}^r d_j K^{j/2} \frac{(-1)^{r-j}}{(r-j)!(16\sqrt{K})^{r-j}} \prod_{\ell=1}^{r-j} ((2j+5)^2 - (2\ell-1)^2), \quad (25.6)$$

where the rational numbers d_j are the coefficients in

$$A(e^{-t}) = C t^{3/2} e^{K/t} \left(\sum_{j=0}^{J-1} d_j t^j + O_J(t^J) \right), \quad t \downarrow 0, \quad (25.7)$$

valid for every fixed J . An empty product in (25.6) is one. In particular,

$$c_1 = -\frac{\sqrt{K}}{6} - \frac{3}{2\sqrt{K}}, \quad c_2 = -\frac{35K}{72} + \frac{1}{2} + \frac{3}{4K}.$$

The theorem is an all-orders Poincaré expansion of the dominant exponential scale. It does not assert a complete transseries of secondary exponential contributions. Such contributions remain a separate problem.

[write] Added 5 October 2026, batch 101. Theorem 25.1 is the same theorem as Part II's Theorem 14.1, written in the unshifted index n instead of $N^{\text{II}} = n - 1/6$; Remark 24.1 converts the coefficients, and its table shows $c_1, c_2, c_3 = -d, e, f$ of Part I's Theorem 1.1. Here C is the real-axis amplitude $1/(4(2\pi)^{3/2})$, not Part II's C , which is δ . What is new in Part III is the proof: Section 28 carries out the coefficient transfer in full (Section 24.3).

26 An exact product decomposition

Throughout this section $0 < q < 1$. Put $p_n = (q; q)_n$, $p_\infty = (q; q)_\infty$, and extend $u_n = f_{n-1}$ by $u_0 = 1$. Thus

$$u_0 = u_1 = 1, \quad u_{n+1} = 2u_n - (1 - q^n)^2 u_{n-1}. \quad (26.1)$$

Introduce an auxiliary sequence with the same recurrence,

$$Q_0 = 1, \quad Q_1 = 2, \quad Q_{n+1} = 2Q_n - (1 - q^n)^2 Q_{n-1}. \quad (26.2)$$

These are a specialization of the Al-Salam–Chihara polynomials, but the proof below needs only (26.2).

[write] Added 5 October 2026, batch 101. This sentence is proved in Remark 26.3 at the end of this section.

26.1 The product generating function and its slope

Dividing (26.2) by p_n and summing gives

$$(1 - z)^2 P(z) = (1 + qz^2)P(qz), \quad P(z) = \sum_{n \geq 0} \frac{Q_n z^n}{p_n}, \quad P(0) = 1.$$

Iteration therefore yields

$$P(z) = \frac{(-qz^2; q^2)_\infty}{(z; q)_\infty^2} \quad (|z| < 1). \quad (26.3)$$

Let $V_n = 2q^n - q^{2n}$. The recurrence implies

$$\Delta Q_{n+1} = \Delta Q_n + V_n Q_{n-1}, \quad \Delta Q_1 = 1.$$

Hence $Q_n \geq n + 1$ and its increments increase. Also $Q_{n-1} \leq n \Delta Q_n$, so

$$\Delta Q_{n+1} \leq (1 + n V_n) \Delta Q_n.$$

Since $\sum n V_n < \infty$, the increments have a finite limit $D > 0$, and

$$n + 1 \leq Q_n \leq D(n + 1), \quad Q_n/n \longrightarrow D. \quad (26.4)$$

Comparison of the double pole of (26.3) with $Q_n/p_n \sim Dn/p_\infty$ gives

$$D = \frac{(-q; q^2)_\infty}{p_\infty}. \quad (26.5)$$

This pole comparison is an elementary Abelian statement: if $b_n/n \rightarrow L$, then $(1 - z)^2 \sum b_n z^n \rightarrow L$ as $z \uparrow 1$.

26.2 Reduction of order

A discrete Wronskian calculation, initialized by $u_0/Q_0 = 1$ and $u_1/Q_1 = 1/2$, gives

$$\frac{u_n}{Q_n} = 1 - \sum_{j=0}^{n-1} \frac{p_j^2}{Q_j Q_{j+1}}. \quad (26.6)$$

Define

$$h(q) = 1 - \sum_{j \geq 0} \frac{p_j^2}{Q_j Q_{j+1}}. \quad (26.7)$$

For every $J \geq 0$, (26.4) gives the useful explicit bound

$$0 \leq \sum_{j \geq J} \frac{p_j^2}{Q_j Q_{j+1}} \leq p_J^2 \sum_{j \geq J} \frac{1}{(j+1)(j+2)} = \frac{p_J^2}{J+1}. \quad (26.8)$$

In particular, $u_n/Q_n \rightarrow h$. To see strict positivity, observe that u_n is positive and convex, with $\Delta u_1 = 0$ and $\Delta u_2 = V_1 > 0$. Its increasing increments are bounded because $u_n \leq Q_n$ by (26.6). They therefore have a finite positive limit D_u , and $h = D_u/D > 0$.

26.3 Duality and the square of the multiplier

Let

$$B_Q(z) = \sum_{n \geq 0} \frac{Q_n z^n}{p_n^2}, \quad A_Q = B_Q(q) - B_Q(q^2), \quad M = \frac{D}{p_\infty^2} = \frac{(-q; q^2)_\infty}{p_\infty^3}.$$

The normalized recurrence has generating equation

$$(1-z)^2 B_Q(z) = 2B_Q(qz) - B_Q(q^2 z). \quad (26.9)$$

Its double-pole coefficient at $z = 1$ is M , so

$$M = 2B_Q(q) - B_Q(q^2). \quad (26.10)$$

For $r \geq 0$, the sequence $w_r = B_Q(q^{r+1})$ satisfies the same recurrence as Q_r and u_r . Matching $r = 0, 1$ in (26.10) proves

$$w_r = M u_r - A_Q Q_r.$$

As $r \rightarrow \infty$, $w_r \rightarrow 1$ whereas $Q_r \rightarrow \infty$. Division by Q_r yields

$$A_Q = M h, \quad u_r - h Q_r = \frac{B_Q(q^{r+1})}{M}. \quad (26.11)$$

Proposition 26.1. *The following decomposition and bounds are exact for $0 < q < 1$:*

$$A(q) = 1 + M(q)h(q)^2 + R(q), \quad (26.12)$$

$$R(q) = \frac{1}{M(q)} \sum_{n \geq 1} \frac{q^n B_Q(q^{n+1})}{p_{n-1} p_n}, \quad (26.13)$$

$$0 \leq R(q) \leq D(q) \frac{q}{1-q^2}, \quad \frac{R(q)}{M(q)} \leq \frac{q p_\infty^2}{1-q^2}. \quad (26.14)$$

Proof. By (25.1), $A - 1 = \sum_{n \geq 1} q^n (1 - q^n) u_n / p_n^2$. Substitute (26.11) and use $A_Q = M h$ to obtain (26.12)–(26.13). Alternatively, the remainder in (26.6) and (26.8) give

$$0 \leq u_n - h Q_n \leq \frac{Q_n p_n^2}{n+1} \leq D p_n^2.$$

Multiplying by $q^n (1 - q^n) / p_n^2$ and summing gives $R \leq D \sum_{n \geq 1} q^n (1 - q^n) = D q / (1 - q^2)$. $\square$

[write] **Added 5 October 2026, batch 101.** The identity (26.12) is Part II's (16.2): Part III's M is Part II's P (Part I's R^I), its h is Part II's $B = D$ (Part I's L^I/R^I), and its R is Part II's $R^{\text{II}} - 1 = E(q)/(q; q)_\infty$; the last equality was checked at $q = 0.3$ and 0.6 to 40 digits. Part II obtains the identity by Euler, Heine and Fine transformations; the duality argument (26.9)–(26.11), the positive series (26.13) and the bounds (26.14) are new. In Part I's letters, $M = 2C_1^Q - C_2^Q$ is the identity $R^I = 2C_1^Q - C_2^Q$ of Section 4.2, and (26.6) is the Wronskian (5.1) divided by $Q_n Q_{n+1}$.

26.4 An explicit alternating series

The multiplier also has an explicit alternating-series expression.

Proposition 26.2. For $0 < q < 1$,

$$h(q) = \frac{1}{1+q} \sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2} (q; q)_j}{(-q^3; q^2)_j}. \quad (26.15)$$

The absolute values of its summands decrease, and the error after $j = J - 1$ is at most the absolute $j = J$ summand, divided by $1 + q$.

Proof. Euler's expansion of $(q^{n+1}; q)_\infty$ gives

$$A_Q = p_\infty^{-1} \sum_{j \geq 0} \alpha_j (P(q^{j+1}) - P(q^{j+2})), \quad \alpha_j = \frac{(-1)^j q^{j(j+1)/2}}{p_j}.$$

All sums are absolutely convergent for fixed q . Shifting the second sum and using $\alpha_j - \alpha_{j-1} = q^{-j} \alpha_j$ gives

$$A_Q = \frac{P(q)}{p_\infty} \sum_{j \geq 0} \frac{(-1)^j q^{j(j-1)/2} p_j}{(-q^3; q^2)_j},$$

since $P(q^{j+1})/P(q) = p_j^2/(-q^3; q^2)_j$. Finally $P(q)/p_\infty = M/(1+q)$ and $A_Q = Mh$. Consecutive absolute summands have ratio

$$\frac{q^j(1 - q^{j+1})}{1 + q^{2j+3}} < 1,$$

which proves the remainder assertion. $\square$

[write] Added 5 October 2026, batch 101. Since $(-q; q^2)_{j+1} = (1+q)(-q^3; q^2)_j$, the series (26.15) is Part II's series D of (15.3), so Proposition 26.2 gives $h = D^{\text{II}}$; with Part II's Proposition 16.1 and Remark 16.2, h is the partial theta function (16.1) and Part I's $D^{\text{I}} = L^{\text{I}}/R^{\text{I}}$. The proof here (Euler's expansion of A_Q) differs from Part II's (Fine's transformation). The alternating error bound is new.

Remark 26.3 (write, 5 October 2026: the Al-Salam–Chihara specialization). In the normalization of Koekoek, Lesky and Swarttouw [16, §14.8], the Al-Salam–Chihara polynomials $Q_n(x; a, b | q)$ satisfy $Q_0 = 1$, $Q_1 = 2x - (a + b)$ and

$$2xQ_n = Q_{n+1} + (a + b)q^n Q_n + (1 - q^n)(1 - abq^{n-1})Q_{n-1} \quad (n \geq 1).$$

For $0 < q < 1$ and either choice of $\sqrt{q}$,

$$Q_n(q) = Q_n(1; i\sqrt{q}, -i\sqrt{q} | q) \quad (n \geq 0).$$

Indeed, with $a = i\sqrt{q}$ and $b = -i\sqrt{q}$ one has $a + b = 0$ and $ab = q$, so $(1 - q^n)(1 - abq^{n-1}) = (1 - q^n)^2$; at $x = 1$ the recurrence becomes $Q_{n+1} = 2Q_n - (1 - q^n)^2 Q_{n-1}$ with $Q_0 = 1$ and $Q_1 = 2$, which is (26.2). Both sequences are determined by these data, so they coincide. (The recurrence above is the Askey–Wilson recurrence with $c = d = 0$, rescaled by 2^n from its monic form.) As a check independent of the recurrence, the basic hypergeometric definition $Q_n(x; a, b | q) = a^{-n}(ab; q)_n {}_3\phi_2(q^{-n}, ae^{i\theta}, ae^{-i\theta}; ab, 0; q, q)$, $x = \cos \theta$, was evaluated at $x = 1$, $q = 0.37$, $0 \leq n \leq 9$ to 40 digits and agrees with (26.2). The point $x = 1$ is an endpoint of the orthogonality interval. The identification concerns Q_n as a function of x ; it says nothing yet about the zeros of Q_m as a polynomial in q , which Part I's research questions 1 and 4 ask about. The solution u_n of the same recurrence, with $u_1 = 1$ instead of 2, is not covered by this identification.

27 The complete real axis expansion

For fixed J , the bound (26.8) is $O((1-q)^{2J})$. Each finite term is rational and regular at $q = 1$, because $Q_j(1) = 2^j$. Thus (26.7) proves a complete expansion of h in integer powers of $1 - q$. The alternating formula (26.15) provides a second algorithm, with remainder $O((1-q)^J)$ after J terms.

The product M is the eta quotient

$$M(q) = \frac{(q^2; q^2)_\infty^2}{(q; q)_\infty^4 (q^4; q^4)_\infty}. \quad (27.1)$$

The standard modular product transformation, equivalently the modular inversion of Dedekind's eta function [12], yields

$$M(e^{-t}) = \left(\frac{t}{2\pi}\right)^{3/2} \exp\left(\frac{K}{t} - \frac{t}{6}\right) \left(1 + O(e^{-\pi^2/t})\right). \quad (27.2)$$

Indeed $(e^{-at}; e^{-at})_\infty = \sqrt{2\pi/(at)} \exp(-\pi^2/(6at) + at/24)(1 + O(e^{-4\pi^2/(at)}))$ for fixed $a > 0$. The bound (26.14) shows that R/M is exponentially small; $1/M$ is likewise exponentially small. Hence (26.12) proves (25.7), with the explicit formal coefficient rule

$$\sum_{j \geq 0} d_j t^j \sim 4e^{-t/6} h(e^{-t})^2. \quad (27.3)$$

This rule means finite asymptotic truncation, rather than convergence of the displayed formal series. For $d = 1 - q$, the first terms are

$$\begin{aligned} h(q) &= \frac{1}{2} - \frac{d^2}{8} - \frac{5d^4}{32} + \frac{d^5}{8} - \frac{65d^6}{128} + \frac{37d^7}{32} - \frac{2053d^8}{512} + O(d^9), \\ 4h(e^{-t})^2 &= 1 - \frac{t^2}{2} + \frac{t^3}{2} - \frac{41t^4}{48} + \frac{7t^5}{4} - \frac{6317t^6}{1440} + \frac{1529t^7}{120} - \frac{1702513t^8}{40320} + O(t^9). \end{aligned}$$

The normalized coefficients are

$$\begin{aligned} d_0 &= 1, & d_1 &= -\frac{1}{6}, & d_2 &= -\frac{35}{72}, \\ d_3 &= \frac{755}{1296}, & d_4 &= -\frac{29375}{31104}, & d_5 &= \frac{1772639}{933120}, \\ d_6 &= -\frac{31514551}{6718464}, & d_7 &= \frac{3808743271}{282175488}, & d_8 &= -\frac{85931514901}{1934917632}. \end{aligned} \quad (27.4)$$

A real-axis expansion by itself does not justify pointwise coefficient extraction. The next section supplies the missing analytic control explicitly.

[write] Added 5 October 2026, batch 101. This section re-proves Part I's Theorem 8.1 by a third route: $\sum_j d_j t^j = e^{-t/6} \mathcal{F}(t)$ with $\mathcal{F}$ of (1.10), and the expansion of h in $d = 1 - q$ is Part I's expansion of $\mathcal{D}$ in $w = 1 - q$ (Section 8.1), coefficient for coefficient. Part II's coefficients (18.1) are those of $h(e^{-z})$ and $h(e^{-z})^2$. The expansion (27.2) is (6.2) and (18.3), here with the explicit remainder exponent π^2/t from the factor $(q^4; q^4)_\infty$.

28 From real asymptotics to coefficients

The proof uses positivity, a common finite partition factor, and local analytic interpolation. It supplies the needed analytic control directly; the pitfalls of omitting complex hypotheses from Tauberian arguments are discussed in [13].

28.1 A positive representation with a common partition factor

Put $u_0 = u_1 = 1$ and $u_n = f_{n-1}$ for $n \geq 1$. The defining recurrence becomes

$$u_n = 2u_{n-1} - (1 - q^{n-1})^2 u_{n-2} \quad (n \geq 2).$$

Define $v_n = u_n / (q; q)_n$, $v_0 = 1$, and $w_n = v_n - v_{n-1}$. Then

$$(1 - q^n)v_n = 2v_{n-1} - (1 - q^{n-1})v_{n-2},$$

and therefore

$$(1 - q^n)w_n = (1 + q^n)w_{n-1} + (q^n + q^{n-1})v_{n-2}. \quad (28.1)$$

Since $w_1 = q/(1 - q)$, induction proves that all v_n and w_n have nonnegative coefficients. The exact area generating function is

$$A(q) = 1 + \sum_{n \geq 1} \frac{q^n v_n(q)}{(q; q)_{n-1}}. \quad (28.2)$$

For any integer N , split (28.2) into L_N , containing its constant term and $n \leq N$, and H_N , containing $n \geq N + 1$. Both have nonnegative coefficients, and

$$H_N(q) = \frac{G_N(q)}{(q; q)_N}, \quad G_N \text{ has nonnegative coefficients.} \quad (28.3)$$

All series here converge in the open unit disk. For completeness, put $a_j = 2r^j - r^{2j}$ for a fixed real $r < 1$. Summing the recurrence twice gives

$$u_n(r) = 1 + \sum_{j=1}^{n-1} (n - j)a_j u_{j-1}(r).$$

For $m_n = u_n(r)/(n + 1)$, it follows that $m_n \leq 1 + \sum_{j=1}^{n-1} ja_j m_{j-1}$. Discrete Gronwall bounds this by $\prod_{j \geq 1} (1 + ja_j) < \infty$. Thus $u_n(r) = O_r(n + 1)$, giving convergence of (28.2). Nonnegative coefficients give absolute convergence on smaller complex disks.

28.2 The low-index part is exponentially smaller

Take $r = e^{-t}$ and $N = \lfloor 1/t \rfloor$. Positivity and the original recurrence give $0 < u_n(r) \leq 2^n$. Hence

$$L_N(r) \leq 1 + N \frac{2^N}{(r; r)_N^2}.$$

The elementary bound $1 - e^{-x} \geq xe^{-x}$ gives

$$(r; r)_N \geq t^N N! e^{-tN(N+1)/2}.$$

Using $N! \geq (N/e)^N$ and $tN = 1 + O(t)$, we obtain

$$L_N(r) = O\left(t^{-1} \exp\left(\frac{E}{t}\right)\right), \quad E = 3 + \log 2 < K = \frac{13\pi^2}{24}. \quad (28.4)$$

Changing the absolute constant accommodates the harmless $O(1)$ in the exponential. Because L_N is coefficientwise nonnegative, the same upper bound applies to $|L_N(re^{i\theta})|$.

28.3 Uniform minor-arc damping

From (28.3),

$$|H_N(re^{i\theta})| \leq H_N(r) D_N(t, \theta), \quad D_N(t, \theta) = \prod_{j=1}^N \frac{1 - r^j}{|1 - r^j e^{ij\theta}|}. \quad (28.5)$$

There is an absolute $c > 0$, independent of sufficiently small t , such that

$$D_N(t, \theta) \leq \exp\left[-c \min\left(\frac{\theta^2}{t^3}, \frac{1}{t}\right)\right] \quad (-\pi \leq \theta \leq \pi). \quad (28.6)$$

Here is a proof. Restrict the logarithm of the product to $N/2 \leq j \leq N$. On this range r^j is bounded above and below by fixed constants strictly between zero and one. Thus

$$\log D_N \leq -c_1 \sum_{N/2 \leq j \leq N} (1 - \cos j\theta).$$

The sum is bounded below by $c_2 N \min\{1, N^2 \theta^2\}$:

- If $|\theta| \leq 1/N$, use $1 - \cos x \gg x^2$ for $|x| \leq 1$.
- For $1/N \leq |\theta| \leq C_0/N$, put $\lambda = N|\theta|$. Uniform Riemann-sum convergence on the compact interval $1 \leq \lambda \leq C_0$, and strict positivity of $\int_{1/2}^1 (1 - \cos \lambda x) dx$, give the claim.
- For $C_0/N \leq |\theta| \leq \pi$, the geometric-sum bound $|\sum e^{ij\theta}| \leq 1/|\sin(\theta/2)| \leq \pi/|\theta|$, with a sufficiently large fixed C_0 , gives the claim.

Consequently,

$$|A(re^{i\theta})| \leq L_N(r) + A(r) \exp\left[-c \min\left(\frac{\theta^2}{t^3}, \frac{1}{t}\right)\right]. \quad (28.7)$$

This is the needed quantitative exclusion of secondary arcs, including neighborhoods of other roots of unity.

[write] Added 5 October 2026, batch 101. Compare Part I's (6.3) and Part II's Lemma 17.3: the same three-range argument, there for the infinite product $R^I = P^{II}$, here for the finite partition factor $(q; q)_N$ common to every term of H_N . Together with the trivial bound $|L_N(re^{i\theta})| \leq L_N(r)$, the bound (28.7) needs no estimate of h , of M or of the remainder R off the real axis; it replaces Part II's whole-circle Lemma 17.1. In (28.6) the three ranges are complete as written: on $N/2 \leq j \leq N$ one has $e^{-1} \leq r^j \leq e^{-(1-t)/2}$, and every factor of D_N is at most one, since $|1 - r^j e^{ij\theta}|^2 = (1 - r^j)^2 + 2r^j(1 - \cos j\theta)$.

28.4 Real all-orders expansions propagate to the entire saddle window

We use the following elementary analytic lemma.

Interpolation lemma. If F is holomorphic on a neighborhood of $|w| \leq 4$, $|F(w)| \leq M$ there, and $|F(x)| \leq \varepsilon$ for $-1 \leq x \leq 1$, then, for every positive integer k ,

$$\sup_{|w| \leq 1} |F(w)| \leq 4^{k-1} \varepsilon + \frac{4}{3} M \left(\frac{2}{3} \right)^k. \quad (28.8)$$

To prove this, interpolate at the k roots of the Chebyshev polynomial T_k . Let ω_k be the monic polynomial with those roots. On $|w| \leq 1$, $|\omega_k(w)| \leq 2^k$; on $|\zeta| = 4$, $|\omega_k(\zeta)| \geq 3^k$. The contour remainder formula gives the second term in (28.8). Moreover $|\omega'_k(x_j)| = 2^{1-k} |T'_k(x_j)| \geq 2^{1-k} k$; bounding the other $k-1$ numerator factors by 2 gives a total Lagrange interpolation norm at most 4^{k-1} , which proves the first term.

Fix any positive constant B , and set

$$R_t = t^{3/2} \sqrt{B \log(1/t)}, \quad z = t + R_t w.$$

For any fixed truncation index L , apply (28.8) to

$$F_t(w) = C^{-1} z^{-3/2} e^{-K/z} A(e^{-z}) - \sum_{j=0}^{L-1} d_j z^j.$$

The powers use the branch analytic in the right half-plane. Nonnegative coefficients imply $|A(e^{-z})| \leq A(e^{-\Re z})$. For $z = x + iy$ with $|w| \leq 4$, $x \asymp t$, and the real leading asymptotic in (25.7) gives

$$|z^{-3/2} e^{-K/z} A(e^{-z})| \ll \exp \left(\frac{Ky^2}{x(x^2 + y^2)} \right) \leq \exp(O(R_t^2/t^3)) = t^{-O(B)}. \quad (28.9)$$

The polynomial subtraction remains bounded for fixed L . On the real interval $-1 \leq w \leq 1$, (25.7) gives $F_t(w) = O_L(t^L)$, uniformly. In (28.8), choose $k = \lceil D \log(1/t) \rceil$, with D sufficiently large; then choose L still larger. For any prescribed P , this makes both terms $O(t^P)$. Dropping the excess polynomial terms proves, for every fixed P ,

$$A(e^{-z}) = C z^{3/2} e^{K/z} \left(\sum_{j=0}^{P-1} d_j z^j + O_P(t^P) \right) \quad (28.10)$$

uniformly on $|z - t| \leq R_t$. Constants may depend on B, P , which is exactly what is needed. This step only requires a complete real-axis expansion, analyticity, and positivity.

28.5 Cauchy saddle and all-orders conclusion

Set $t = \sqrt{K/n}$. Cauchy's formula reads

$$a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} A(e^{-t+i\theta}) e^{nt-in\theta} d\theta.$$

For any requested algebraic accuracy, choose B sufficiently large. Equations (28.4) and (28.7) show that $|\theta| > R_t$ contributes less than the main scale by that power of t . Indeed the high-index part has factor at most $e^{-cB \log(1/t)}$, while the low-index part is exponentially smaller by $e^{-(K-E)/t}$. The main scale is $t^3 e^{2K/t}$, up to a fixed constant, and the crude length bound loses only the fixed power $t^{-3/2}$ relative to $A(r)e^{nt}$.

On $|\theta| \leq R_t$, (28.10) applies. In particular,

$$|e^{K/z+nz}| = \exp\left(\frac{2K}{t} - \frac{K\theta^2}{t(t^2 + \theta^2)}\right) \leq e^{2K/t} e^{-K\theta^2/(2t^3)}$$

for all sufficiently small t . Thus a bracket error $O(t^P)$ in (28.10) integrates to an error $O(t^P)$ relative to the main coefficient scale, without a logarithmic loss. Expanding the saddle phase $K/z + nz$ about $z = t$, with $\theta = t^{3/2}u$, gives Gaussian factor e^{-Ku^2} . Taylor expansion to arbitrary fixed order and Gaussian tail bounds justify integration term by term. Equivalently, the contribution of each $Cd_j z^{3/2+j} e^{K/z}$ is represented asymptotically by

$$Cd_j \left(\frac{K}{n}\right)^{5/4+j/2} L_{-5/2-j}(2\sqrt{Kn}). \quad (28.11)$$

Formula (28.11) is an asymptotic shorthand for the saddle expansion, rather than a claim of an exact coefficient identity. The standard large-positive-argument Bessel expansion [12] yields

$$a_n \sim \frac{CK}{2\sqrt{\pi}} n^{-3/2} e^{2\sqrt{Kn}} \sum_{r \geq 0} c_r n^{-r/2}, \quad c_0 = 1, \quad (28.12)$$

where

$$c_r = \sum_{j=0}^r d_j K^{j/2} \frac{(-1)^{r-j}}{(r-j)!(16\sqrt{K})^{r-j}} \prod_{\ell=1}^{r-j} ((2j+5)^2 - (2\ell-1)^2). \quad (28.13)$$

An empty product is 1. Since the Bessel orders are half-integral, the product is zero whenever $r-j > j+2$; this can simplify calculation. For example,

$$c_1 = d_1 \sqrt{K} - \frac{3}{2\sqrt{K}}, \quad c_2 = d_2 K - 3d_1 + \frac{3}{4K}.$$

Precisely, after retaining $r = 0, \dots, m-1$, the error in the bracket in (28.12) is $O_m(n^{-m/2})$. Finally,

$$\frac{CK}{2\sqrt{\pi}} = \frac{13\sqrt{2}}{768}, \quad 2\sqrt{Kn} = \pi\sqrt{13n/6},$$

which proves Theorem 25.1.

[write] **Added 5 October 2026, batch 101.** The step from the saddle window to (28.12) is stated above as “Taylor expansion . . . and Gaussian tail bounds” and, equivalently, by the shorthand (28.11). Lemma 28.1 and the paragraph after it carry it out exactly: on the window, each term is a Hankel integral equal to the Bessel function in (28.11), up to an error that is a large power of t once the constant B is large. This is Part II’s Hankel–Bessel step (19.2), with the contour deformation written out.

Lemma 28.1 (write, 5 October 2026: the saddle segment is part of a Hankel contour). *Let α be real, $n \geq K$, $t = \sqrt{K/n}$, $\lambda = \sqrt{Kn} = K/t$ (so $\lambda \geq 1$), and $0 < \rho \leq 1$. Let S be the segment $z = t + iy$, $|y| \leq \rho t$, oriented upwards. Then*

$$\frac{1}{2\pi i} \int_S z^\alpha e^{K/z + nz} dz = t^{\alpha+1} (I_{-\alpha-1}(2\lambda) + \varepsilon), \quad |\varepsilon| \leq 2^{(|\alpha|+1)/2} e^{2\lambda - \lambda \rho^2/2} + \pi^{-1} \Gamma(|\alpha| + 1) e^{1-\lambda}. \quad (28.14)$$

Proof. Substitute $z = ts$. Since $K/z = \lambda/s$ and $nz = \lambda s$, the left side of (28.14) is $t^{\alpha+1} \Phi_S$ with $\Phi_\Gamma = \frac{1}{2\pi i} \int_\Gamma s^\alpha e^{\lambda(s+1/s)} ds$ and S now the segment $s = 1 + i\sigma$, $|\sigma| \leq \rho$; powers are principal in the plane cut along $(-\infty, 0]$. Put $\rho_1 = \sqrt{1 + \rho^2}$ and $\psi_0 = \arctan \rho$, so that the ends of S are $\rho_1 e^{\pm i\psi_0}$. Let $\mathcal{H}$ run from $-\infty$ to $-\rho_1$ along the lower edge of the cut, along $|s| = \rho_1$ from $\arg s = -\pi$ to $-\psi_0$, up S , along $|s| = \rho_1$ from ψ_0 to π , and back to $-\infty$ along the upper edge. On $\mathcal{H}$ one has $|s| \geq 1$, so $e^{\lambda/s} = \sum_{k \geq 0} \lambda^k s^{-k}/k!$ converges uniformly there, and $\int_{\mathcal{H}} |s^{\alpha-k} e^{\lambda s}| |ds| < \infty$. Integrating term by term and using Hankel's formula $\frac{1}{2\pi i} \int_{\mathcal{H}} s^{-w} e^{\lambda s} ds = \lambda^{w-1}/\Gamma(w)$ (for $\lambda > 0$ and every complex w ; the contour may be deformed to the standard one since the integrand decays on the edges),

$$\Phi_{\mathcal{H}} = \sum_{k \geq 0} \frac{\lambda^k}{k!} \frac{\lambda^{k-\alpha-1}}{\Gamma(k-\alpha)} = \sum_{k \geq 0} \frac{\lambda^{2k-\alpha-1}}{k! \Gamma(k-\alpha)} = I_{-\alpha-1}(2\lambda),$$

by the power series of I_ν with $\nu = -\alpha - 1$ (terms with $1/\Gamma(k-\alpha) = 0$ vanish on both sides). It remains to bound the pieces of $\mathcal{H}$ outside S . On the arcs $s = \rho_1 e^{i\psi}$, $\psi_0 \leq |\psi| \leq \pi$,

$$\Re\left(s + \frac{1}{s}\right) = \left(\rho_1 + \frac{1}{\rho_1}\right) \cos \psi \leq \left(\rho_1 + \frac{1}{\rho_1}\right) \frac{1}{\rho_1} = 2 - \frac{\rho^2}{1 + \rho^2} \leq 2 - \frac{\rho^2}{2},$$

$|s^\alpha| \leq \rho_1^{|\alpha|} \leq 2^{|\alpha|/2}$, and the two arcs have total length at most $2\pi\rho_1 \leq 2\sqrt{2}\pi$; after division by 2π they contribute at most $2^{(|\alpha|+1)/2} e^{\lambda(2-\rho^2/2)}$. On the edges $s = xe^{\pm i\pi}$, $x \geq \rho_1 \geq 1$, the modulus of the integrand is $x^\alpha e^{-\lambda(x+1/x)} \leq x^{|\alpha|} e^{-x} e^{-(\lambda-1)x} \leq x^{|\alpha|} e^{-x} e^{1-\lambda}$; the two edges contribute at most $\frac{1}{\pi} \Gamma(|\alpha| + 1) e^{1-\lambda}$. Hence $|\Phi_S - I_{-\alpha-1}(2\lambda)|$ is bounded as stated. $\square$

(Dated note, 5 October 2026: the edge term in (28.14) first read $\Gamma(|\alpha| + 1) e^{1-\lambda}$, without the factor π^{-1} that the last step of the proof gives. Both bounds are true; the sharper one stands since the independent check recorded at the end of Section 34. The extraction step below uses only the factor $e^{-\lambda}$ of this term, so nothing else changes.)

The extraction step, completed (write, 5 October 2026). Take $\rho = R_t/t = \sqrt{tB \log(1/t)}$, which is at most one for small t . Then $\lambda \rho^2 = KB \log(1/t)$, so the error in (28.14) is at most $c_\alpha t^{KB/2} e^{2\lambda}$, while $I_{-\alpha-1}(2\lambda) \asymp e^{2\lambda} \lambda^{-1/2} \asymp e^{2\lambda} t^{1/2}$. In Cauchy's formula the window $|\theta| \leq R_t$ is the segment S (with $z = t - i\theta$ and $a_n = \frac{1}{2\pi i} \int A(e^{-z}) e^{nz} dz$ along the upward segment). By (28.10) with P terms, the window contributes

$$\sum_{j < P} C d_j t^{5/2+j} (I_{-5/2-j}(2\lambda) + \varepsilon_j) + O(t^{P+3} e^{2K/t}),$$

where the last term is the integrated bracket error, by the Gaussian bound for $|e^{K/z + nz}|$ displayed above and $|z|^{3/2} \leq 2t^{3/2}$. The ε_j contribute $O(t^{3+KB/2-1/2} e^{2K/t})$, which is $O(t^{P+3} e^{2K/t})$ once $KB/2 \geq P + 1/2$. The main scale is $t^{3/2} e^{2K/t}$. Finally, for half-integral order the expansion [12, 10.40.1] terminates:

$I_{-5/2-j}(x) = e^x (2\pi x)^{-1/2} (\sum_{k=0}^{j+2} (-1)^k a_k x^{-k} + O(x^{-j}))$ for every J , with $a_k = \prod_{\ell=1}^k ((2j+5)^2 - (2\ell-1)^2)/(k! 8^k)$ [12, 10.17.1]. With $x = 2\lambda$ and $t^{5/2+j}\lambda^{-1/2} = K^{1+j/2}n^{-3/2-j/2}$, the j th term is

$$\frac{CK}{2\sqrt{\pi}} n^{-3/2} e^{2\sqrt{Kn}} d_j K^{j/2} n^{-j/2} \sum_{k=0}^{j+2} \frac{(-1)^k \prod_{\ell=1}^k ((2j+5)^2 - (2\ell-1)^2)}{k! (16\sqrt{K})^k} n^{-k/2} (1 + O(n^{-J})).$$

Collecting $j+k=r$ gives (28.13); the terms with $r \geq P$ and all errors are $O(t^P) = O(n^{-P/2})$ relative to the main scale. This is Theorem 25.1 with $m = P$.

29 Coefficient expansion and exact inversion

Put

$$K = \frac{13\pi^2}{24}, \quad \beta = 2\sqrt{K} = \pi\sqrt{\frac{13}{6}}, \quad \delta = \frac{13\sqrt{2}}{768}, \quad z = \beta\sqrt{n}, \quad \mathcal{A} = \delta\beta^3 = \frac{169\sqrt{39}\pi^3}{13824}.$$

The coefficient-transfer theorem, applied to the all-orders real-axis expansion, gives, for every fixed integer $R \geq 0$,

$$a_n = \delta n^{-3/2} e^{\beta\sqrt{n}} \left(\sum_{r=0}^R \frac{b_r}{z^r} + O(z^{-R-1}) \right). \quad (29.1)$$

Here $b_0 = 1$. If d_j denotes the coefficient of t^j in the normalized real-axis expansion, the finite coefficient rule is

$$b_r = \sum_{j=\max(0, \lceil (r-2)/2 \rceil)}^r d_j (2K)^j \frac{(-1)^{r-j} (r+2)!}{2^{r-j} (r-j)! (2j+2-r)!}. \quad (29.2)$$

In particular,

$$\begin{aligned} b_1 &= -3 - \frac{K}{3}, \\ b_2 &= 3 + 2K - \frac{35}{18}K^2, \\ b_3 &= -5K + \frac{175}{9}K^2 + \frac{755}{162}K^3, \\ b_4 &= 5K - \frac{175}{2}K^2 - \frac{3775}{54}K^3 - \frac{29375}{1944}K^4. \end{aligned}$$

Thus, in the alternative convention with powers $n^{-1/2}$, the correction coefficient is $c_r = b_r/\beta^r$.

For completeness, the rule follows from the dominant saddle contributions

$$C \sum_j d_j (K/n)^{(j+5/2)/2} I_{-j-5/2}(2\sqrt{Kn}), \quad C = \frac{1}{8\sqrt{2}\pi^{3/2}}.$$

For $m = j+2$, the dominant exponential part of the half-integral Bessel expansion is

$$I_{-m-1/2}(z) \sim \frac{e^z}{\sqrt{2\pi z}} \sum_{k=0}^m \frac{(-1)^k (m+k)!}{k! (m-k)! (2z)^k}.$$

Collecting $j + k = r$ gives (29.2). The leading constant is $CK/(2\sqrt{\pi}) = \delta$. These Bessel expressions describe the dominant saddle: they do not claim a convergent infinite Bessel sum or determine exponentially small contributions from other parts of the contour. The coefficient-transfer proof is needed to establish (29.1); a positive real-axis asymptotic by itself would not suffice. An independent Gaussian-saddle expansion checks (29.2) through order z^{-4} in the supplied symbolic code.

29.1 The logarithmic expansion

Define λ_r by the formal series

$$\log \left(1 + \sum_{r \geq 1} b_r u^r \right) = \sum_{r \geq 1} \lambda_r u^r.$$

A finite recursion, requiring no convergence assumption, is

$$\lambda_r = b_r - \frac{1}{r} \sum_{k=1}^{r-1} k \lambda_k b_{r-k}.$$

The first four coefficients are

$$\begin{aligned} \lambda_1 &= -3 - \frac{K}{3}, \\ \lambda_2 &= -2K^2 + K - \frac{3}{2}, \\ \lambda_3 &= 4K^3 + \frac{251}{18}K^2 - K, \\ \lambda_4 &= -\frac{47}{3}K^4 - \frac{148}{3}K^3 - \frac{125}{3}K^2 - K + \frac{9}{4}. \end{aligned}$$

For every fixed R this gives

$$\log a_n = z - 3 \log z + \log \mathcal{A} + \sum_{r=1}^R \frac{\lambda_r}{z^r} + O(z^{-R-1}). \quad (29.3)$$

29.2 Strict increase and the meaning of an inverse

We count L-convex polyominoes up to translation. L-convexity implies column convexity: two cells in the same column can be joined by a path with at most one turn only if the intervening vertical segment lies in the polyomino. The same argument gives row convexity, so each occupied row has an interval of columns. These row intervals are pairwise nested. Indeed, if two were incomparable, choose a cell in each interval lying outside the other. Neither of the two possible one-turn corner cells is present, contradicting L-convexity.

Choose the topmost widest row and append one cell to its right end. Nestedness means that the chosen row meets every occupied column. The new cell can therefore be joined to any old cell by following the chosen row and then the old cell's column. Column convexity proves that this path stays inside. Old connections are unchanged, so the result remains L-convex. The added cell is the sole cell in a new rightmost column; deleting that column recovers the original translation class.

This is an injection from area n to area $n + 1$. The vertical column of area $n + 1$ is outside its image. Consequently

$$a_{n+1} \geq a_n + 1 \quad (n \geq 1).$$

The empty polyomino is a separate initial case, with $a_0 = a_1 = 1$. Thus for $y > 1$ in the exact range there is a unique positive integer n with $a_n = y$. For arbitrary real $y > 1$ define the threshold

$$N(y) = \min\{n \geq 1 : a_n \geq y\}.$$

Exact-range inversion and this step-valued threshold must be distinguished.

[write] Added 5 October 2026, batch 101. Part II's Corollary 20.1 uses the same injection, mirrored (a cell adjoined on the left of the row), and concludes only that a_n is nondecreasing, with strict increase for large n from the asymptotics. The observation that the vertical bar is not in the image sharpens this to $a_{n+1} \geq a_n + 1$ for every $n \geq 1$. In the shipped data $a_{n+1} - a_n \geq 1$ for $1 \leq n < 8000$, with equality only at $n = 1$ ($a_2 = 2 = a_1 + 1$).

29.3 A Lambert model and its corrections

The leading smooth model is $y = \mathcal{A}e^z/z^3$. On its increasing branch,

$$z_0(y) = -3W_{-1}\left(-\frac{1}{3}(\mathcal{A}/y)^{1/3}\right), \quad n_0(y) = \frac{z_0(y)^2}{4K}.$$

It is real for $y \geq \mathcal{A}e^3/27$, and $z_0 \geq 3$. On the exact range $y = a_n \rightarrow \infty$, inversion of (29.3) gives

$$\begin{aligned} z &= z_0 - \frac{\lambda_1}{z_0} - \frac{3\lambda_1 + \lambda_2}{z_0^2} - \frac{\lambda_1^2 + 9\lambda_1 + 3\lambda_2 + \lambda_3}{z_0^3} + O(z_0^{-4}), \\ n &= \frac{1}{4K} \left[z_0^2 - 2\lambda_1 - \frac{6\lambda_1 + 2\lambda_2}{z_0} - \frac{\lambda_1^2 + 18\lambda_1 + 6\lambda_2 + 2\lambda_3}{z_0^2} + O(z_0^{-3}) \right]. \end{aligned}$$

There is a finite recursive extension to every order: substitute $z = z_0 + \sum_{r \geq 1} Q_r z_0^{-r}$ in

$$(z - z_0) - 3 \log(z/z_0) + \sum_{r \geq 1} \lambda_r z^{-r} = 0$$

and cancel powers successively. In particular,

$$n - n_0(a_n) \longrightarrow \eta = -\frac{\lambda_1}{2K} = \frac{1}{6} + \frac{36}{13\pi^2} \approx 0.447248.$$

The bound $0 < \eta < 1/2$ follows already from $\pi > 3$. Thus nearest-integer rounding of $n_0(a_n)$ recovers n for all sufficiently large n . This is an eventual, non-effective statement until explicit remainder constants are provided; it does not justify such rounding for arbitrary y or at a claimed finite numerical cutoff.

Remark 29.1 (write, 5 October 2026: the limit η). The limit is proved as follows. Theorem 25.1 with $m = 2$ gives (29.3) with $R = 1$: $\log a_n = g(z) + \log \mathcal{A} + \lambda_1/z + O(z^{-2})$, where $z = \beta\sqrt{n}$ and $g(x) = x - 3 \log x$. For large n , $a_n \geq \mathcal{A}e^3/27$, and $z_0 = z_0(a_n) \geq 3$ satisfies $a_n = \mathcal{A}e^{z_0}/z_0^3$, that is $\log a_n = g(z_0) + \log \mathcal{A}$; this is the defining equation of W_{-1} after the substitution $w = -z_0/3$. Hence $g(z) - g(z_0) = -\lambda_1/z + O(z^{-2})$. Since g is increasing on $[3, \infty)$ and $g(z_0) = \log a_n - \log \mathcal{A} \rightarrow \infty$,

also $z_0 \rightarrow \infty$, and $g' \geq 1/2$ on $[6, \infty)$ gives $z - z_0 = O(1/z)$. By the mean value theorem $g(z) - g(z_0) = (z - z_0)(1 - 3/\xi)$ with ξ between z and z_0 , so $z - z_0 = -\lambda_1/z + O(z^{-2}) = -\lambda_1/z_0 + O(z_0^{-2})$ (the sign agrees with the first correction of the display above). Therefore

$$n - n_0(a_n) = \frac{z^2 - z_0^2}{4K} = \frac{(z - z_0)(z + z_0)}{4K} = -\frac{\lambda_1}{2K} + O(n^{-1/2}),$$

and $-\lambda_1/(2K) = (3 + K/3)/(2K) = 1/6 + 36/(13\pi^2) = \eta$. Finally $\eta < 1/2$ if and only if $\pi^2 > 108/13$, which holds since $\pi > 3$; so $n_0(a_n) \in (n - 1/2, n + 1/2)$ for all large n , and nearest-integer rounding recovers n . Part I's table of the leading inverse in Section 10.3 ("leading inverse minus n ", from -0.5986 at $n = 50$ to -0.4762 at $n = 2000$) is $-(n - n_0(a_n))$ at the same points, since Part I's s_0 is z_0 ; it shows the approach to $-\eta = -0.44725 \dots$ from below.

29.4 All orders power log inversion

Put

$$Y = \log y, \quad T = 3 \log Y - \log \mathcal{A}.$$

For $y = a_n \rightarrow \infty$ the complete inverse has the power-log form

$$z = Y + T + \sum_{r=1}^M \frac{P_r(T)}{Y^r} + O\left(\frac{(\log Y)^{M+1}}{Y^{M+1}}\right) \quad (29.4)$$

for every fixed $M \geq 0$. For $M = 0$ the remainder is understood as $O((\log Y)/Y)$. The first four polynomials are

$$\begin{aligned} P_1(T) &= 3T - \lambda_1, \\ P_2(T) &= -\frac{3}{2}T^2 + (9 + \lambda_1)T - 3\lambda_1 - \lambda_2, \\ P_3(T) &= T^3 - \left(\lambda_1 + \frac{27}{2}\right)T^2 + (27 + 9\lambda_1 + 2\lambda_2)T - \lambda_1^2 - 9\lambda_1 - 3\lambda_2 - \lambda_3, \\ P_4(T) &= -\frac{3}{4}T^4 + \left(\lambda_1 + \frac{33}{2}\right)T^3 - \left(\frac{33}{2}\lambda_1 + 3\lambda_2 + 81\right)T^2 \\ &\quad + (3\lambda_1^2 + 54\lambda_1 + 15\lambda_2 + 3\lambda_3 + 81)T \\ &\quad - \frac{15}{2}\lambda_1^2 - 3\lambda_1\lambda_2 - 27\lambda_1 - 9\lambda_2 - 3\lambda_3 - \lambda_4. \end{aligned}$$

Here is a finite rule for all orders. Let $V_{m-1}(u) = T + \sum_{r=1}^{m-1} P_r(T)u^r$. Then

$$P_m(T) = -[u^m] \left\{ V_{m-1}(u) - T - 3 \log(1 + uV_{m-1}(u)) + \sum_{j=1}^m \lambda_j u^j (1 + uV_{m-1}(u))^{-j} \right\}.$$

Each use requires only finite Taylor expansions. This also proves by induction that P_m is a polynomial of degree at most m . Substitution cancels the logarithmic equation to the asserted order. The finite smooth logarithmic model has derivative $1 + O(z^{-1})$ with respect to z , so the mean value theorem converts the residual into the inverse remainder in (29.4). No derivative of an unspecified asymptotic remainder is taken.

Squaring gives a particularly useful version directly for the area:

$$n = \frac{1}{4K} \left[Y^2 + 2TY + T^2 + 6T - 2\lambda_1 + \frac{3T^2 + 18T - 6\lambda_1 - 2\lambda_2}{Y} + \frac{-T^3 + (54 + 6\lambda_1 + 2\lambda_2)T - \lambda_1^2 - 18\lambda_1 - 6\lambda_2 - 2\lambda_3}{Y^2} \right] + O\left(\frac{(\log Y)^4}{Y^3}\right).$$

For example, retaining only the first line gives an $o(1)$ error, namely $O((\log Y)^2/Y)$, and therefore eventually recovers the exact-range inverse by nearest-integer rounding. These are power-log asymptotic expansions, often called logarithmic transseries; no assertion of convergence or of complete exponentially small sectors is being made.

29.5 Rigorous threshold brackets and limitations

For fixed $M \geq 1$, define

$$F_M(x) = \beta\sqrt{x} - \frac{3}{2}\log x + \log \delta + \sum_{r=1}^M \frac{\lambda_r}{(\beta\sqrt{x})^r}.$$

This is strictly increasing for all sufficiently large x . Let $r_M(y)$ be its large real root $F_M(r_M(y)) = \log y$. Equation (29.3), together with $F'_M(x) \sim \beta/(2\sqrt{x})$, shows that for some constants $C_M, Y_M > 0$,

$$\lceil r_M(y) - C_M(\log y)^{-M} \rceil \leq N(y) \leq \lceil r_M(y) + C_M(\log y)^{-M} \rceil \quad (\log y \geq Y_M). \quad (29.5)$$

To justify the bracket uniformly, use the logarithmic remainder on a fixed-size neighborhood of $r_M(y)$, where $x \asymp (\log y)^2$. Increasing or decreasing x by $C_M(\log y)^{-M}$ then dominates the error in $\log a_x$ at integer arguments. The upper comparison produces a coefficient at least y ; the lower comparison excludes every smaller integer. Strict coefficient monotonicity completes the argument.

On the exact range, the stronger unrounded conclusion is $n = r_M(a_n) + O((\log a_n)^{-M})$. For general y , a step-valued threshold cannot have an unqualified $o(1)$ expansion by a smooth function. The bracket is often a single integer, and otherwise gives two consecutive possibilities once its width is less than one. An explicit power-log approximation may replace r_M , using its own remainder bound: for example the displayed expansion through Y^{-2} approximates $r_3(y)$ (and likewise $r_M(y)$ for $M \geq 3$) with error $O((\log Y)^4/Y^3)$ in the corresponding bracket. The roots r_1 and r_2 omit terms needed at this accuracy. The constants and the onset Y_M in these asymptotic assertions have not been made explicit. Thus the results prove eventual exact-range recovery and asymptotic threshold bounds, not certified finite-data cutoffs.

Remark 29.2 (write, 5 October 2026: step-valued thresholds). The sentence above that a step-valued threshold has no unqualified $o(1)$ expansion by a smooth function holds for every function g continuous on some $[y_1, \infty)$: $N(y) - g(y)$ does not tend to zero. Proof: by the strict increase $a_{n+1} \geq a_n + 1$ ($n \geq 1$), $N(a_n) = n$ and $N(y) = n + 1$ for $a_n < y \leq a_{n+1}$. If $N(y) - g(y) \rightarrow 0$, choose n with $a_n \geq y_1$ and $|N(y) - g(y)| < 1/3$ for all $y \geq a_n$. Then $g(a_n) < n + 1/3$, while $g(y) > n + 2/3$ for $a_n < y \leq a_{n+1}$; letting $y \downarrow a_n$, continuity gives $g(a_n) \geq n + 2/3$, a contradiction. (Only the jumps of N are used, so nondecreasing $a_n \rightarrow \infty$ would suffice: the same argument works at any point where N jumps.)

[write] Added 5 October 2026, batch 101: an instance of repository results. As for Parts I and II (the notes in Sections 9.2 and 20), this inversion is an instance of the transseries-and-inversion volume [8], and no novelty is claimed for the method: the core $z_0 - 3 \log z_0 = \log(y/\mathcal{A})$ is `p0:thm:lambert-core` with $a = 1$, $b = -3$ and the branch W_{-1} ; the corrected Lambert expansion and the power-log inverse (29.4) are the reversion of `p0:thm:perturbed-inversion`, in the Lambert-centred form `p0:thm:lambert-centered`; and the brackets (29.5) are part (2) of `p0:thm:staircase`. They are Part II's Corollary 20.1 with the order shifted by one: the model F_M here contains $\lambda_1, \dots, \lambda_M$, that is $M + 1$ coefficients, and its width $(\log y)^{-M} \asymp x^{-M/2}$ is Part II's δ_{M+1} . Part III's z_0 is Part II's y_0 (20.4) and Part I's s_0 at the same target, and its λ_r are the unshifted counterparts of Part II's ℓ_r (20.3). New here are the limit η and the eventual rounding rule of Remark 29.1, which Part I (rounding of the interpolation ν) and Part II (brackets) do not state.

30 Exact checks and numerical diagnostics

The accompanying code makes three logically distinct checks. First, an integer coefficient recurrence reconstructs all 2001 coefficients $a_0, \dots, a_{2000}$ in the public OEIS b-file [4], with exact agreement. Second, a literal expansion of (25.1)–(25.2) independently agrees through $n = 400$. Third, exact recurrence generation extends the sequence through $n = 8000$. The additional terms are generated data, not independently verified OEIS data.

For reproducibility, the retained normalized b-file snapshot (one index/value pair per line) has SHA-256

d036795c5c67b45836e4dbd76036ef0af520
baf4751aaf88421f853e5cedf027.

The normalized polynomial recurrence used for exact computation is

$$b_n(q) = \frac{2b_{n-1}(q) - b_{n-2}(q)}{(1 - q^n)^2}, \quad b_{-1}(q) = b_0(q) = 1,$$

where $b_n(q) = u_n(q)/(q; q)_n^2$ and the symbol $b_n(q)$ here denotes a generating series, distinct from the numerical asymptotic constants b_r in the preceding section. Since

$$A(q) = 1 + \sum_{n \geq 1} q^n (1 - q^n) b_n(q),$$

truncated integer power series suffice. Multiplication by $(1 - q^n)^{-2}$ uses the exact coefficients $j + 1$ at powers q^{nj} .

The leading normalized ratios are

n	$a_n / (\delta n^{-3/2} e^{\beta \sqrt{n}})$
100	0.88579013894172337144
500	0.95077637637757003576
1000	0.96568896870264176064
2000	0.97602525341243592074
4000	0.983206124724057642
8000	0.988210257959352301

Retained correction	$n = 2000$	$n = 8000$
z^{-1}	$+8.723657094 \cdot 10^{-4}$	$+2.308154832 \cdot 10^{-4}$
z^{-2}	$-1.309262065 \cdot 10^{-4}$	$-1.691475405 \cdot 10^{-5}$
z^{-4}	$-1.443558234 \cdot 10^{-6}$	$-4.735525978 \cdot 10^{-8}$
z^{-8}	$-1.184484504 \cdot 10^{-9}$	$-2.517837511 \cdot 10^{-12}$

Table 5: Exact-data diagnostics for the algebraic coefficient expansion.

Table 5 uses relative prediction error $\hat{a}_n/a_n - 1$ after retaining the stated order in z^{-1} , where $z = \beta\sqrt{n}$.

The leading Lambert inverse biases $n - n_0(a_n)$ at $n = 2000, 4000, 8000$ are, respectively,

$$0.47624742230330, \quad 0.46806132666914, \quad 0.46212708952736,$$

consistent with the proven limit $0.447248405983909726 \dots$. The fourth-order corrected Lambert expression has bias approximately $-1.44654 \cdot 10^{-6}$ at $n = 8000$; the power-log inverse through Y^{-3} has bias approximately $-7.80091 \cdot 10^{-6}$ there.

A finite scan finds that nearest-integer rounding of the leading Lambert model succeeds for every $556 \leq n \leq 8000$, fails for $2 \leq n \leq 555$, and has no real large-branch value at $n = 1$. This is a diagnostic observation only. It does not prove that 556 is a global cutoff; the theorem proves eventual recovery without an effective onset.

The source package includes exact coefficient generation, the normalized b-file snapshot, high-precision diagnostics, a symbolic expansion of the regular multiplier, and an independent Gaussian-saddle calculation. The scripts use explicit checks that remain active under Python's optimized mode. Numerical agreement is supplementary evidence; the asymptotic theorem rests on the estimates in Sections 2–4.

[write] Added 5 October 2026, batch 101: files and rerun. In the repository the scripts are `code/03-transfer-*`, the recorded outputs are `data/03-transfer-*.json`, and the exact-checks note is `03-transfer-exact-checks-README.md`; the report's README maps every delivered name and gives a rerun recipe on a scratch copy in the delivered layout. The b-file snapshot (`b126764_web.txt`) is not shipped: it is byte-identical to the live OEIS b-file (checked at intake, 5 October 2026) and equals Part II's fixture `data/02-all-orders-coefficients-2000.txt` plus a final newline. All 2001 terms of that b-file therefore agree with the coefficients computed in each of the three parts. The terms $2001 \leq n \leq 8000$ were also reproduced at intake by Part I's `code/verify.py`, a different implementation, so they rest on two programs, though still on the one cited generating function. The OEIS data are licensed CC BY-SA 4.0.

31 Scope of the result and further questions

The exact area generating function is established in the 2007 enumeration paper [2]; the 2023 numerical study [3] explicitly presents the leading equivalent as a conjecture. The argument here supplies an exact product decomposition, proves that equivalent, and gives its all-orders algebraic refinement and inverse expansions. A targeted public-literature search through 2 October 2026 found no subsequent published proof of this equivalent. That is a bounded search statement, not a claim of exhaustive coverage of unpublished work. In particular, the independently derived area generating function in W. R. G. James's 2005 Melbourne thesis, cited by the 2007 paper, was not checked in full.

A separate unpublished manuscript, *L-convex polyominoes and a colored-partition comparison*, prepared for Vladimir Reshetnikov with ChatGPT and dated 1 October 2026 [11], also proves the leading equivalent and three coefficient corrections, and develops an all-orders radial expansion and inverse asymptotics. It uses the same auxiliary product and eta quotient, and its radial coefficients through t^8 agree with those above. Its coefficientwise colored-partition comparison is a different route to finite-order coefficient estimates; that manuscript explicitly leaves the all-orders pointwise coefficient expansion outside its proved scope. The present common-partition-factor minor-arc bound and Chebyshev interpolation into the saddle window supply that additional coefficient-transfer step. Accordingly, the distinction claimed here is the all-orders pointwise coefficient theorem, rather than the first proof of the leading equivalent.

Remark 31.1 (write, 5 October 2026: a stale novelty sentence). The sentence “Accordingly, the distinction claimed here is the all-orders pointwise coefficient theorem, rather than the first proof of the leading equivalent” was accurate against Part I’s manuscript, which leaves that theorem open (Section 8.2, research question 1, the claim ledger). It is stale against this report. Part II, a manuscript dated 1 October 2026 and written into the report on 2 October 2026, proves the all-orders coefficient theorem (Theorem 14.1); Part III’s manuscript is dated a day later and does not know it. The corrected reading: Theorem 25.1 is the same theorem as Theorem 14.1 (Remark 24.1), and the distinction of Part III is its route, a fully written coefficient transfer (Section 28 with Lemma 28.1) that needs no complex estimate of h , of M or of the remainder, where Part II sketches its whole-circle Lemmas 17.1 and 17.3. The manuscript’s statement that it is not the first proof of the leading equivalent stands; that equivalent is now proved three times in this report (Section 24.3). The same sentence appears in the delivery README (not shipped) and in the key `known_related_unpublished_manuscript` of the shipped `data/03-transfer-PROVENANCE.json`, which stays byte-identical.

The proof is specific to L-convex area enumeration. It does not depend on the distinct 2024 Z-convex area series, whose data and ensuing analysis were later flagged as incorrect on [Tony Guttmann’s publication list](#). Nor does a perimeter generating function resolve the area problem.

[write] Added 5 October 2026, batch 101. Checked at intake on 5 October 2026: Tony Guttmann’s publication list annotates the 2024 paper *Asymptotics of Z-convex polyominoes* (cited here as [6]) with the remark that its series data are wrong, so the subsequent analysis is too. The remark is the author’s own; no erratum was examined. Part I cites that paper only as a research target (question 7 of Section 11, with a dated note) and Part II for the conjectural status of the L-convex asymptotic, which the note does not affect.

Several natural extensions remain.

1. Determine the genuinely exponentially small terms beyond (25.5). The exact remainder (26.13), modular corrections to (27.2), and neighborhoods of roots of unity can all contribute. The minor-arc estimate proves their irrelevance to every algebraic order but does not identify their individual leading terms.
2. Make remainder constants effective. This would turn the eventual inverse-rounding results into certified finite cutoffs and provide rigorous finite- n error bars for the coefficient expansion.
3. Study the large-order growth of the rational d_j . The finite recurrence for h and the alternating series (26.15) make such a study computationally accessible, but factorial growth, Borel singularities, and optimal truncation are not proved here.
4. Introduce a width or height parameter in the auxiliary recurrence, and investigate joint area–shape asymptotics. The double-pole product and its dual recurrence suggest a route, but a bivariate theorem requires its own uniform estimates.

Two other sequence families remain distinct possible research directions rather than consequences of this result. The normalized growth constant for 000-avoiding ascent sequences, OEIS A202058, is conjectural in the 2022 numerical study [14]; its finer stretched-exponential scale is not sufficiently settled to assume the present method applies. For OEIS A229741 and the late coefficients A260879, factorial-series composition already falls under general asymptotic machinery [15]; a defensible new target would concern late-term growth or optimal truncation, with an additional focused prior-art check.

[write] Added 5 October 2026, batch 101. Under Vladimir’s standing rule these questions, and the two sequence families of the preceding paragraph, are restated with their status in Section 33, where they are matched with Part I’s research questions and Part II’s open problems. The preceding paragraph is partly stale against the repository: the growth constant of A202058 is proved, and a stretched-exponential scale excluded, in the report `a202058-ascent-000-growth`, and the late-coefficient equivalent of A260879 is proved in the transseries collection (Section 33, item 5, gives the details).

32 A Lambert series for the multiplier (the manuscript’s Appendix A)

This identity is not needed for the proof of the main theorem, but can be useful for further analytic work. Let

$$S(z) = \sum_{n \geq 0} \frac{u_n z^n}{p_n}, \quad S(z) = P(z)H(z).$$

The recurrence gives $(1 - z)^2 S(z) = (1 + qz^2)S(qz) - z$. Division by the homogeneous product and iteration yield

$$H(z) = 1 - \sum_{j \geq 0} \frac{q^j z (q^{j+1} z; q)_\infty^2}{(-q^{2j+1} z^2; q^2)_\infty}.$$

Since $H(1) = h$, this gives

$$h = 1 - \frac{p_\infty^2}{(-q; q^2)_\infty} \sum_{j \geq 0} \frac{q^j (-q; q^2)_j}{p_j^2}.$$

Take $a = i\sqrt{q}$, $b = -i\sqrt{q}$. Heine’s first transformation [12], with parameters $a, b, c = q$ and argument q , cancels the infinite products and gives

$$h(q) = 1 - \sum_{n \geq 0} \frac{b^n}{1 - aq^n}. \quad (32.1)$$

The series converges absolutely for $0 < |q| < 1$ with either choice of $\sqrt{q}$; swapping the square root interchanges a, b and leaves the value unchanged by the symmetric double-series expansion. The value at $q = 0$ is obtained by continuation. Thus (32.1) is an analytic representation in the open unit disk, not a choice-dependent extension. Its use for detailed secondary exponential analysis would still require appropriate uniform estimates.

[write] Added 5 October 2026, batch 101. This is Part II’s (16.3), by the same Heine transformation; Part II’s Lemma 15.1 gives the formula for $H(z)$. Here $S(z)$ is Part II’s $H(z)$ (both are $\sum f_{n-1} z^n / (q; q)_n$), $P(z)$ is Part II’s $G(z)$, and Part III’s $H(z)$ is the bracket of (15.2). Part II goes on to the partial theta form (16.1), so that $h = 1 - \sum_{k \geq 0} q^{k(k+1)} (1 - q^{2k+1}) / (1 + q^{2k+1})$.

33 Part III: further questions and research

This section was written when Part III was added. Under Vladimir's standing rule of 4 October 2026, every claim of the manuscript that it does not prove is stated here as an open question, with its source, the argument sketched for it and what is missing. Two sketched statements of the manuscript are proved in this write and are listed for the record. No claim of the manuscript was found to be false; one sentence is stale (Remark 31.1).

1. **Exponentially small terms.** (Section 31, question 1.) Identify the exponentially small contributions beyond (25.5): from the remainder R of (26.13), from modular corrections to (27.2), and from neighbourhoods of other roots of unity. The minor-arc bound (28.7) proves only that they do not affect any algebraic order; it does not give their leading terms. The real bound $R/M \leq qp_\infty^2/(1 - q^2)$ of (26.14) is a starting point on the radius, not on the circle. This is Part II's open problem 2 and Part I's research question 5.
2. **Effective constants and the onset 556.** (Section 31, question 2, and Section 30.) Make the constants of Section 28 explicit: c in (28.6), the implied constants in (28.4) and (28.9), and the remainder constants of the real expansion. That would give certified error bars for a_n and turn the eventual rounding rule of Remark 29.1 into a theorem with an explicit starting index. Concretely: is nearest-integer rounding of $n_0(a_n)$ correct for every $n \geq 556$? The finite scan supports it through $n = 8000$: the bias $n - n_0(a_n)$ is 0.4999927 at $n = 556$ and 0.4621 at $n = 8000$, its smallest value in the scan, but nothing proves that it stays below $1/2$ beyond the scanned range. This is Part II's open problem 1 and Part I's research question 2.
3. **Large-order behaviour.** (Section 31, question 3.) Determine the growth of the rational d_j (and so of c_r): factorial growth, Borel singularities, optimal truncation. The tail bound (26.8) and the alternating series (26.15) make the computation accessible; nothing is proved. This is Part II's open problem 3.
4. **Area and shape.** (Section 31, question 4.) Introduce a width or height parameter into the auxiliary recurrence and prove joint area–shape asymptotics. The source suggests the double-pole product (26.3) and the duality (26.11) as a route; a bivariate theorem needs its own uniform estimates. This continues Part II's open problem 4 and Part I's research question 7 in another direction.
5. **Two other sequence families.** (Section 31.) The growth constant of 000-avoiding ascent sequences (OEIS A202058) is conjectural in [14], and its stretched-exponential scale is not settled enough to assume that the present method applies. For OEIS A229741 and the late coefficients A260879, the general machinery of [15] already covers factorial-series composition; a new target would concern late-term growth or optimal truncation, after a focused prior-art check. The source offers both as directions, not results.

Status in the repository (dated 5 October 2026, this write). Both families are now treated elsewhere in the repository, so the item is re-scoped. The report `oeis-sequence-asymptotics/a202058-ascent-000-growth` proves the normalized growth constant $\mu = 8/(3\pi^2)$ of A202058 conjectured in [14] (its Part I, Theorem 1.1), excludes every stretched-exponential factor $\exp(cn^\sigma)$ with $c, \sigma > 0$ (its Corollary 9.2), and proves $\limsup_n \log c_n / (\log n)^2 = 2/3$ for $c_n = a_n \mu^{-n} / n!$ (its Part III); the pointwise law $\log c_n \sim \frac{2}{3}(\log n)^2$ is open there. The volume *Late coefficients of the factorially forced Catalan recurrence* of the series-and-transseries collection (directory `Late_Coefficients_Factorially_Forced_Catalan_Recurrence` under `Analysis/Transseries/docs/series-and-transseries`) proves the late-coefficient equivalent

conjectured in A260879, $c_k \sim \Gamma(k)/(\log 2)^k$, with every fixed algebraic correction, and asserts no optimal truncation or Borel continuation. What remains of the source's suggestion is the pointwise law for A202058 and optimal truncation for A229741 and A260879; neither is connected with the method of this report.

6. **The finite denominators.** (This write, from Remark 26.3.) $Q_m(q)$ is the Al-Salam–Chihara polynomial $Q_m(1; i\sqrt{q}, -i\sqrt{q} | q)$. Does that identification help to locate the zeros of Q_m as polynomials in q , the question left open by Part I's research questions 1 and 4? Nothing is claimed.

Sketched statements proved in this write.

- The remark that Q_n is a specialization of the Al-Salam–Chihara polynomials, stated without proof in Section 26: Remark 26.3.
- The side remark of Section 29 that a step-valued threshold has no unqualified $o(1)$ expansion by a smooth function: Remark 29.2, for every continuous function.

The two steps that the manuscript states briefly, the limit η and the passage from the saddle window to Bessel functions, are written out in Remark 29.1 and Lemma 28.1.

34 Provenance of Part III, and where the write had to choose

This section was written when Part III was added.

The manuscript. Part III (Sections 25–32) is bundle Report 95, “All Order Area Asymptotics for L Convex Polyominoes”, dated 2 October 2026, batch 101 of the intake: archive L_Convex_Polyomino_Area_Asymptotics_Source.zip, arrival commit 60f54ea06, placed in f7c612c72. It contributed the duality proof of the decomposition, the transfer proof of the all-orders theorem, the strict increase, the limit η , the inverse in $z = \beta\sqrt{n}$ and $Y = \log y$, and exact coefficients through $n = 8000$. It names no ProveIt commit, so it has no pin. The report is now built from three manuscripts; Parts I and II are described in Section 23.

Where the write had to choose.

- *Placement.* Part III is an addition to this report, not a separate report: it re-proves the report's main theorem by a third route. It is printed after Part II's closing Section 23 and before Part I's appendices, so no number of Parts I or II moves. The manuscript's Appendix A is printed as Section 32.
- *Letters.* The manuscript's letters are kept, with a clash table (Section 24.4) and superscripts I, II in write notes. Its macro `\ee` is the report's; its `\dd` (a spaced differential, unused in its text) and its unused `\N`, `\Z`, `\R`, `\qP` were dropped; the pandoc command `\tightlist` was replaced by an `itemize` option.
- *Labels.* The manuscript's 54 labels carry the prefix `lcp:tr:` (so its `tr:2` is `lcp:tr:tr:2`); the write added section labels, two tables, the conversion remark and equation, four further remarks, a lemma and its equation, all with the same prefix.

- *Duplicated results.* Results proved in Parts I or II keep Part III's proof as well, with cross-references (Section 24.3); no proof was shortened. Part III's coefficients are unshifted, and Remark 24.1 converts them.
- *Bibliography.* The manuscript's entries for Castiglione–Frosini–Munarini–Restivo–Rinaldi, Guttmann–Kotěšovec and OEIS A126764 cite the same works as [2, 3, 1, 4] and were merged into them (its page reference for the first is noted in that entry). Its three DLMF entries became one entry [12]; its entries for Bringmann–Jennings-Shaffer–Mahlburg, Conway et al. and Borinsky are [13, 14, 15]; its entry for Part I's manuscript is [11], with a pointer to Part I. The write added [16] for Remark 26.3.
- *Additions.* Section 24, the two closing sections, the remarks and the lemma headed “write”, and the [write] notes are editorial. No statement, proof or number of the manuscript was changed.
- *Not shipped.* The manuscript's L^AT_EX sources, all printed here: the main file `article.tex`, its inputs `transfer.tex`, `inverse.tex`, `diagnostics.tex`, `context.tex` and `references.tex`, and the expanded `article_standalone.tex`. Also not shipped: its 16-page PDF, its delivery README, `verify_package.py` (a SHA-256 check against PROVENANCE.json), `build_local.sh` (a PDF build of the unshipped sources), the b-file snapshot (a copy of OEIS data that Part II's fixture already carries) and a duplicate of the symbolic results. The archive survives in the arrival commit; retrieve it with `git show 60f54ea06:docs/incoming/L_Convex_Polyomino_Area_Asymptotics_Source.zip`.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of Part III, made by the intake after the write (e8055d890), reread the five items of the write that carry a proof, Remarks 24.1 (with its table), 26.3, 29.1 and 29.2 and Lemma 28.1 with the extraction step after it, and spot-checked two steps of the manuscript: the Chebyshev interpolation inequality (28.8) with its application, and the strict increase $a_{n+1} \geq a_n + 1$ of Section 29. It found all seven items valid, no counterexample, and no gap in any proof chain. Three refinements of wording were adopted where they stand. The edge term of Lemma 28.1 now carries the factor π^{-1} that its proof gives (dated note after the proof). Remark 29.1 now writes out the step $g(z_0) = \log a_n - \log \mathcal{A} \rightarrow \infty$ behind $z_0 \rightarrow \infty$. Remark 29.2 now notes that only the jumps of N are used. No other statement or number changed.

The check used neither the delivered programs nor the write's. In exact SymPy arithmetic, with $\sqrt{K}$ symbolic, it expanded $h(e^{-t})$ from the alternating series (26.15) and recovered $d_0, \dots, d_8$ of (27.4) and Part II's b_m through t^8 . It confirmed that the conversion (24.1), built from Part II's c_s^{II} , gives the c_r of (25.6) for $r \leq 5$ (the table stops at $r = 4$). Every entry of the table, the two displayed conversions, the exponent series, $c_r = b_r/\beta^r$, $b_1, \dots, b_4$, $\lambda_1, \dots, \lambda_4$, $\delta = CK/(2\sqrt{\pi})$, $\mathcal{A} = \delta\beta^3$ and the closed form of η agree exactly. For Remark 26.3 it rederived the quoted recurrence from the monic Askey–Wilson recurrence at $c = d = 0$. It then compared the ${}_3\phi_2$ definition with that recurrence at the generic point $(x, a, b, q) = (0.3, 0.4, -0.25, 0.6)$, where they agree to 10^{-189} at 200 digits. At $a = -b = i\sqrt{q}$, with both choices of $\sqrt{q}$, it compared the definition with Q_n for $n \leq 11$ and $q = 0.05, 0.37, 0.81$, with agreement to 10^{-122} or better. For Lemma 28.1 it computed the segment integral Φ_5 by mpmath quadrature in 312 cases: 13 values of α from -7.3 to 12 , six values of λ from 1 to 60 , and $\rho \in \{0.05, 0.2, 0.5, 1\}$. The error $|\Phi_5 - I_{-\alpha-1}(2\lambda)|$ never exceeds the bound. Its largest ratio to the bound is 0.287 for the bound as first printed and 0.900 for the sharpened one, both at $\alpha = 9.5$, $\lambda = 1$. The whole contour $\mathcal{H}$ reproduces $L_{-7/2}(6)$ to 30 digits. For Remark 29.1 it evaluated $n - n_0(a_n)$, with W_{-1} at 50 digits, for every shipped term with $1 \leq n \leq 8000$. The bias decreases strictly on $2 \leq n \leq 8000$ and stays above $\eta = 0.44724840598\dots$. It matches the three

values printed in Section 30 to all 14 digits, and Part I's six values in Section 10.3 are its negatives to every printed digit. The quantity $(n - n_0(a_n) - \eta)\sqrt{n}$ rises from 1.07 at $n = 50$ to 1.33 at $n = 8000$, consistent with the next term of the expansion, about 1.37. There is no real W_{-1} value at $n = 1$; rounding fails for $2 \leq n \leq 555$, last with bias 0.500036, and succeeds for every $556 \leq n \leq 8000$. The observed onset 556 is thus confirmed; it remains an observation. Finally, the check recomputed $a_0, \dots, a_{1500}$ from scratch by another recurrence, for $y_n = u_n/p_n^2$, and found exact agreement with `data/03-transfer-exact-exact_coefficients_8000.json`, whose smallest increment for $1 \leq n < 8000$ is 1, attained only at $n = 1$. For (28.8), the Lebesgue sum on $|w| = 1$ stays far below 4^{k-1} for $k \leq 15$, and the whole inequality, tested on εT_m with $m \leq 13$ and $k \leq 24$, reaches at most 0.22 of its bound. This was a careful reading with numerical tests, not a formal verification or an external review.

A Reproduction commands and files

The archive contains this standalone L^AT_EX source, the compiled PDF, two verification scripts, exact coefficient data, and their verification logs. Build the document with

```
pdflatex -interaction=nonstopmode -halt-on-error lconvex_asymptotics.tex
pdflatex -interaction=nonstopmode -halt-on-error lconvex_asymptotics.tex
```

Run the checks from the archive directory with

```
python verify.py --nmax 2000 --out checks
python verify_symbolic.py
```

The first command requires only the Python standard library. The second additionally requires SymPy and mpmath. The quadratic number of arithmetic operations in the coefficient checker is not a quadratic bit-complexity claim: the integer sizes grow with n . The code uses $O(N)$ large-integer storage for each of a fixed number of truncated series. No external data download is required for reproduction.

[write] Added 2 October 2026, batch 75. In the repository the source is `article.tex`, the scripts are in `code/`, and the contents of the delivered `checks/` directory are in `data/`; the compiled PDF of the archive is not shipped. `verify.py` writes into `-out` relative to the working directory, while `verify_symbolic.py` reads `coefficients.csv` from, and writes into, a `checks/` directory beside itself. The commands above therefore work only in the delivered layout; the report's README recreates it on a scratch copy. On 2 October 2026 that recipe reproduced `coefficients.csv` and both tables byte for byte and the other outputs apart from line endings.

B A short review checklist

A reader auditing the proof can concentrate on four potential failure points. First, retain the slope term when $z \rightarrow 1$ in (3.4); setting $(1 - z)^2 C(z)$ to zero would give a false identity. Second, distinguish coefficientwise positivity from positivity on the real interval; the proof of (4.3) establishes the stronger statements explicitly. Third, the finite-cut error is estimated on the *whole* coefficient circle using positive tails and the controlled full-circle integral of Q_3^{-2} . Fourth, the real-axis all-orders series is not used as a substitute for this complex estimate. These checks isolate the substantive logical steps of the argument.

References

- [1] The OEIS Foundation Inc., *A126764: Number of L-convex polyominoes with n cells*. Definition, fixed-polyomino convention, and Guttmann–Kotěšovec conjecture dated June 9, 2021. <https://oeis.org/A126764>; internal entry <https://oeis.org/A126764/internal>. Accessed October 1, 2026.
- [2] G. Castiglione, A. Frosini, E. Munarini, A. Restivo, and S. Rinaldi, *Combinatorial aspects of L-convex polyominoes*, *European Journal of Combinatorics* **28** (2007), 1724–1741. <https://doi.org/10.1016/j.ejc.2006.06.020>. The exact recurrence used here is also explicitly reproduced in [3], equation (1.1). Part III cites its equations (24)–(25), pp. 1738–1739 (note added in batch 101).
- [3] A. J. Guttmann and V. Kotěšovec, *A numerical study of L-convex polyominoes and 201-avoiding ascent sequences*, *Séminaire Lotharingien de Combinatoire*, **87B** (2023), Lattice Path Combinatorics special issue, article 2, 11 pp.; arXiv:2109.09928, version 3 (2023). <https://arxiv.org/abs/2109.09928>. Journal page: <https://www.mat.univie.ac.at/~slc/wpapers/lattpath/2.html>.
- [4] V. Kotěšovec, *Table of $n, a(n)$ for $n = 0, \dots, 2000$* , b-file for OEIS A126764. <https://oeis.org/A126764/b126764.txt>. Selected anchor values were checked against this source.
- [5] NIST Digital Library of Mathematical Functions, *Section 27.14: Unrestricted partitions*, especially 27.14.12–27.14.14 on Dedekind’s eta function and its transformation. <https://dlmf.nist.gov/27.14>. Accessed October 1, 2026.
- [6] A. J. Guttmann and P. Massazza, *Asymptotics of Z-convex polyominoes*, *RAIRO–Theoretical Informatics and Applications* (2024). <https://www.rairo-ita.org/articles/ita/pdf/2024/01/ita220046.pdf>. Cited as a further research target, not as an input to the proof. Part II cites its p. 19, which repeats that the L-convex asymptotic is conjectural (note added in batch 77P5). In batch 101 (checked 5 October 2026) the first author’s publication list was found to annotate this paper: its series data are wrong, and so is the analysis based on them (note in Section 31).
- [7] V. Reshetnikov, *ProveIt*, mathematical research and formalization repository. <https://github.com/VladimirReshetnikov/ProveIt>. Repository overview and targeted A126764 search inspected October 1, 2026.
- [8] *Transseries and inversion*, a volume of the ProveIt repository: `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (theorems `p0:thm:lambert-core`, `p0:thm:lambert-centered`, `p0:thm:staircase`; Section `p3:sec:lambert`, `p3:thm:core-correction`, `p3:thm:staircase`). Added in writing, batch 75. Part II also cites `p0:thm:perturbed-inversion` (batch 77P5).
- [9] *Finite-prefix corrections and dual certificates for polyomino growth*, a report of the ProveIt research-report collection: `SetTheory/Cardinals/docs/reports/enumerative-combinatorics/polyomino-growth-finite-prefix-corrections`. Added in writing, batch 75.

- [10] NIST Digital Library of Mathematical Functions, version 1.2.8, 15 September 2026. Heine and Fine transformations, §17.6; eta transformation, §23.18; modified Bessel asymptotics, §10.40. Inspected 1 October 2026. Cited by Part II (batch 77P5).
- [11] *L-convex polyominoes and a colored-partition comparison: A proof of the A126764 asymptotic, three correction terms, an all-orders radial expansion, and inverse asymptotics*. Separate unpublished research manuscript prepared for Vladimir Reshetnikov with ChatGPT, 1 October 2026, 19 pp. Cited by Part III as printed; this manuscript is Part I of the present report (note added in batch 101).
- [12] NIST Digital Library of Mathematical Functions: §23.18, modular transformations, and §27.14, Dedekind eta products, <https://dlmf.nist.gov/23.18>, <https://dlmf.nist.gov/27.14>; equations 10.40.1 and 10.17.1, modified Bessel asymptotics and coefficient polynomials, <https://dlmf.nist.gov/10.40.E1>, <https://dlmf.nist.gov/10.17.E1>; equation 17.6.6, Heine's first transformation, <https://dlmf.nist.gov/17.6.E6>. Cited by Part III (the manuscript's three DLMF entries, merged in batch 101).
- [13] K. Bringmann, C. Jennings-Shaffer and K. Mahlburg, *On a Tauberian theorem of Ingham and Euler–Maclaurin summation*. <https://arxiv.org/abs/1910.03036>. See the complex hypothesis and counterexamples discussed in Theorem 1.1 and Section 3.2. Cited by Part III.
- [14] A. R. Conway, M. Conway, A. Elvey Price and A. J. Guttmann, *Pattern-avoiding ascent sequences of length 3*, *Electronic Journal of Combinatorics* **29**(4) (2022), P4.25. doi:10.37236/11266; <https://arxiv.org/abs/2111.01279>. Cited by Part III.
- [15] M. Borinsky, *Generating asymptotics for factorially divergent sequences*, *Electronic Journal of Combinatorics* **25**(4) (2018), P4.1. <https://arxiv.org/abs/1603.01236>. Cited by Part III.
- [16] R. Koekoek, P. A. Lesky and R. F. Swarttouw, *Hypergeometric Orthogonal Polynomials and Their q -Analogues*, Springer Monographs in Mathematics, Springer, Berlin, 2010; §14.8, the Al-Salam–Chihara polynomials. Added in writing, batch 101 (Remark 26.3).